Air Pollution, Spending and Mobility in Seoul: Districts

Slides of the district paper, results draft 9

APC project team

Working draft, 26 October 2026

What this deck covers

Question. On more polluted days, do Seoul residents spend less, spend differently, and stay closer to home?

Design. Seoul's 25 districts, each with its own monitor and weather station. Residents' card spending (486 days, 2018–19) and presence (243 days, 2019).

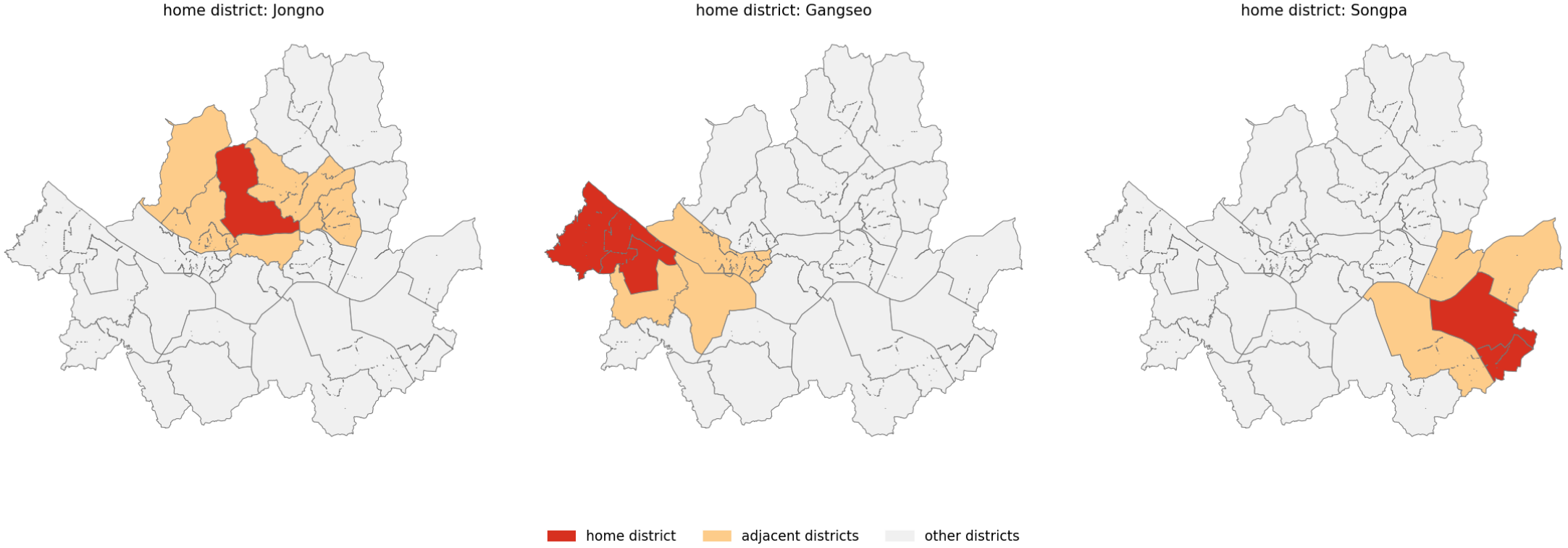
Order of the results (as in the paper):

1. The city as one daily time series: OLS, distributed lags, 2SLS/LIML, local projections
2. Weekends and holidays vs. weekdays
3. Geography within a district-day: where spending and time go
4. The size of purchases
5. Pre-specified categories
6. Air-quality alerts

Like the paper, the slides report estimates without interpreting them.

Data and specifications

Definition of home, adjacent, and other districts



Home district; adjacent districts (sharing a boundary; 2–7 per district); other districts. Online spending is a fourth destination. "Outside the home district" = adjacent + other.

Data: card spending (Shinhan Card)

- **Period**: January–August 2018 and 2019, 486 days.
- **Coverage**: Shinhan was Korea's largest card company, about 22% of credit-card sales in 2019. We keep cardholders resident in Seoul and merchants in Seoul or online.
- **Cells**: day × time slot (06–10, 10–18, 18–22, 22–06h) × merchant category (62, plus online) × merchant neighborhood × cardholder sex × ten-year age group × cardholder residence. Amount and number of transactions per cell.
- **Scaling**: amounts and counts are Shinhan's own figures times one factor per month (4.89–4.96 in 2018, 5.01–5.07 in 2019), the same for every category, place and day of a month. A count of 5 is one Shinhan transaction; shares, elasticities and amounts per transaction are unaffected; year-month fixed effects absorb the factor.
- **No cardholder identifier**: individuals cannot be followed.
- **Aggregated to** residence district × merchant district (or online) × day. City means: all spending 206 bn KRW a day, offline 175, online 31.

Data: mobility (KT inflow population)

- **Period**: January–August 2019, 243 days.
- **What it measures**: the estimated number of people present in each Seoul administrative neighborhood each hour, by home neighborhood, sex and five-year age band (*saenghwal idong*).
- **Source**: LTE signals of KT subscribers (about 31% of mobile lines), scaled to the population. Cells below about 2.9 people are suppressed and count as zero.
- **Aggregated to** people-hours over all 24 hours (and by the card data's time slots), by home district and district of presence; Seoul residents present in Seoul.
- **Different people from the card data**: Shinhan cardholders by registered address vs. KT subscribers by habitual location; the paper compares their age and district composition.

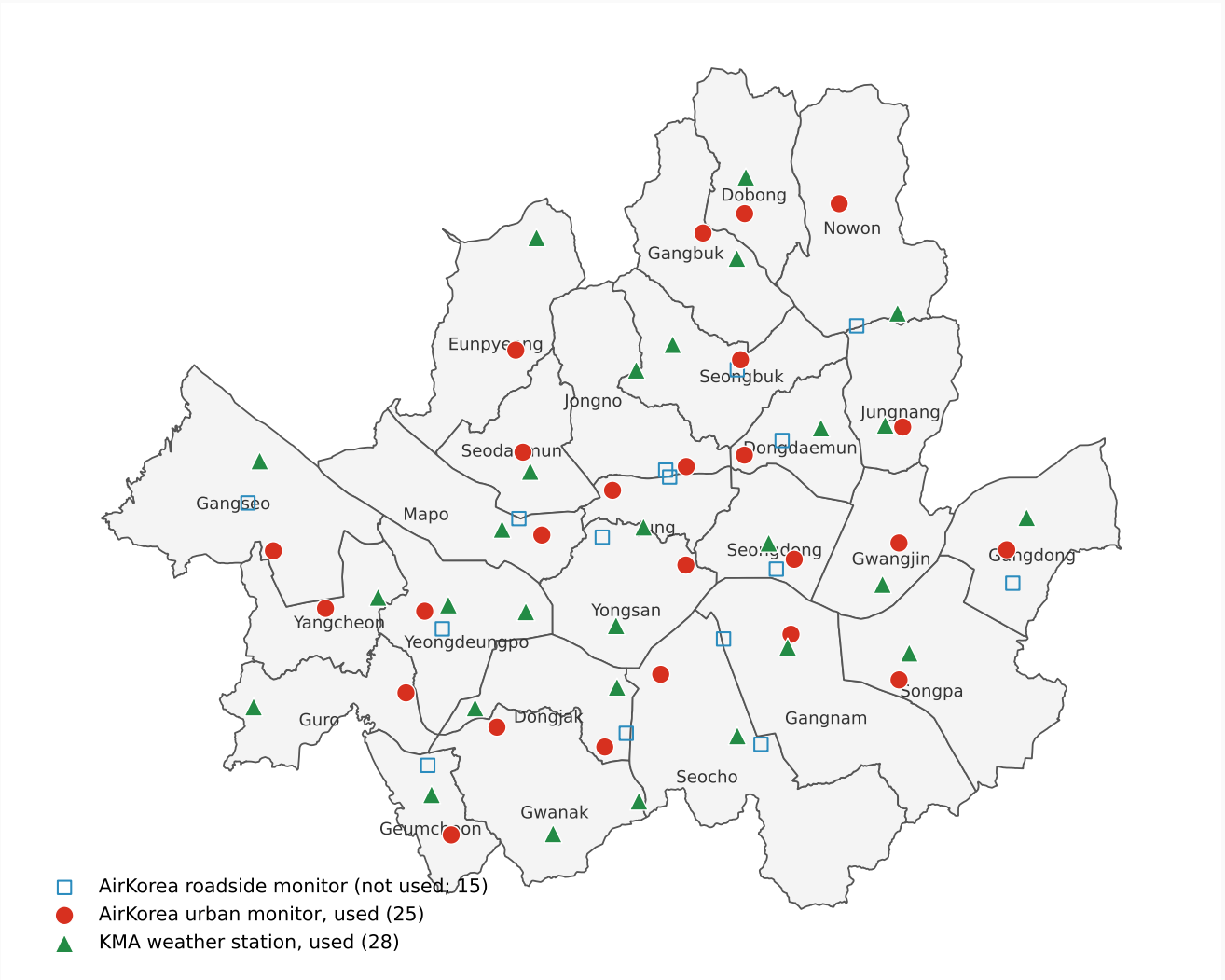
Data: pollution (AirKorea)

- **Monitors**: Seoul's urban network has exactly one monitor in each of the 25 districts; each district's PM2.5 (and other pollutants) is the daily value of its own monitor.
- **Excluded**: roadside monitors (in 15 districts), as is usual when the target is residents' general exposure.
- **Gaps**: missing days filled from neighboring monitors; all 25 monitors report on more than 80% of days.
- **City value**: mean of the 25 district monitors; mean 26 $\mu g/m^3$, standard deviation 18, range 3–135 on sample days.
- **Variation**: almost entirely from day to day, common to the whole city; differences between districts on the same day are small.

Data: weather (KMA automatic weather stations)

- **Variables**: hourly temperature, humidity, precipitation, wind speed and wind direction.
- **Stations**: each district's own station; the mean of two stations in Gwanak, Dongjak and Yeongdeungpo. Jongno's station reports on fewer than 80% of days, so Jongno takes the nearest station; humidity is measured at only 16 stations and comes from the nearest one in 11 districts.
- **Daily values**: mean, maximum and minimum temperature; mean humidity and wind speed; total precipitation.
- **Wind-direction instrument**: direction of the day's mean hourly wind vector, in eight 45° sectors (calm days separate); at the district's station, or over the districts for the city. Wind speed is a control.
- **Not available yet**: sunshine, cloud cover, air pressure, visibility (these need KMA's synoptic (ASOS) station data).

Data: where the monitors and weather stations are



Pollution: each of the 25 districts contains exactly one AirKorea urban monitor (red dots), and it is the one the district uses.

Weather: the KMA stations each district uses (green triangles) all lie inside that district; two in Gwanak, Dongjak and Yeongdeungpo.

Not used: roadside monitors (blue squares).

Data: thermal inversions (Osan radiosonde)

- **Source**: NOAA's Integrated Global Radiosonde Archive (IGRA v2.2), Osan Air Base, about 45–50 km south of Seoul City Hall.
- **Soundings**: a weather balloon carries a radiosonde that records temperature, pressure, humidity and wind as it rises; up to four a day (03, 09, 15, 21 KST).
- **Inversion**: a layer where temperature rises with height, which traps pollution near the ground.
- **Instrument**: the number of soundings (21 KST the previous day, 03 and 09 KST) with a temperature rise within the lowest 100 hPa, and the strength of the 03 KST inversion (°C).
- **One value for the whole city**: the same for all districts.

Data: air-quality alerts (Seoul)

- **Source**: Seoul's advisory and warning history, 2018–19.
- **PM alerts**: 25 PM2.5 and 14 PM10 alerts, all city-wide. Documented PM2.5 cutoff: 90 μg/m^3 before 1 July 2018, 75 from then; issuance actually jumps 5.5 μg/m^3 above it.
- **Ozone alerts**: 83, issued by area (five groups of whole districts: Downtown, Northeast, Northwest, Southwest, Southeast); cutoff 0.12 ppm.
- **Timing**: an alert is in effect from its issue hour to its clear hour; matched to the card data's time slots.

Two sets of specifications: city level and district level

City level

$$Y_t = \beta P_t + \text{CAL}_t + W_t'\delta + \varepsilon_t$$

- t: day (486 days; 243 for mobility).
- Y_t: outcome summed over Seoul residents.
- P_t: mean PM2.5 of the 25 district monitors, $\mu\text{g/m}^3$; coefficients per 10 $\mu\text{g/m}^3$.
- CAL_t: day of week, year-month (year-week in robustness), one effect per public holiday, bridge days, the days around Seollal.
- W_t: cubic in temperature; max and min temperature; rain indicator; quadratics in precipitation, humidity, wind speed.
- 2SLS and LIML: P_t instrumented by wind sectors or inversions.
- Standard errors clustered by year-week.

District level: one regression per share

$$s_{rkt} = \beta_k P_{rt} + \alpha_{rk} + \lambda_{k,\text{cal}(t)} + \lambda_{k,\text{wc}(rt)} + \lambda_{k,\text{hd}(rt)} + \varepsilon_{rkt}$$

- r: home district (25); k: band (home, adjacent, other districts, online).
- s_{rkt}: share of district r's spending (or people-hours) on day t in band k, in pp.
- P_{rt}: PM2.5 at district r's own monitor, $\mu\text{g/m}^3$; coefficients per 10 $\mu\text{g/m}^3$, as in the city series (Poisson models: log PM2.5).
- α_{rk}: district $\times$ band effects; $\lambda_{k,\text{cal}}$: the calendar terms of CAL_t, by band.
- $\lambda_{k,\text{wc}}$: weather cells (3°C temperature bin $\times$ precipitation category $\times$ wind-speed decile); $\lambda_{k,\text{hd}}$: humidity deciles; both at the district's own station, by band.
- OLS weighted by mean spending (people-hours); 2SLS and LIML. Clustered by district and date.
- The same equation with a district-day outcome gives counts, amounts per transaction and distances.

1. The city time series

City series: least squares

$$Y_t = \beta P_t + \text{CAL}_t + W_t'\delta + \varepsilon_t$$

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
A. Least squares							
Year-month FE (main)	**-1.64****	**-0.78****	**-0.87**	**-0.24**	**0.09**	**-0.06*****	**-0.005****
	(0.81)	(0.36)	(0.64)	(0.23)	(0.08)	(0.02)	(0.002)
Year-week FE	**-0.04**	**-0.39**	**0.35**	**0.15**	**-0.04**	**-0.04**	**-0.003**
	(0.93)	(0.34)	(0.68)	(0.23)	(0.08)	(0.03)	(0.003)
Year-month FE, weather $t \pm 2$	**-1.41***	**-0.61***	**-0.80**	**-0.23**	**0.08**	**-0.06****	**-0.005****
	(0.86)	(0.35)	(0.67)	(0.24)	(0.08)	(0.02)	(0.002)
Year-week FE, weather $t \pm 2$	**-0.14**	**-0.35**	**0.21**	**0.09**	**-0.03**	**-0.04**	**-0.003**
	(0.95)	(0.36)	(0.67)	(0.23)	(0.08)	(0.03)	(0.003)
Mean of outcome	206.1	175.2	30.9	14.2	43.5	18.6	1.61

With year-month fixed effects, all and offline spending are lower on more polluted days; online spending, the online share and the share outside the home district do not move significantly. With year-week fixed effects the spending coefficients are close to zero. Time outside the home district and distance from home are lower.

City series: distributed lag

$$Y_t = \sum_{j=-2}^{3} \beta_j P_{t-j} + \mathrm{CAL}_t + W_t'\delta + \varepsilon_t$$

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
B. Distributed lag ($P_{t+2}, \ldots, P_{t-3}$)							
Year-month FE: coefficient on P_t	**-1.43***	**-0.93****	**-0.50**	**-0.11**	**0.08**	**-0.07***	**-0.006***
	(0.78)	(0.42)	(0.59)	(0.20)	(0.08)	(0.04)	(0.004)
Leads P_{t+1}, P_{t+2}: joint p	0.72	0.22	0.56	0.39	0.42	0.00	0.00
Sum of $P_{t+2}, \ldots, P_{t-3}$	**-2.26**	**-1.07**	**-1.18**	**-0.32**	**0.09**	**-0.12*****	**-0.011*****
	(1.40)	(0.75)	(0.97)	(0.33)	(0.13)	(0.03)	(0.003)
Year-week FE, weather $t \pm 2$: coefficient on P_t	**-0.39**	**-0.48**	**0.09**	**0.06**	**-0.01**	**-0.06***	**-0.006***
	(0.92)	(0.44)	(0.67)	(0.23)	(0.09)	(0.03)	(0.003)
Leads P_{t+1}, P_{t+2}: joint p	0.36	0.48	0.35	0.27	0.04	0.10	0.11
Sum of $P_{t+2}, \ldots, P_{t-3}$	**0.10**	**-1.02**	**1.12**	**0.48**	**-0.15**	**-0.14***	**-0.012**
	(1.75)	(0.66)	(1.34)	(0.44)	(0.12)	(0.08)	(0.008)
Mean of outcome	206.1	175.2	30.9	14.2	43.5	18.6	1.61

Spending leads are jointly insignificant and the same-day coefficients on all and offline spending stay negative; the sum over the window is negative but insignificant. Mobility leads are jointly significant; the mobility sums are negative and significant.

City series: 2SLS and LIML

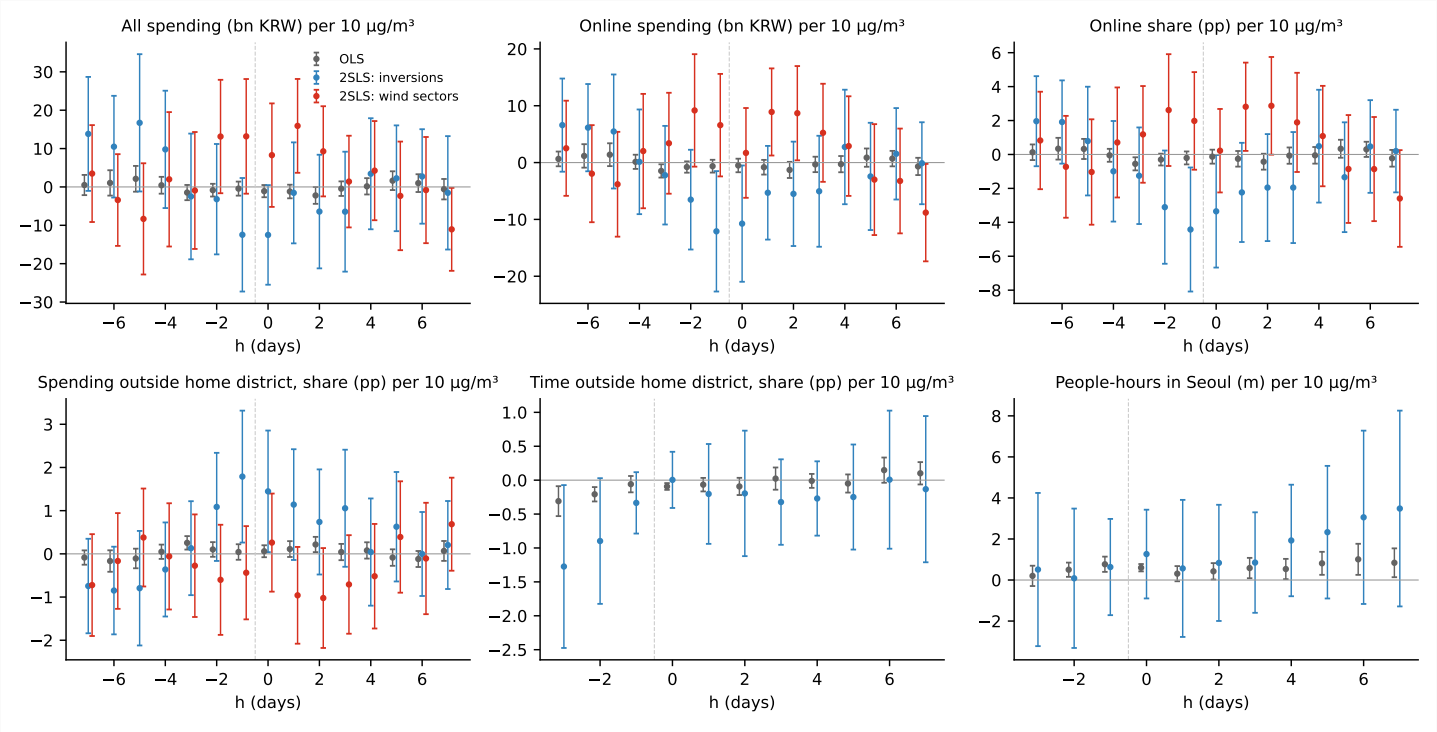
$$Y_t = \beta P_t + \mathrm{CAL}_t + W_t'\delta + \varepsilon_t, \qquad P_t = Z_t'\pi + \mathrm{CAL}_t + W_t'\eta + v_t$$

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
C. Two-stage least squares and LIML							
2SLS, wind direction (8 sectors)	**6.52**	**4.94***	**1.58**	**0.17**	**0.07**	**-0.26**	**-0.027**
	(5.32)	(2.56)	(3.46)	(1.13)	(0.47)	(0.22)	(0.022)
First-stage F (Kleibergen–Paap)	3.2	3.2	3.2	3.2	3.2	1.3	1.3
Effective F (Montiel Olea–Pflueger)	2.3	2.3	2.3	2.3	2.3	1.7	1.7
LIML, wind direction	**12.53**	**7.37***	**9.56**	**4.50**	**0.06**	**-0.97**	**-0.129**
	(10.18)	(4.14)	(17.33)	(17.89)	(1.09)	(2.19)	(0.356)
2SLS, thermal inversions	**-8.23****	**-0.86**	**-7.37****	**-2.33****	**0.94****	**0.00**	**-0.001**
	(3.88)	(1.79)	(2.96)	(0.96)	(0.38)	(0.18)	(0.019)
First-stage F (Kleibergen–Paap)	7.8	7.8	7.8	7.8	7.8	3.4	3.4
Effective F (Montiel Olea–Pflueger)	9.5	9.5	9.5	9.5	9.5	4.2	4.2
LIML, inversions	**-8.50****	**-0.88**	**-7.39****	**-2.34****	**0.94****	**0.01**	**0.001**
	(4.07)	(2.01)	(2.97)	(0.97)	(0.39)	(0.21)	(0.023)
2SLS, inversions, year-week FE, weather $t \pm 2$	**-4.43**	**-0.73**	**-3.70**	**-0.97**	**0.62**	**-0.10**	**-0.007**
	(4.06)	(1.57)	(3.10)	(0.98)	(0.41)	(0.24)	(0.025)
First-stage F (Kleibergen–Paap)	5.4	5.4	5.4	5.4	5.4	2.5	2.5
Effective F (Montiel Olea–Pflueger)	8.8	8.8	8.8	8.8	8.8	3.3	3.3
Mean of outcome	206.1	175.2	30.9	14.2	43.5	18.6	1.61

Wind is weak (effective F 2.3). Inversions are stronger (9.5 for spending) but below the 23.1 critical value: all and online spending and the online share are lower, the share outside the home district higher; LIML agrees; smaller and insignificant with year-week FE and weather leads.

City series: local projections

$$Y_{t+h} = \beta_h P_t + \gamma_{1h} P_{t-1} + \gamma_{2h} P_{t-2} + \mathrm{CAL}_t + W_t' \delta_h + \varepsilon_{t+h}$$



OLS and 2SLS (inversions, wind sectors), per 10 μg/m^3; 95% intervals, year-week clusters.

2. Weekends and holidays

Weekends and holidays: city series

$$Y_t = \beta P_t + \delta P_t D_t + \text{CAL}_t + W_t'\eta + \varepsilon_t, \qquad D_t = \text{weekend or holiday}$$

	Card spending, 2018–19					Mobility, 2019, 24 hours			
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home	People-hours in Seoul (m)	Time, 60+ (pp)
Year-month FE									
Weekdays	**-1.86****	**-0.87****	**-1.00**	**-0.28**	**0.09**	**-0.05*****	**-0.005*****	**0.370*****	**-0.05*****
	(0.95)	(0.41)	(0.77)	(0.28)	(0.09)	(0.02)	(0.002)	(0.086)	(0.02)
Weekends and holidays	**-1.39**	**-0.47**	**-0.92**	**-0.29**	**0.11**	**-0.15*****	**-0.014*****	**0.688*****	**-0.11*****
	(0.97)	(0.49)	(0.64)	(0.20)	(0.08)	(0.05)	(0.005)	(0.157)	(0.03)
Difference	**0.48**	**0.40**	**0.08**	**-0.00**	**0.02**	**-0.09****	**-0.009****	**0.318****	**-0.06***
	(1.08)	(0.53)	(0.77)	(0.26)	(0.10)	(0.04)	(0.004)	(0.157)	(0.03)
Year-week FE, weather $t \pm 2$									
Weekdays	**-0.26**	**-0.41**	**0.15**	**0.06**	**-0.02**	**-0.03**	**-0.002**	**-0.331*****	**-0.04****
	(0.96)	(0.32)	(0.71)	(0.25)	(0.09)	(0.02)	(0.002)	(0.127)	(0.02)
Weekends and holidays	**0.05**	**0.01**	**0.04**	**0.04**	**-0.02**	**-0.15****	**-0.014***	**-0.326**	**-0.13****
	(1.64)	(0.70)	(1.07)	(0.32)	(0.10)	(0.07)	(0.007)	(0.272)	(0.06)
Difference	**0.31**	**0.42**	**-0.11**	**-0.02**	**0.00**	**-0.12***	**-0.012***	**0.005**	**-0.09***
	(1.60)	(0.69)	(1.05)	(0.32)	(0.11)	(0.07)	(0.007)	(0.245)	(0.05)

Spending responds no differently on weekends and holidays. Time outside the home district and distance from home fall more on weekends and holidays, and people-hours in Seoul rise more.

Weekends and holidays: city series, 2SLS

P_t and $P_t D_t$ instrumented by Z_t and $Z_t D_t$

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
2SLS, thermal inversions							
Weekdays	**-10.35****	**-2.06**	**-8.29****	**-2.61****	**0.96****	**-0.20**	**-0.019**
	(4.57)	(2.06)	(3.44)	(1.12)	(0.44)	(0.12)	(0.012)
Weekends and holidays	**-7.44****	**-0.21**	**-7.23****	**-2.30****	**0.91****	**-0.28**	**-0.028**
	(3.61)	(1.76)	(2.82)	(0.92)	(0.37)	(0.17)	(0.019)
Difference	**2.91**	**1.85**	**1.06**	**0.32**	**-0.05**	**-0.08**	**-0.010**
	(2.65)	(1.25)	(1.91)	(0.62)	(0.26)	(0.11)	(0.012)
First-stage F (Kleibergen–Paap, both endogenous)	4.1	4.1	4.1	4.1	4.1	1.8	1.8
2SLS, wind direction (8 sectors)							
Weekdays	**8.11***	**4.85****	**3.26**	**0.69**	**-0.31**	**-0.04**	**-0.005**
	(4.64)	(2.33)	(3.00)	(0.96)	(0.40)	(0.09)	(0.009)
Weekends and holidays	**7.37***	**4.66***	**2.71**	**0.74**	**-0.33**	**-0.15**	**-0.019**
	(4.40)	(2.75)	(2.32)	(0.74)	(0.33)	(0.16)	(0.017)
Difference	**-0.74**	**-0.19**	**-0.55**	**0.05**	**-0.03**	**-0.11**	**-0.014**
	(3.66)	(2.23)	(2.10)	(0.67)	(0.29)	(0.12)	(0.012)
First-stage F (Kleibergen–Paap, both endogenous)	3.8	3.8	3.8	3.8	3.8	1.7	1.7

With inversions or wind, the weekend–weekday differences are not significant for any outcome. Both first stages are weak once the weekend interactions are instrumented (KP F about 4 for spending, below 2 for mobility).

Weekends and holidays: district panel

$$s_{rkt} = \beta_k P_{rt} + \delta_k P_{rt} D_t + \text{FE} + \varepsilon_{rkt}$$

Outcome	PM2.5 (weekdays)	PM2.5 × weekend or holiday
Share of spending, home district (pp)	**0.156** (0.119)	**0.159** (0.134)
Share of spending, adjacent districts (pp)	**0.028** (0.038)	**0.082**** (0.037)
Share of spending, other districts (pp)	**0.020** (0.029)	**0.033** (0.038)
Online share (pp)	**-0.204** (0.175)	**-0.274** (0.182)
Share of time, home district (pp)	**0.032*** (0.018)	**0.106***** (0.034)
Share of time, adjacent districts (pp)	**-0.019**** (0.008)	**-0.064***** (0.018)
Share of time, other districts (pp)	**-0.013** (0.011)	**-0.042**** (0.020)
Spending outside home district, share (pp)	**0.048** (0.064)	**0.115*** (0.070)
Time outside home district, share (pp)	**-0.032*** (0.018)	**-0.106***** (0.034)
Km from home	**-0.003** (0.002)	**-0.009**** (0.003)
Time outside home district, 60+ (pp)	**-0.044***** (0.013)	**-0.055**** (0.024)
All spending (PPML)	**-0.010** (0.008)	**-0.006** (0.008)
Offline spending (PPML)	**-0.007** (0.005)	**0.008** (0.007)
Online spending (PPML)	**-0.062*** (0.034)	**-0.052** (0.044)
People-hours in Seoul (PPML)	**0.005***** (0.002)	**0.007**** (0.003)

On weekdays, no spending share moves; time shifts slightly toward the home district (10%). On weekends and holidays the shift of time toward home is larger (shares of time, time outside the home district, distance: differences significant); of the spending shares only the adjacent share (5%) and the share outside the home district (10%) rise more.

Weekends and holidays: district panel, 2SLS

P_{rt} and $P_{rt}D_t$ instrumented by Z and ZD_t

	2SLS, wind		2SLS, inversions		KP F	
Outcome	weekdays	weekend − weekday	weekdays	weekend − weekday	wind	inv.
Spending outside home district, share (pp)	**-0.942**** (0.400)	**0.727**** (0.348)	**0.480*** (0.284)	**0.177** (0.164)	3.6	10.4
Online share (pp)	**3.308***** (0.900)	**-1.856***** (0.621)	**-1.118*** (0.675)	**-0.433** (0.396)	3.6	10.4
Time outside home district, share (pp)	**0.979*** (0.560)	**-0.806** (0.503)	**0.095** (0.152)	**0.064** (0.110)	4.3	3.8
Km from home	**0.023** (0.074)	**-0.008** (0.074)	**0.010** (0.015)	**0.005** (0.011)	4.3	3.8

With wind, on weekdays the share outside the home district is lower and the online share and time outside the home district higher; the weekend differences in the spending shares are of the opposite sign. With inversions, the share outside the home district is higher and the online share lower (10% level), no weekend difference. First stages are weak (KP F 3.6–10.4).

3. Geography within a district-day

Where spending and time go: band shares

$$s_{rkt} = \beta_k P_{rt} + \alpha_{rk} + \lambda_{k,\text{calendar}} + \lambda_{k,\text{weather cell}} + \lambda_{k,\text{humidity}} + \varepsilon_{rkt}$$

Outcome	Mean	OLS	2SLS wind	2SLS inversions	LIML wind	KP F wind	KP F inv.
A. Share of each band (pp), one regression per share							
Spending, home district	41.2	**0.201****	**-0.644**	**0.927****	**-1.126**	3.0	17.1
		(0.095)	(0.621)	(0.415)	(1.025)		
Spending, adjacent districts	21.1	**0.051**	**-0.414**	**0.400****	**-0.470**	3.0	17.1
		(0.032)	(0.267)	(0.174)	(0.303)		
Spending, other districts	24.2	**0.029**	**-0.067**	**0.286****	**-0.074**	3.0	17.1
		(0.026)	(0.213)	(0.133)	(0.229)		
Spending, online	13.5	**-0.281****	**1.124**	**-1.613****	**1.642**	3.0	17.1
		(0.142)	(0.975)	(0.666)	(1.373)		
Time present, home district	80.9	**0.056*****	**0.263**	**-0.046**	**0.367**	1.2	5.6
		(0.017)	(0.267)	(0.148)	(0.425)		
Time present, adjacent districts	10.2	**-0.034*****	**0.024**	**0.020**	**0.042**	1.2	5.6
		(0.006)	(0.105)	(0.055)	(0.140)		
Time present, other districts	9.0	**-0.023****	**-0.288**	**0.027**	**-0.402**	1.2	5.6
		(0.011)	(0.229)	(0.093)	(0.368)		

Without an instrument, the home-district spending share is higher and the online share lower (5%); the share of time at home is higher and in the other bands lower. With inversions (KP F 17.1), the online share is lower and all three physical bands higher; with wind (F 3.0), no share moves. Instrumented mobility shares are imprecise.

PM2.5 in levels (since 27 October; was log PM2.5): the same unit as the city series and most of the pollution literature; the share stays in levels so that the spending shares' coefficients sum to zero. The log version was a holdover from the Poisson models, which keep it (elasticities).

Transactions, distances and mobility

$$Y_{rt} = \beta P_{rt} + \alpha_r + \text{FE} + \varepsilon_{rt}$$

Outcome	Mean	OLS	2SLS wind	2SLS inversions	LIML wind	KP F wind	KP F inv.
B. Transactions and mobility							
Number of transactions (per district-day)	314,893	**-653.8****	**2316.2**	**549.2**		2.3	16.9
		(264.5)	(2196.6)	(1316.2)			
Amount per transaction (KRW)	25,888	**-197.3*****	**1353.8****	**-862.3****		3.0	17.1
		(71.6)	(590.7)	(378.8)			
Km, offline, transaction-weighted	4.62	**0.002**	**-0.007**	**0.006**		3.0	17.1
		(0.003)	(0.020)	(0.012)			
Km, offline, spending-weighted	4.78	**-0.009***	**0.045**	**-0.036**		3.0	17.1
		(0.005)	(0.035)	(0.023)			
People-hours in Seoul (thousand)	6,818	**14.83*****	**60.78**	**31.13**		1.2	5.5
		(3.81)	(54.07)	(32.46)			
Time outside home district (pp)	19.13	**-0.056*****	**-0.263**	**0.046**		1.2	5.6
		(0.017)	(0.267)	(0.148)			
Km from home	1.62	**-0.005*****	**-0.038**	**0.004**		1.2	5.6
		(0.002)	(0.033)	(0.015)			

By OLS, fewer transactions (5%) and a smaller amount per transaction (1%), a slightly shorter spending-weighted distance (10%), more people-hours in Seoul, less time outside the home district and a shorter distance from home (1%). The amount per transaction is higher with wind and lower with inversions; other instrumented estimates are imprecise.

Do we want the instrument here?

- Wind at the district's station is weak as eight sectors (first-stage F 2.3) and as the modal direction (3.9).
- The westerly indicator is the strongest wind instrument (F 9.4 with the held-out arc) but gives imprecise share estimates. All first stages are weaker with PM2.5 in levels than in logs (5.6, 7.7, 32).
- The city reduced forms show wind moving spending directly $\Rightarrow$ doubt on the exclusion restriction of every wind instrument.
- Inversions are city-wide: F 16.9 for spending (below 23.1), 5.5 for mobility.
- OLS relies on the controls (weather cells, humidity deciles, calendar FE, all measured at the district's own station) for what moves both pollution and where residents spend.

Why does the wind IV have the wrong sign? Reduced form over first stage

City series. $\hat{\beta}_{IV}$ = reduced form / first stage, so its sign = sign(reduced form) × sign(first stage). Per sector: days with that wind direction vs. calm days and 270–315° (west-northwest).

	First stage	Reduced form (bn KRW)		Implied IV, all spending
Wind sector	PM2.5 (μg/m^3)	All spending	Offline	(per 10 μg/m^3)
1: 0–45° (N–NE)	**-5.6*** **	**-6.3** **	**-3.5** **	**+11.2**
2: 45–90° (NE–E)	**-6.3*** **	**-2.5**	**-2.1**	**+4.0**
3: 90–135°	**-3.8***	**+1.1**	**-1.8**	**-2.9**
5: 180–225°	**-2.4**	**+5.0**	**+0.7**	**-20.4**
4, 6, 8	≈ 0	noisy	noisy	± 70 to 240
2SLS, all sectors (OLS)				**+6.5** (**-1.6** *)

- **The first stage has the expected sign**: every sector is cleaner than the westerly base.
- **The sign flip is in the reduced form of sectors 1 and 2**, which have the strongest first stage and so the most weight in 2SLS: on northerly days the air is cleaner *and* residents spend less.
- **So wind moves spending through something other than PM2.5** (an exclusion failure). Cold northerly outbreaks are not the explanation: the sign is the same in January–April and May–August (next slide).
- **And the instrument is weak** (effective F 2.3 vs. 23.1), which magnifies any such violation (bias = Cov(Z, error) / Cov(Z, PM2.5)).

Inversions: the sign matches OLS, but the pattern is suspicious

City series; inversion count (21 KST, 03, 09 KST soundings) and 03 KST strength (°C).

	First stage	Reduced form (bn KRW)		
	PM2.5 (μg/m^3)	All spending	Offline	Online
Inversion strength, 03 KST	**+2.2*** **	**-1.3**	**+0.2**	**-1.5** **
Inversion count	**+0.7**	**-1.7**	**-0.9** **	**-0.8**
2SLS per 10 μg/m^3 (OLS)		**-8.2** ** (**-1.6** **)	**-0.9** (**-0.8** **)	**-7.4** ** (**-0.9**)

- **First stage**: inversion strength raises PM2.5, as expected; the count does not move it.
- **No sign flip for all spending**, but the 2SLS estimate is five times OLS, and **all of it is online spending**: offline 2SLS equals OLS.
- **Two warning signs**: (i) strong inversions lower online spending and raise the share spent outside the home district, the opposite of pollution pushing spending online; (ii) the count lowers offline spending although it does not move PM2.5, a reduced form without a first stage.
- **Candidate channel**: inversions form on clear, calm, high-pressure nights, so they also mark fair weather, which can draw people out by itself.
- Effective F 9.5, below the 23.1 critical value.

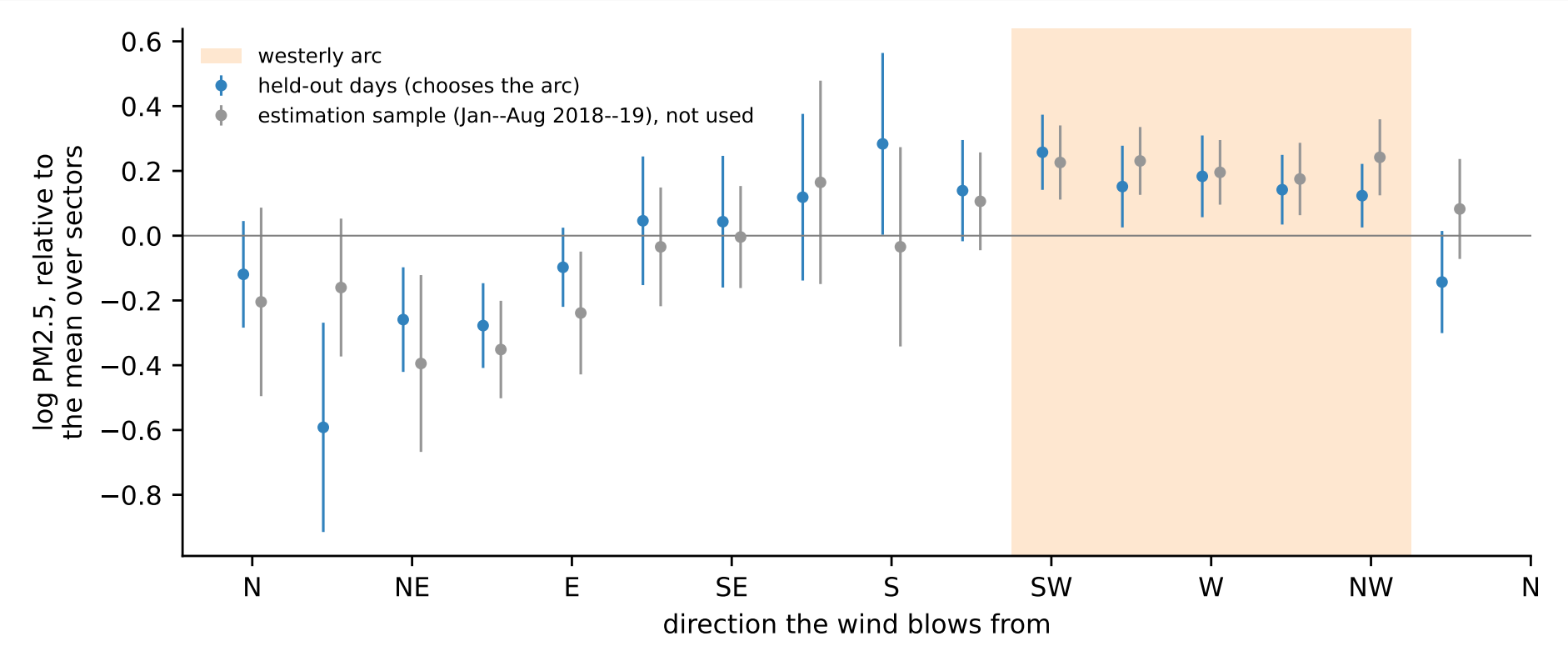
Checks of both instruments

City series, 2SLS of all spending per 10 μg/m^3 unless stated [effective F]; OLS -1.6**. Details: `output/draft9/district/iv_checks.md`.

Check	Wind (7 sectors)	Inversions (count + strength)
Base	**+6.5** [2.3]	**-8.2**** [9.5]; online **-7.4****
Jan–Apr / May–Aug only	**+4.2** [1.3] / **+13.4*** [3.6]	**-12.9**** [3.9] / **+1.7** [6.7]
Wind without sectors 1 and 8	**+5.6** [3.2]	
Strength only, count as a control		**-5.9** [12.8]; online **-6.9***
Placebo: reduced form for spending 2 (3) days *before*, joint p	0.03 (0.03); online 0.00 (0.01)	0.92 (0.68)
Weather still predicted given the base controls ($p < 0.05$)	temperature change, yesterday's temperature, tomorrow's rain, temp. $\times$ wind, temp. $\times$ humidity	tomorrow's temperature, temp. $\times$ humidity
Adding 8 weather sets one at a time	**+4.2** to **+9.0**, all $F < 5$	**-7.1** to **-8.3** (except yesterday's weather: F 3.7)
All 8 sets together	**+6.0** [1.6]	**-3.8** [10.2]; online **-5.6****

- **Wind**: positive under every control set, in both halves of the year and without the northerly sectors, and it predicts spending on the days *before* $\Rightarrow$ wind days differ in ways the controls miss; not usable here.
- **Inversions**: no pre-trend; stable to single weather controls; the effect is in Jan–Apr and online. With all controls, all spending halves (insignificant), online stays.
- **Missing**: sunshine, cloud and pressure (fair weather): KMA ASOS data, being requested.

Which wind directions bring PM2.5



Effect of the day's wind direction (16 sectors) on log city PM2.5, relative to the mean over the sectors, with the weather terms; 95% intervals. Blue: held-out days, which choose the westerly arc (shaded); grey: the estimation sample.

4. The size of purchases

Amount per transaction: by channel, band and time slot

$$Y_t = \frac{\text{spending}_t}{\text{transactions}_t} = \beta P_t + \text{CAL}_t + W_t'\delta + \varepsilon_t \qquad \text{(KRW per 10 } \mu\text{g/m}^3\text{)}$$

Amount per transaction	Mean (KRW)	OLS (KRW)	OLS (% of mean)	OLS, year-week, weather $t \pm 2$	DLM, P_t	2SLS inversions	DLM leads p
All	26,094	**-122.2**	**-0.47**	**48.4**	**-57.4**	**-839.2****	0.82
		(94.7)		(111.0)	(83.4)	(426.8)	
Offline	22,737	**-35.0**	**-0.15**	**9.0**	**-25.7**	**44.7**	0.83
		(35.9)		(32.7)	(34.3)	(158.4)	
Online	166,988	**-2902.8**	**-1.74**	**1870.1**	**454.9**	**-41743.9*****	0.22
		(3256.6)		(3945.6)	(3142.8)	(16063.4)	
All, without single-transaction cells	25,523	**-125.2**	**-0.49**	**45.5**	**-55.4**	**-852.3****	0.82
		(93.4)		(109.7)	(82.2)	(424.8)	
Offline, home district	21,875	**-4.8**	**-0.02**	**8.7**	**-14.7**	**230.2**	0.61
		(42.8)		(28.2)	(41.1)	(152.9)	
Offline, adjacent districts	26,908	**-51.3**	**-0.19**	**5.9**	**-26.1**	**-14.1**	0.80
		(40.3)		(44.7)	(43.1)	(183.7)	
Offline, other districts	21,431	**-78.4**	**-0.37**	**5.8**	**-44.1**	**-203.5**	0.96
		(52.7)		(68.8)	(44.7)	(241.4)	
06–10h	19,661	**-126.7**	**-0.64**	**188.4**	**35.1**	**-1844.8***	0.39
		(235.3)		(273.2)	(236.5)	(985.1)	
10–18h	29,625	**-171.1**	**-0.58**	**72.7**	**-88.6**	**-1237.1****	0.88
		(137.4)		(157.2)	(114.0)	(623.4)	
18–22h	24,587	**-64.8**	**-0.26**	**-12.3**	**-45.2**	**-54.4**	1.00
		(41.6)		(48.9)	(42.3)	(231.8)	
22–06h	19,047	**-3.1**	**-0.02**	**-8.9**	**-9.5**	**-112.9**	0.44
		(18.0)		(21.1)	(24.6)	(107.5)	

Amount per transaction: by category

$$Y_t = \frac{\text{spending}_t}{\text{transactions}_t} = \beta P_t + \text{CAL}_t + W_t'\delta + \varepsilon_t$$

Amount per transaction	Mean (KRW)	OLS (KRW)	OLS (% of mean)	OLS, year-week, weather $t \pm 2$	DLM, P_t	2SLS inversions	DLM leads p
Health care	46,696	**-216.0**	**-0.46**	**108.1**	**-95.6**	**-2264.8****	0.83
		(180.3)		(234.0)	(170.3)	(946.8)	
Pharmacies (masks)	18,355	**-167.0**	**-0.91**	**410.9**	**156.1**	**-1919.2***	0.93
		(282.4)		(319.4)	(220.6)	(1147.3)	
Air purifiers	204,242	**-4063.5**	**-1.99**	**-2172.2**	**-1475.0**	**-16569.9**	0.75
		(4098.5)		(4487.7)	(4498.8)	(18826.3)	
Outdoor recreation	61,830	**-5.7**	**-0.01**	**252.8**	**-81.2**	**-33.2**	0.72
		(252.9)		(244.3)	(236.1)	(950.4)	
Indoor entertainment	7,420	**-28.9***	**-0.39***	**-15.8**	**-13.7**	**-172.4**	0.51
		(17.4)		(18.5)	(18.5)	(118.7)	
Delivery food	13,860	**4.4**	**0.03**	**2.0**	**-5.3**	**42.5**	0.30
		(10.3)		(13.6)	(15.3)	(53.7)	
Taxi	7,817	**-6.5**	**-0.08**	**-3.2**	**-3.2**	**-43.4**	0.22
		(4.7)		(4.4)	(7.0)	(33.1)	
Bus and subway	8,266	**-10.3****	**-0.12****	**-5.9**	**-11.6***	**-63.2***	0.23
		(4.8)		(5.1)	(6.5)	(35.5)	

By OLS the average purchase does not change for any channel, band or slot; bus and subway (5%) and indoor entertainment (10%) transactions are slightly smaller. With inversions, the average purchase is smaller for all and online purchases, in the daytime slots, and in health care and pharmacies.

5. Categories that pollution should move

Pre-specified categories: definitions

Group	Predicted sign	Shinhan code	English (Shinhan code list)	Korean (our translation)	Mean, bn KRW per day (% of all)
Health care	+	51	Medical Services	병 · 의원 등 의료기관	24.3 (11.8)
		54	Public Health Clinics	보건소	
		55	Oriental Medicine	한의원	
		56	Pharmacy	약국	
		57	Medical Products and Centers	의료기기 · 의료용품	
		not 52, 53	*Plastic Surgeon, Dermatology*	*성형외과, 피부과 (제외)*	
Pharmacies (masks)	+	56	Pharmacy	약국	4.5 (2.2)
Air purifiers	+	16	Air Control Machines	공기조절기기 (공기청정기 등)	0.09 (0.04)
Outdoor recreation	−	31	Recreational/Leisure Activities	레저 · 여가 활동	5.0 (2.4)
Indoor entertainment	?	36	Cinemas, Exhibits, etc.	영화관 · 전시 · 공연	1.3 (0.6)
		37	PC Cafes and Arcades	PC방 · 오락실	
Delivery food	+	24	Fried Chicken, Pizza, and Burger	치킨 · 피자 · 햄버거	3.5 (1.7)
Taxi	+	48	Taxi	택시	2.4 (1.2)
Bus and subway	−	46	Bus and Subway	버스 · 지하철	2.3 (1.1)

- **Codes**: Shinhan's merchant category (`MCT_BR`: 60 offline, 99 miscellaneous, 999 online). Seoul residents at Seoul merchants; online (999) is in no group, so app-ordered delivery is not counted. Pharmacies (56) are also in health care.
- **Korean names**: our translation; Shinhan's own Korean names are not in our files (to confirm). Means: city series, Jan–Aug 2018–19 (all spending 206 bn KRW a day).

Pre-specified categories

City: $Y_t = \beta P_t + \text{CAL}_t + W_t'\delta + \varepsilon_t$ (% of mean), PPML, DLM. District: $E[Y_{rt}] = \exp(\beta \log P_{rt} + \alpha_r + \text{FE})$, PPML and PPML-IV (wind).

	Predicted	City × day				District × day		DLM leads
Category	sign	OLS (% of mean)	OLS, year-week (%)	PPML	DLM, P_t (%)	PPML	PPML-IV	joint p
Health care	up	**-0.28**	**0.75**	**-0.003**	**-0.28**	**-0.008**	**0.050***	0.69
		(0.47)	(0.53)	(0.015)	(0.42)	(0.008)	(0.027)	
Pharmacies (masks)	up	**-0.62**	**3.13***	**-0.007**	**1.09**	**-0.023**	**0.240****	0.92
		(1.70)	(1.77)	(0.053)	(1.35)	(0.030)	(0.095)	
Air purifiers	up	**-0.26**	**1.07**	**-0.051**	**-0.09**	**-0.010**	**0.218**	0.29
		(1.76)	(1.49)	(0.045)	(1.92)	(0.029)	(0.139)	
Outdoor recreation	down	**-0.36**	**-0.07**	**-0.005**	**-0.66**	**-0.005**	**0.071*****	0.38
		(0.43)	(0.46)	(0.012)	(0.41)	(0.008)	(0.025)	
Indoor entertainment	ambiguous	**-1.02****	**-1.06*****	**-0.034****	**-0.50**	**-0.027*****	**-0.025**	0.87
		(0.43)	(0.40)	(0.017)	(0.44)	(0.009)	(0.025)	
Delivery food	up	**-0.23**	**-0.31**	**-0.006**	**-0.51**	**-0.006**	**0.023**	0.06
		(0.20)	(0.31)	(0.007)	(0.34)	(0.005)	(0.016)	
Taxi	up	**0.00**	**-0.10**	**0.004**	**0.08**	**0.000**	**0.003**	0.17
		(0.13)	(0.14)	(0.003)	(0.16)	(0.002)	(0.009)	
Bus and subway	down	**-0.25**	**-0.06**	**-0.003**	**-0.21**	**-0.004**	**0.004**	0.36
		(0.18)	(0.20)	(0.005)	(0.20)	(0.003)	(0.010)	

No category moves in its predicted direction in the city series with year-month FE. Indoor entertainment (sign not predicted) falls in the city series and the district PPML. With the wind control function, health care (10%), pharmacies and outdoor recreation rise; outdoor recreation has the opposite of its predicted sign.

6. Air-quality alerts

Alerts, given measured pollution

$$Y_{ts} = \beta A_{ts} + f(P_{ts}) + \alpha_{s,\text{calendar}(t)} + W'_{ts}\delta + \varepsilon_{ts} \qquad \text{(day } t,\ \text{slot } s\text{)}$$

	Card spending, 2018–19				Mobility, 2019, 24 hours	
	All (bn KRW)	All (%)	Online (%)	Online share (pp)	Time outside home district (pp)	Hours outside home district (%)
PM2.5 alert in effect, city slots	**-1.48**	**-2.4**	**-3.5**	**-1.26**	**-0.16**	**-1.0**
	(1.52)	(2.7)	(10.9)	(1.08)	(0.16)	(0.9)
Ozone alert in effect, area slots (date × slot FE)	**-0.91***	**-3.4**	**-15.9****	**-1.71**	**0.10**	**0.1**
	(0.55)	(2.3)	(7.2)	(1.06)	(0.11)	(0.3)

Given measured pollution, PM2.5 alerts do not change all spending, online spending, the online share or time outside the home district. Ozone alerts (area × day × slot, date × slot FE) lower online spending of the alert area's residents; the estimate depends on the Downtown area (-4.5%, $p = 0.59$, without it).

Alerts without pollution controls

$$Y_{ts} = \beta A_{ts} + \alpha_{s,\text{calendar}(t)} + W'_{ts}\delta + \varepsilon_{ts}$$

	Card spending, 2018–19				Mobility, 2019, 24 hours	
	All (bn KRW)	All (%)	Online (%)	Online share (pp)	Time outside home district (pp)	Hours outside home district (%)
PM2.5 alert, city slots, slot × calendar FE	**-2.71*** **	**-5.0*** **	**-14.5** **	**-1.67** **	**-0.32** **	**-1.5** **
	(1.05)	(1.8)	(6.7)	(0.83)	(0.13)	(0.6)
Ozone alert, area slots, date × slot FE	**-0.76**	**-3.5**	**-16.7** **	**-1.81***	**0.16**	**0.2**
	(0.55)	(2.2)	(6.9)	(1.02)	(0.11)	(0.3)
PM2.5 alert day (city × day)	**-10.76*** **			**-1.71** **	**-0.23**	
	(3.67)			(0.84)	(0.15)	

Without pollution controls, slots under a PM2.5 alert have lower spending (all and online), a lower online share and fewer people-hours outside the home district; alert days have lower daily spending and online share. Ozone alerts lower online spending with or without pollution controls.

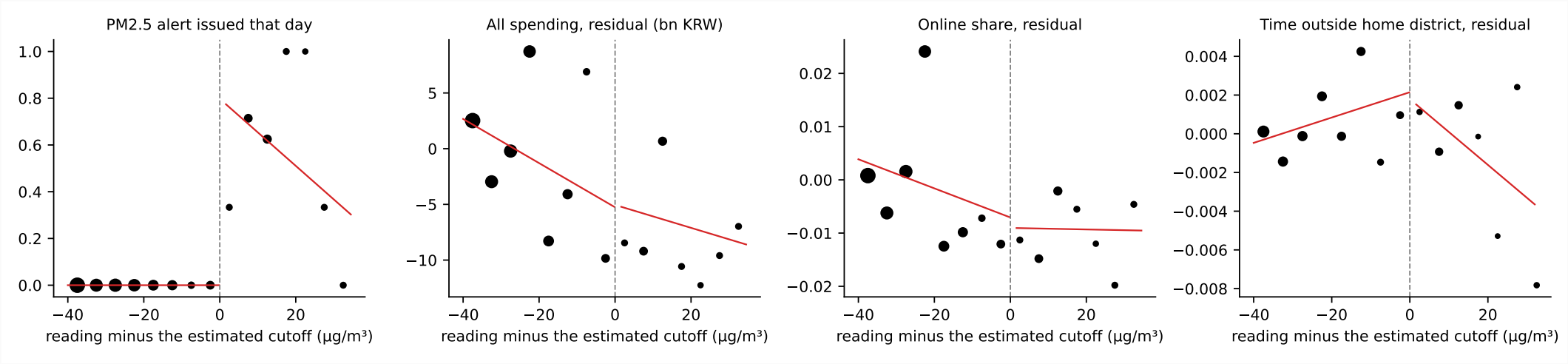
Days just above vs. just below the estimated PM2.5 cutoff

Fuzzy RD: running variable = the day's highest two-hour reading minus the estimated cutoff (where issuance actually jumps); treatment = alert issued that day; local linear.

	Fuzzy RD at the estimated PM2.5 cutoff, bandwidth				2018-07-01 change:
Outcome	MSE-optimal	10 μg/m^3	20	30	80.5–95.5 $\times$ after, reduced form
First stage: alert issued that day	**0.63** (0.32)	**0.62** (0.49)	**0.58** (0.29)	**0.56** (0.22)	**0.64***** (0.08)
alert in effect at some time that day	**0.62**** (0.22)	**0.89***** (0.26)	**0.64***** (0.24)	**0.61**** (0.20)	
Days below / above the cutoff	20/18	11/10	35/21	76/26	
All spending (bn KRW)	**-9.737** [0.48]	**2.278** [0.59]	**-6.086** [0.52]	**0.930** [0.54]	**11.882** [0.23]
Online (bn KRW)	**-4.243** [0.64]	**1.856** [0.45]	**-4.098** [0.74]	**2.153** [0.56]	**6.782** [0.33]
Online share	**-0.009** [0.80]	**0.010** [0.39]	**-0.009** [0.89]	**0.005** [0.75]	**0.013** [0.58]
Share outside home district	**0.008** [0.56]	**-0.008** [0.15]	**0.010** [0.70]	**0.005** [0.52]	**0.002** [0.87]
Time outside home district (share)	**0.001** [0.64]	**0.002** [0.52]	**-0.001** [0.10]	**-0.002** [0.88]	**-0.001** [0.67]
People-hours in Seoul (m)	**-0.339** [0.80]	**-4.766***** [0.00]	**-0.264** [0.37]	**0.564** [0.75]	**-0.285** [0.85]
All spending, next day	**5.240** [0.79]	**-24.482** [0.81]	**9.275** [0.21]	**18.619** [0.79]	**28.340***** [0.01]
Time outside home district, next day	**0.010** [0.35]	**-0.000** [0.94]	**0.005** [0.44]	**0.004** [0.34]	**0.002** [0.75]

Alert probability jumps by about 0.6 at every bandwidth; no outcome jumps (people-hours only at the 10 μg/m^3 bandwidth). 10–26 days above the cutoff: at most suggestive. After the 1 July 2018 rule change, next-day spending is higher (1% level) on days between the two cutoffs. Columns: bandwidth; MSE-optimal = chosen from the data to minimize the mean squared error of the jump; 10, 20, 30 = days within that many μg/m^3 of the cutoff.

Alert issuance and outcomes around the PM2.5 cutoff



Area-slots just above vs. just below the ozone cutoff

Fuzzy RD on area × day × slot, May–August (4,920). Running variable = the slot's highest hourly ozone at any monitor in the area minus the cutoff where issuance jumps (estimated 0.119 ppm; documented 0.12); treatment = ozone alert in effect; local linear.

Outcome	MSE-optimal	0.01 ppm	0.02 ppm
First stage: ozone alert in effect	**0.76***** (0.14)	**0.84**** (0.19)	**0.76***** (0.13)
All spending (bn KRW)	**-1.4753** [0.42]	**-0.8152** [0.86]	**-1.4213** [0.64]
Online (bn KRW)	**-1.6467** [0.15]	**-1.5324** [0.61]	**-1.5254** [0.43]
Online share	**-0.0394** [0.13]	**-0.0645** [0.41]	**-0.0463** [0.34]
Time outside home district (share)	**0.0032** [0.22]	**0.0015** [0.92]	**0.0031** [0.83]
People-hours in Seoul (m)	**-0.0207** [0.92]	**0.0180** [0.25]	**-0.0169** [0.99]

- **What it compares**: area-slots whose peak ozone just crossed the cutoff with those just below it; if nothing else changes at the cutoff, the difference is the effect of the alert.
- **First row**: just above the cutoff, an alert is 76–84 percentage points more likely to be in effect.
- **Other rows**: jump in the outcome divided by the jump in alert probability = effect of an alert for area-slots that get one because they cross; [robust bias-corrected p]. Outcomes residualized on date × slot, area × slot, area × day-of-week, area × year-month effects.
- **Columns = bandwidth**, i.e. how close to the cutoff an area-slot must be to be used. *MSE-optimal*: chosen from the data to minimize the mean squared error of the estimated jump (Calonico, Cattaneo and Titiunik 2014). *0.01 ppm, 0.02 ppm*: fixed bandwidths, area-slots with peak ozone within 0.01 (0.02) ppm of the cutoff.
- **Result**: no outcome jumps in any column.

Summary of what the tables show

- **City OLS**: all and offline spending lower on more polluted days with year-month FE, close to zero with year-week FE; residents spend less time outside the home district.
- **City IV**: wind is weak; inversions (below the critical value) give lower online spending and a higher share outside the home district.
- **Weekends**: no difference for spending in the city series; in the panel, time moves toward home more on weekends and holidays, spending shares mostly do not.
- **Geography**: by OLS, a higher home-district and lower online share of spending, fewer and smaller transactions, more time at home; IV results depend on the instrument.
- **Purchase size and categories**: mostly unchanged; no category moves in its predicted direction by OLS.
- **Alerts**: no effect given measured pollution (PM2.5); ozone alerts lower online spending, driven by Downtown; no jumps at the cutoffs.

Air Pollution, Spending and Mobility in Seoul

Results draft for internal review — batches 1–9, districts

APC project*

October 27, 2026

About this draft

What changed since draft 8. This draft is compared with draft 8 (commit `1f641d7`, 25 October 2026). Text, captions, bullets and table notes that are unchanged are set in gray; new or edited text is in black; deleted text is not shown. A table or figure whose numbers or content changed is set in black and marked "new or changed in this draft"; an unchanged one is in gray (the bold type of the estimates and the renaming of PPML-CF to PPML-IV alone do not count as a change). An all-black version is `district_paper_clean.pdf`. *Change of 27 October 2026:* the linear district-panel regressions (shares, counts, distances, amounts per transaction, their weekend versions and first stages) now use PM2.5 in levels instead of log PM2.5 (Section 2); their tables and the sentences below them are re-estimated; the Poisson and log-outcome models keep log PM2.5.

This draft repeats the analysis of the neighborhood draft with Seoul's 25 districts (gu) as the spatial unit: residents' spending and presence are grouped by home district, each district's pollution and weather are those measured at its own AirKorea monitor and KMA station, and spending and time are located in the home district, in districts bordering it, or in the rest of the city. It reports estimates without interpreting them: each results table is preceded by the equation it estimates and followed by the conclusions it supports, stated qualitatively, and every table note states the estimator, how zeros are handled and how many observations are used. Districts are the unit of Kim and Trusov (2025) and of most district-level pollution studies of Seoul, which has one urban monitor per district; both the card data's legal neighborhoods and the phone data's administrative neighborhoods nest in them, so no data are split or imputed.

Conventions. Amounts are in Korean won (KRW; about 1,100–1,170 KRW per US dollar in 2018–19). The card amounts and counts are Shinhan Card's own figures multiplied by one factor per month, about 5 (Appendix A.2), so levels are Shinhan's sales scaled to the size of the card market; shares, elasticities and amounts per transaction are unaffected. "First-stage F" is the Kleibergen–Paap rk Wald F, computed with the same clustering as the standard errors; every 2SLS

*Prepared with Claude Code, 25 October 2026, at Rae's request (24 October 2026) to run the current analysis, with the draft-8 and draft-9 changes (`notes/2026-10-24_draft8_changes.md`, `notes/2026-10-25_draft9_changes.md`), at the level of Seoul's 25 districts, each with its own monitor and weather station. Specifications: `output/draft9/spec_batch9.pdf`; estimates: `output/draft9/district/results_batch9.csv`. The companion draft at the neighborhood level is `output/spatial/paper/spatial_paper.pdf`. Not for circulation outside the team.

table also reports the Montiel Olea–Pflueger effective F (critical value 23.1 for a worst-case 2SLS bias of 10%). City estimates and the district panel's linear regressions (shares, counts, distances, amounts per transaction) are per 10 μg/m^3 of PM2.5; the district Poisson models (stacked bands, levels, categories) and log-outcome regressions are elasticities, per log point of PM2.5. In every table, point estimates are in bold and other statistics (standard errors, first-stage F, p-values, means, observations) in regular type, and the note ends with a "Reading" sentence: the unit of the coefficient and the size of one example, relative to the outcome's mean and to a typical day-to-day change in PM2.5. The spatial units are, from small to large: the *neighborhood* (dong), the *district* (gu; 25), the *area* (the five groups of districts for which Seoul issues ozone alerts: Downtown, Northeast, Northwest, Southwest, Southeast), and the *city*.

Related literature. Kim and Trusov (2025) use Korean card transactions of individual Seoul cardholders, with spending located in the 25 districts, and a wind-direction instrument, and find that spending rises with pollution, most in hedonic categories; Jung (2020) relates Korean card expenditures to fine particulate matter; Barwick et al. (2024) show that China's 2013 disclosure of pollution information raised avoidance behavior and purchases of protective goods. Avoidance can also be intertemporal (Graff Zivin and Neidell, 2009) and can show up as defensive purchases (Zhang and Mu, 2018; Ito and Zhang, 2020). The card-spending study closest in design, Barwick et al. (2018), uses health-care card spending across 367 Chinese cities. Appendix G compares the fixed effects used in this literature with ours.

1 Data

Districts. Seoul has 25 districts, of 10 to 47 km^2 (median 24), each made of 3 to 87 legal neighborhoods and 10 to 27 administrative neighborhoods (Appendix Table A.1). Every legal and administrative neighborhood code begins with its district's five-digit code, and the crosswalk between the two kinds of neighborhood never links two districts, so the card data (legal neighborhoods) and the phone data (administrative neighborhoods) are aggregated to districts by code, without splitting any neighborhood. Two districts are *adjacent* when their boundaries touch (55 pairs, 10 of them across the Han River; each district has 2 to 7 neighbors).

Card spending. Shinhan Card records, January–August 2018 and 2019 (486 days). Shinhan was Korea's largest card company, with about 22% of credit-card sales in 2019 (Appendix A.3). The provider aggregates transactions to cells of day × time slot (06–10, 10–18, 18–22, 22–06h) × merchant category (62, plus online) × merchant neighborhood × cardholder sex × ten-year age group × cardholder residence; Appendix Table A.2 lists the columns. The amounts and counts are Shinhan's own amounts and counts multiplied by one factor per month (4.89–4.96 in 2018, 5.01–5.07 in 2019; its inverse, about 20%, is Shinhan's market share) and rounded; the factor is the same in every category, place and day of a month (Appendix A.2). A count of 5 is therefore a single Shinhan transaction, not a suppression floor, and a zero is a cell with no Shinhan transaction. The data contain no cardholder or card identifier, so individuals cannot be followed and individual fixed effects, as in Kim and Trusov (2025), cannot be estimated. We keep cardholders resident in Seoul and merchants in Seoul or online, and sum to residence district × merchant district (or online) ×

day.

Mobility. KT inflow population data (*saenghwal idong*), January–August 2019 (243 days): the estimated number of people present in each Seoul administrative neighborhood each hour, by home neighborhood, sex and five-year age band, from LTE signals of KT subscribers (about 31% of mobile lines) scaled to population. Cells below about 2.9 people are suppressed and count as zero. We sum residents' presence into people-hours over all 24 hours (and by the card data's time slots), by home district and district of presence. The sample is Seoul residents' presence in Seoul.

Who is covered. Card and mobility data both describe Seoul residents within Seoul, but different people: Shinhan cardholders, by registered address, and KT subscribers, by habitual location. Appendix Table A.4 compares their age composition and their distribution across districts; the card users' file (`num_ccu`) is an index (first cell = 100), not a count.

Pollution and weather. Each district's PM2.5 (and other pollutants) is the daily value of its own AirKorea urban-network monitor: Seoul's urban network has exactly one monitor in each district (Appendix Table A.1); roadside monitors (in 15 districts) are excluded, as is usual when the target is residents' general exposure. Missing days are filled from neighboring monitors (as in the neighborhood draft); all 25 urban monitors report on more than 80% of days. Weather (temperature, humidity, precipitation, wind speed and direction) is the district's own KMA automatic weather station, the mean of two stations in Gwanak, Dongjak and Yeongdeungpo; Jongno's station reports on fewer than 80% of days, so Jongno takes the nearest station, and humidity, measured at only 16 stations, comes from the nearest station that measures it in 11 districts. City values are means over the 25 districts. Pollution varies almost entirely from day to day (Appendix A.4).

Instruments. *Wind direction*: the daily direction of the mean hourly wind vector at the district's station (or, for the city, over the districts), in eight 45° sectors; wind speed is a control. Two other codings are compared in Appendix B.1: the *modal* direction, the day's most frequent hourly sector, and a *westerly* indicator, wind from the arc between south-west and north-west (213.75–326.25°) where PM2.5 is significantly higher; the arc is chosen from the data on days outside the estimation sample (September–December 2018 and 2019, January–August 2020), so that the instrument is not selected for its first-stage strength on the days it is used for (Appendix Figure B.1).[1] *Thermal inversions*: from vertical temperature profiles of the air above Osan Air Base, about 45–50 km south of Seoul City Hall, in NOAA's Integrated Global Radiosonde Archive (IGRA v2.2). Each profile is a *sounding*: a weather balloon carrying a radiosonde, a small instrument package that records temperature, pressure, humidity and wind as it rises. Osan's balloon profiles are taken up to four times a day (03, 09, 15 and 21 KST); we use those of 21 KST the previous day and 03 and 09 KST. An inversion is a layer where temperature rises with height, which traps pollution near the ground. The main measure counts the balloon profiles with a temperature rise within the lowest 100 hPa and adds the strength of the 03 KST inversion; it is the same for all districts.

[1] Wind speed instruments pollution in Mink (2025) and Zhang et al. (2020); we use direction, as Deryugina et al. (2019), because speed is tied to the weather system and to the attractiveness of going out.

Alerts. Seoul's advisory and warning history: in 2018–19, 25 PM2.5 and 14 PM10 alerts, all city-wide, and 83 ozone alerts issued by area (each area is a group of whole districts). An alert is in effect from its issue hour to its clear hour (Appendix E.1).

Outcomes. Table 1 defines the outcomes and Table 2 the exposure, control and instrument variables; Tables 3 and 4 summarize them. The geography around home has three bands: the *home district*, *adjacent districts* and *other districts* (Figure 1); "outside the home district" means adjacent or other districts. Rings by the number of district borders crossed are in Appendix D. Mobility outcomes use the same bands and count people-hours over all 24 hours. The average amount per transaction is total spending divided by the number of transactions. The distance of offline spending is measured two ways, between district centroids (home district = 0 km): weighted by spending and, new in this draft, weighted by the number of transactions, so that a few large purchases far away do not dominate it. The main district outcomes are the shares of each band: the share of all spending in the home district, adjacent districts, other districts and online, and the share of people-hours in each band.

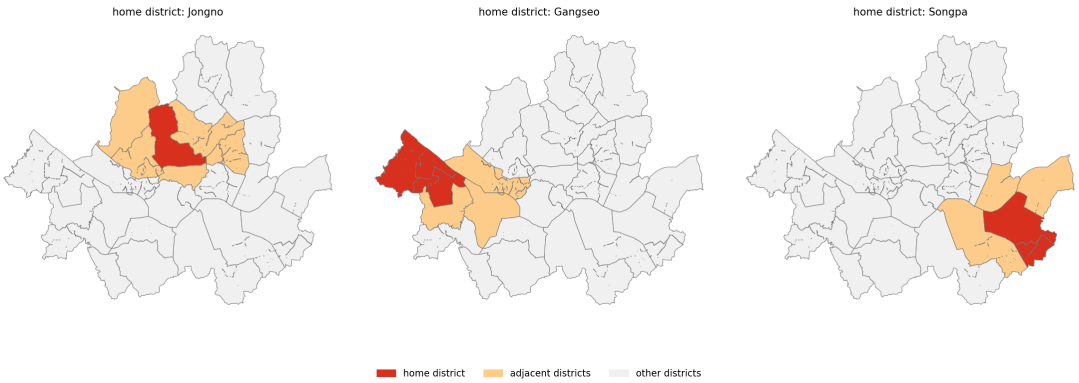


Figure 1: The three bands around three example home districts: home district, adjacent districts (sharing a boundary), other districts.

Table 1: Outcome variables

Variable	Definition	Unit	Used in
A. City × day			
All card spending	Card spending of Seoul residents at Seoul merchants and online	bn KRW/day	main
Offline spending	Spending at merchants in Seoul	bn KRW/day	main
Online spending	Spending at online merchants (no location)	bn KRW/day	main
Spending, the home district	Offline spending at merchants in the cardholder's home district	bn KRW/day	main
Spending, adjacent districts	Offline spending in districts bordering the home district	bn KRW/day	main
Spending, other districts	Offline spending in all other districts	bn KRW/day	main
Online share	Online spending / all spending	%	main
Share outside the home district	Offline spending outside the home district / all spending	%	main
Transactions	Number of card transactions	million/day	appendix
Amount per transaction	All spending / all transactions (ratio of sums)	KRW	main
Amount per transaction, offline	Offline spending / offline transactions	KRW	appendix
Amount per transaction, online	Online spending / online transactions	KRW	appendix
Distance, offline	Spending-weighted mean centroid distance from the home district to the merchant district (own district = 0)	km	main
Distance, offline, transaction-weighted	The same, weighted by the number of transactions instead of spending, so that a few large purchases far away do not dominate it	km	main
People-hours in Seoul	Residents' people-hours present in Seoul, all 24 hours (KT, 2019)	million/day	main
Time outside the home district	Share of residents' people-hours (24 hours) outside the home district	%	main
Time outside the home district, 06–22h	As above, 06–22h only (robustness)	%	appendix
Time outside the home district, 60+	As above (24 hours), residents aged 60 and over	%	main
Distance from home	People-hour-weighted mean centroid distance from the home district (home = 0), 24 hours	km	main
Health care	Categories 51, 54–57 (medical services, clinics, oriental medicine, pharmacies, medical products)	bn KRW/day	main
Pharmacies	Category 56	bn KRW/day	main
Air purifiers	Category 16 (air control machines)	bn KRW/day	main
Outdoor recreation	Category 31 (recreational and leisure activities)	bn KRW/day	main
Indoor entertainment	Categories 36–37 (cinemas, exhibits; PC cafes, arcades)	bn KRW/day	main
Delivery food	Category 24 (fried chicken, pizza, burgers)	bn KRW/day	main
Taxi	Category 48	bn KRW/day	main
Bus and subway	Category 46	bn KRW/day	main
B. District × day			
All card spending	Card spending of the district's residents	m KRW/day	main
Spending, the home district	At merchants in the home district	m KRW/day	main
Spending, adjacent districts	At merchants in bordering districts	m KRW/day	main
Spending, other districts	At merchants in all other districts	m KRW/day	main
Online spending	Online merchants	m KRW/day	main
Share outside the home district	Offline spending outside the home district / all spending	%	main
Share of spending, the home district	Spending the home district / all spending (online included)	%	main
Share of spending, adjacent districts	Spending in adjacent districts / all spending	%	main
Share of spending, other districts	Spending in other districts / all spending	%	main
Online share	Online spending / all spending	%	main
Transactions	Card transactions of the district's residents (as reported: Shinhan's count times the month's factor of about 5)	per day	main
Amount per transaction	All spending / all transactions	KRW	main
Distance, offline	Spending-weighted mean centroid distance to the merchant district	km	main

Table 2: Exposure, controls and instruments

Variable	Definition	Unit	Used in
A. City × day			
PM2.5	Mean over the 25 districts of each district's urban AirKorea monitor	$\mu g/m^3$	main
PM2.5, alternative	Mean over districts of the legal-neighborhood inverse-distance PM2.5	$\mu g/m^3$	appendix
Temperature	Daily mean, KMA stations	°C	control
Humidity	Daily mean relative humidity	%	control
Precipitation	Daily total	mm	control
Wind speed	Daily mean	m/s	control
Wind direction	Sector (1–8, 45°) of the daily mean wind vector; 0 = calm	sector	instrument
Wind direction, modal	The day's most frequent hourly sector (1–8); 0 = calm	sector	instrument
Westerly wind	Daily wind from the south-west to north-west arc where PM2.5 is higher on held-out days (Appendix Figure B.1)	0/1	instrument
Inversion strength, 03 KST	Warmest temperature in the lowest 100 hPa minus the surface temperature (Osan balloon profile)	°C	instrument
Inversions	Balloon profiles with an inversion in the lowest 100 hPa (21 KST $t-1$, 03, 09 KST)	0–3	instrument
B. District × day			
PM2.5	Daily PM2.5 at the district's urban AirKorea monitor	$\mu g/m^3$	main
Temperature	The district's KMA station	°C	control
Wind speed	Daily mean	m/s	control
Wind direction	Sector of the daily mean wind vector at the district's station(s)	sector	instrument
Wind direction, modal	The day's most frequent hourly sector at the district	sector	instrument
Westerly wind	Wind from the held-out westerly arc at the district	0/1	instrument

Table 3: Summary statistics: outcomes

Variable	Unit	Mean	SD	Min	Median	Max	Obs.
A. City × day							
All card spending	bn KRW/day	206.1	33.4	60.7	206.4	307.5	486
Offline spending	bn KRW/day	175.2	23.0	56.3	173.0	239.9	486
Online spending	bn KRW/day	30.9	19.8	4.4	26.9	98.4	486
Spending, the home district	bn KRW/day	86.2	11.2	28.0	84.9	124.4	486
Spending, adjacent districts	bn KRW/day	41.5	5.6	12.8	40.8	56.1	486
Spending, other districts	bn KRW/day	47.5	7.1	15.3	47.1	62.2	486
Online share	%	14.2	7.3	4.2	13.5	32.9	486
Share outside the home district	%	43.5	3.2	33.5	43.9	48.9	486
Transactions	million/day	7.87	0.78	3.32	8.01	9.12	486
Amount per transaction	KRW	26,094	2,906	17,902	25,439	35,545	486
Amount per transaction, offline	KRW	22,737	1,649	17,049	22,481	28,893	486
Amount per transaction, online	KRW	166,988	91,421	35,482	148,354	553,999	486
Distance, offline	km	4.72	0.21	3.89	4.71	5.33	486
Distance, offline, transaction-weighted	km	4.68	0.25	4.00	4.74	5.09	486
People-hours in Seoul	million/day	170.4	11.1	147.5	168.8	188.9	243
Time outside the home district	%	18.6	2.9	12.0	20.1	22.5	243
Time outside the home district, 06–22h	%	24.9	4.6	14.2	27.5	30.0	243
Time outside the home district, 60+	%	14.6	1.6	9.3	15.2	17.1	243
Distance from home	km	1.61	0.28	0.97	1.77	1.96	243
Health care	bn KRW/day	24.3	10.4	1.7	27.5	43.1	486
Pharmacies	bn KRW/day	4.51	2.84	0.19	4.34	13.41	486
Air purifiers	bn KRW/day	0.092	0.046	0.001	0.092	0.291	486
Outdoor recreation	bn KRW/day	4.97	0.80	1.59	4.96	7.57	486
Indoor entertainment	bn KRW/day	1.30	0.36	0.71	1.25	2.49	486
Delivery food	bn KRW/day	3.47	0.59	2.00	3.41	5.61	486
Taxi	bn KRW/day	2.43	0.24	1.24	2.43	2.96	486
Bus and subway	bn KRW/day	2.31	0.27	1.44	2.31	3.00	486
B. District × day							
All card spending	m KRW/day	8,243	3,885	934	7,583	31,910	12,150
Spending, the home district	m KRW/day	3,447	1,861	289	2,989	11,071	12,150
Spending, adjacent districts	m KRW/day	1,660	776	142	1,568	6,372	12,150
Spending, other districts	m KRW/day	1,900	716	265	1,872	5,235	12,150
Online spending	m KRW/day	1,235	1,376	60	862	16,285	12,150
Share outside the home district	%	45.3	7.6	8.6	45.1	69.5	12,150
Share of spending, the home district	%	41.2	8.0	4.7	41.3	65.8	12,150
Share of spending, adjacent districts	%	21.1	7.6	4.0	20.0	52.0	12,150
Share of spending, other districts	%	24.2	5.0	4.6	24.5	41.7	12,150
Online share	%	13.5	8.2	2.3	12.3	86.3	12,150
Transactions	per day	314,893	115,792	49,986	306,080	650,740	12,150
Amount per transaction	KRW	25,888	5,668	15,601	24,677	153,258	12,150
Distance, offline	km	4.78	0.62	2.82	4.78	9.80	12,150
Distance, offline, transaction-weighted	km	4.62	0.67	3.30	4.50	6.66	12,150
People-hours in Seoul	thousand/day	6,818	2,300	2,108	6,650	13,328	6,075
Share of time, the home district	%	80.9	3.9	72.4	80.5	91.9	6,075
Share of time, adjacent districts	%	10.2	2.7	3.7	10.1	18.7	6,075
Share of time, other districts	%	8.95	2.34	3.75	9.20	15.11	6,075
Time outside the home district	%	19.1	3.9	8.1	19.5	27.6	6,075
Distance from home	km	1.62	0.34	0.79	1.65	2.59	6,075

Panel A: days of the sample (486 for 2018–19; 243 for the 2019 mobility outcomes). Panel B: district × day.

Table 4: Summary statistics: exposure, controls and instruments

Variable	Unit	Mean	SD	Min	Median	Max	Obs.
A. City × day							
PM2.5	$\mu g/m^3$	26.2	18.0	2.7	22.3	135.3	486
PM2.5, alternative	$\mu g/m^3$	26.1	18.0	2.7	22.2	135.5	486
Temperature	°C	14.4	11.4	-14.4	16.4	33.3	486
Humidity	%	62.1	15.7	27.0	61.6	96.1	486
Precipitation	mm	3.37	11.37	0.00	0.00	107.19	486
Wind speed	m/s	1.67	0.51	0.76	1.53	4.04	486
Wind direction	sector	5.34	2.06	1.00	6.00	8.00	486
Wind direction, modal	sector	5.14	2.12	1.00	6.00	8.00	486
Westerly wind	0/1	0.650	0.477	0.000	1.000	1.000	486
Inversion strength, 03 KST	°C	0.811	1.252	0.000	0.000	6.800	484
Inversions	0–3	1.47	1.05	0.00	1.00	3.00	478
B. District × day							
PM2.5	$\mu g/m^3$	26.2	18.5	1.0	22.0	152.8	12,150
Temperature	°C	14.4	11.4	-16.9	16.4	34.4	12,150
Wind speed	m/s	1.67	0.69	0.10	1.53	6.76	12,150
Wind direction	sector	5.19	2.18	0.00	6.00	8.00	12,150
Wind direction, modal	sector	4.44	2.57	0.00	5.00	8.00	12,150
Westerly wind	0/1	0.525	0.499	0.000	1.000	1.000	12,150

Panel A: days of the sample (486). Panel B: district × day. Wind direction is a sector code, so its mean is not meaningful.

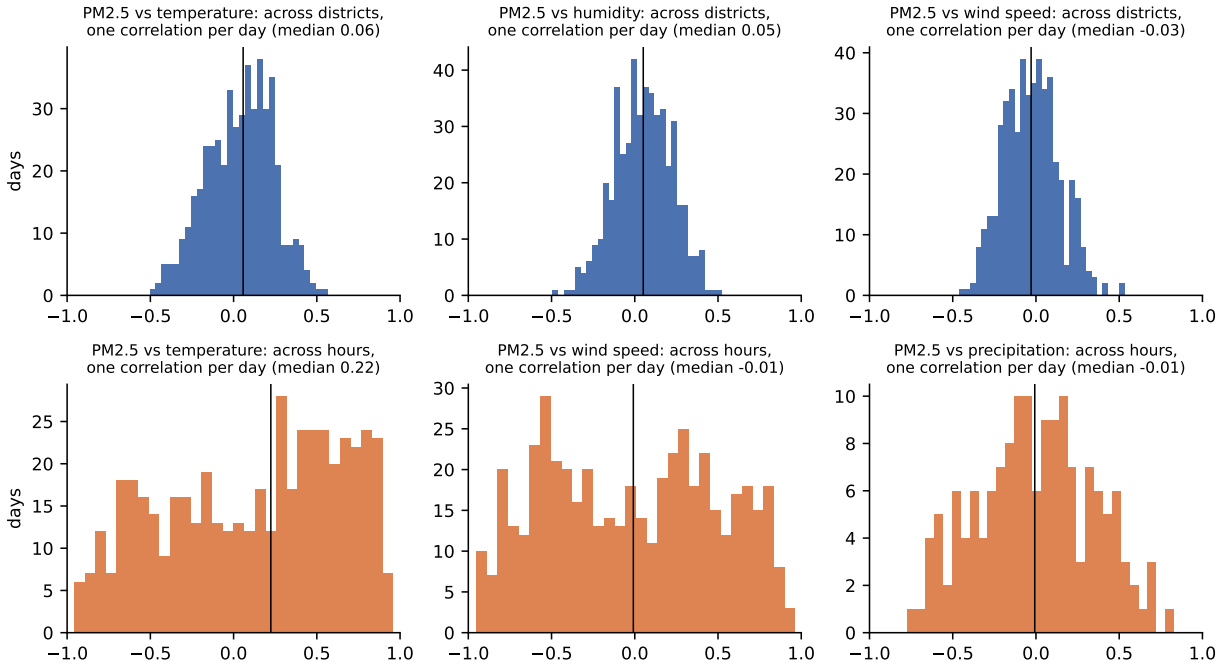


Figure 2: How PM2.5 and weather co-move. Top: for each day, the correlation across the 25 districts between PM2.5 and temperature, humidity or wind speed (histogram over days; line: median). Bottom: for each day, the correlation across the 24 hours between city PM2.5 and city temperature, wind speed or precipitation.

What the figure shows.

- Across districts on the same day, PM2.5 is only weakly correlated with temperature, humidity and wind speed (medians near zero), unlike across neighborhoods, where the interpolation from shared stations correlates them; the correlations vary widely from day to day.
- Within a day, hourly PM2.5 and temperature are positively correlated on most days; hourly correlations with wind speed and precipitation fall on both sides of zero. Example maps and hourly profiles are in Appendix Figures A.2–A.3.

2 Specifications

Controls, defined once. In the city series (day t), the calendar terms are $\alpha_{\text{dow}(t)} + \alpha_{\text{ym}(t)} + \alpha_{\text{hol}(t)} + \gamma_1 B_t + \gamma_2 S_t$: day-of-week fixed effects, year-month fixed effects (year-week in robustness rows), one fixed effect per public holiday, an indicator B_t for bridge days and an indicator S_t for the day before and after Seollal. The weather terms W_t are a cubic in mean temperature, daily maximum and minimum temperature, a rain indicator, and quadratics in precipitation, humidity and wind speed; "weather $t\pm2$" adds temperature and precipitation at $t-2, \ldots, t+2$. In the district panel, the same calendar terms enter as fixed effects together with weather cells (3°C temperature bin × precipitation category × wind-speed decile) and humidity deciles of the district's own station, and district fixed effects α_r. Year-month fixed effects also absorb the card data's monthly scaling factor.

City series. One observation per day; Y_t is an outcome summed over Seoul residents and P_t is the mean PM2.5 of the 25 district monitors:

$$Y_t = \beta P_t + \alpha_{\text{dow}(t)} + \alpha_{\text{ym}(t)} + \alpha_{\text{hol}(t)} + \gamma_1 B_t + \gamma_2 S_t + W_t'\delta + \varepsilon_t. \tag{1}$$

Distributed lag: $\sum_{j=-2}^{3} \beta_j P_{t-j}$ replaces βP_t. Local projections: $Y_{t+h} = \beta_h P_t + \gamma_{1h} P_{t-1} + \gamma_{2h} P_{t-2} + \alpha_{\text{dow}(t)} + \alpha_{\text{ym}(t)} + \alpha_{\text{hol}(t)} + \gamma_1 B_t + \gamma_2 S_t + W_t'\delta_h + \varepsilon_{t+h}$. 2SLS and LIML: P_t instrumented by Z_t (wind-sector dummies, or the inversion count and strength). *Why LIML.* With several weak instruments, such as the seven wind-sector dummies, 2SLS is biased toward OLS, and the bias grows with the number of instruments; limited-information maximum likelihood (LIML) is approximately median-unbiased in that case. An LIML estimate close to the 2SLS one is therefore a check that many-weak-instrument bias is not driving the 2SLS result, as in Deryugina et al. (2019); with a single instrument the two coincide. Standard errors are clustered by year-week. The city series differs from the neighborhood draft's only in the exposure (the mean of the 25 district monitors rather than of the 467 neighborhood interpolations) and in the geography of the share outcomes (outside the home district rather than beyond 2 km).

Leads. Coefficients on future pollution are not treated as placebos: anticipation from forecasts or visible haze shows up as leads, possibly of the opposite sign, while slow-moving confounding implies leads and lags of the same sign (Graff Zivin and Neidell, 2009). The tables report the joint test of the leads and the sum over the window.

District panel: one regression per share. Residence district r, day t, exposure P_{rt}, the PM2.5 of the district's monitor in μg/m^3. The main geography specification regresses each band's share directly on pollution, one regression per share:

$$s_{rkt} = \beta_k P_{rt} + \alpha_{rk} + \lambda_{k,\mathrm{ym}(t)} + \lambda_{k,\mathrm{dow}(t)} + \lambda_{k,\mathrm{hol}(t)} + \lambda_{k,B(t)} + \lambda_{k,S(t)} + \lambda_{k,\mathrm{wc}(rt)} + \lambda_{k,\mathrm{hd}(rt)} + \varepsilon_{rkt}, \quad (2)$$

where s_{rkt} is the share of district r's residents' spending on day t in band k (0: home district, 1: adjacent districts, 2: other districts, online; shares of all spending, so they sum to one) or the share of their people-hours in band k (0–2). β_k is the change in the share, in percentage points, per 10 μg/m^3 of PM2.5. *Why PM2.5 in levels.* As in the city series and most of the air-pollution literature, pollution enters in levels, so the district and city estimates are in the same unit and can be compared directly; the share itself stays in levels so that the four spending shares' coefficients sum to zero (a log or log-odds share would not add up). Earlier drafts used $\log P_{rt}$, a holdover from the Poisson models of spending levels, where log pollution gives elasticities; those models (the stacked bands below, the levels and the categories) keep $\log P_{rt}$. Estimated by OLS, weighted by the district's mean spending (or people-hours), and by 2SLS and LIML; standard errors are clustered by district and date. The OLS coefficients of the spending shares sum to zero. The same equation with a district-day outcome Y_{rt} gives the count and mobility rows of the geography table: the number of transactions, the amount per transaction, the two distances of offline spending, people-hours in Seoul, the share of time outside the home district and the distance from home.

The stacked model (appendix). The earlier drafts' main geography specification, now in Appendix C, stacks the bands k in one Poisson pseudo-maximum-likelihood regression:

$$E[Y_{rkt}] = \exp\Big(\textstyle\sum_{j\neq 0} \beta_j \, \mathbf{1}[k=j] \, \log P_{rt} + \mu_{rt} + \psi_{rk} + \lambda_{k,\mathrm{ym}(t)} + \lambda_{k,\mathrm{dow}(t)} + \lambda_{k,\mathrm{hol}(t)} + \lambda_{k,B(t)} + \lambda_{k,S(t)} + \lambda_{k,\mathrm{wc}(rt)} + \lambda_{k,\mathrm{hd}(rt)} \Big) \quad (3)$$

with district × day fixed effects μ_{rt}, district × band fixed effects ψ_{rk}, and band-specific calendar, weather-cell and humidity-decile effects. Standard errors are clustered by district and date; with 25 districts, the p-values in the results files also use a t distribution with 24 degrees of freedom.

What stacking does. With one row per district, day and band and one Poisson regression on all rows, the district × day fixed effects absorb everything common to all bands of a district on a day, including the effect of pollution on its total spending; β_j is the change in band-j spending relative to the home district. With μ_{rt}, the model is a fractional multinomial logit of the band shares weighted by the district-day's total spending, so β_j is the change in $\log(s_j/s_0)$; for small changes $\Delta s_j \approx s_j(\beta_j - \sum_l s_l \beta_l)$ (Appendix Table C.2). It answers "does pollution move spending out of, or into, the home district?", robust to day-level confounders, but not whether total spending changes. One regression per band (Appendix Table D.3) gives the level effect in each band instead; it cannot include μ_{rt} and relies on the calendar and weather controls. If pollution cuts spending equally in every band, the per-band elasticities are equal and negative and the stacked β_j are zero; if it moves spending into the home district with the total unchanged, the stacked β of the other bands are negative. When the controls deal with day-level confounding, $\beta_j \approx$ (level elasticity of band j) − (level elasticity in the home district).

Instruments in Poisson models: PPML with a control function (PPML-IV). Step 1: OLS of $\log P_{rt}$ on the instruments (wind sectors at the district's station, or the city-wide inversion count

and strength) and the same fixed effects and controls; its residual $\hat{v}_{rt}$ is the part of pollution the instruments do not explain. Step 2: the Poisson regression with $\hat{v}_{rt}$ added (by band in the stacked model) (Wooldridge, 2015). The pollution coefficient then uses only instrument-driven variation; the joint test of the $\hat{v}$ terms tests exogeneity. Standard errors include the first stage's sampling error (two-step correction, Murphy and Topel, 1985; Appendix Table D.8 adds a bootstrap). With a weak first stage, PPML-IV estimates are noisy and sensitive.

Do we still need PPML-IV? At the district level no district-day is zero, so zeros are no reason for Poisson. They were never the only one: Santos Silva and Tenreyro (2006) show that elasticities from OLS on logs are biased when the variance of the error depends on the regressors (heteroskedasticity), while Poisson pseudo-maximum likelihood is consistent; and in the stacked band model, Poisson with district × day fixed effects is the share (multinomial) model. With the shares now estimated directly by OLS and 2SLS (equation (2)), PPML-IV is needed only for the elasticities of levels. Appendix Table D.3 compares, for each level, PPML and PPML-IV with log OLS and log 2SLS using the same instruments: where they agree, PPML-IV adds nothing; where they differ, the difference is the heteroskedasticity that PPML is robust to.

Zeros and observations. Every table note says whether zeros are kept and how many observations are used (Appendix H). At the district level, district-day spending is never zero, so the share and per-transaction outcomes lose no observations; PPML drops observations separated by the fixed effects, and every Poisson table note reports how many. *What "separated" means.* A fixed-effect group whose outcome is zero in every observation (for example, a district-day with no spending in some band, in the stacked model) can only be fit by sending its fixed effect to minus infinity, so that the predicted outcome is zero; those observations are fit perfectly whatever pollution does and carry no information about the pollution coefficient. PPML drops them, as it drops singleton groups (Correia et al., 2020); dropping them does not change the estimate, only the count of observations.

Alerts. As in the neighborhood draft: city × day × slot for PM alerts (slot calendar fixed effects; the comparison with other slots of the same day is not estimated) and area × day × slot for ozone alerts (date × slot fixed effects, comparing areas in the same slot), with or without flexible controls for pollution in the same slot; date clusters. *Days just above and below the cutoffs*: the running variable is the day's highest two-hour reading behind the PM2.5 rule minus the cutoff at which issuance actually jumps, the treatment an alert issued that day, estimated by local linear regression (fuzzy) (Calonico et al., 2014). By our readings, alerts are issued above the documented cutoff (90 μg/m^3 before 1 July 2018, 75 from then), so we estimate the cutoff from the data as a threshold: the shift δ of the documented cutoff for which "reading $\geq$ documented cutoff $+\,\delta$" best fits whether an alert was issued, by least squares on a grid of 0.5 μg/m^3, with the likelihood-ratio 95% set of Hansen (2000) (Appendix Table E.2). Issuance jumps $\hat{\delta} = 5.5$ μg/m^3 above the documented cutoff (95% set 5.5–11; 10 before 1 July 2018 and 5.5 after); the threshold change gives a second comparison (readings of 80.5–95.5). For ozone the estimated cutoff is the documented 0.12 ppm ($\hat{\delta} = -0.001$), at the area's highest monitor. The alert analyses use city or area outcomes, so they differ from the neighborhood draft only in the mobility share (outside the home district).

3 Results

3.1 The city time series

Specification.

- Equation (1) (panel A row 1); row 2 year-week fixed effects; rows 3–4 weather $t \pm 2$. Panel B: distributed lag, β_0, the joint test of the leads and $\sum_j \beta_j$. Panel C: 2SLS and LIML, Z_t = seven wind-sector dummies or the inversion count and 03 KST strength.
- City × day; spending 486 days, mobility 243 days. Outcomes: all, offline and online card spending (billion KRW per day); online share and share of spending outside the home district (pp); share of residents' people-hours (24 hours) outside the home district (pp); people-hour-weighted distance from home (km between district centroids). Year-week clusters.

Table 5: City time series: effect of 10 μg/m^3 of PM2.5

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
A. Least squares							
Year-month FE (main)	**-1.64****	**-0.78****	**-0.87**	**-0.24**	**0.09**	**-0.06*****	**-0.005****
	(0.81)	(0.36)	(0.64)	(0.23)	(0.08)	(0.02)	(0.002)
Year-week FE	**-0.04**	**-0.39**	**0.35**	**0.15**	**-0.04**	**-0.04**	**-0.003**
	(0.93)	(0.34)	(0.68)	(0.23)	(0.08)	(0.03)	(0.003)
Year-month FE, weather $t \pm 2$	**-1.41***	**-0.61***	**-0.80**	**-0.23**	**0.08**	**-0.06****	**-0.005****
	(0.86)	(0.35)	(0.67)	(0.24)	(0.08)	(0.02)	(0.002)
Year-week FE, weather $t \pm 2$	**-0.14**	**-0.35**	**0.21**	**0.09**	**-0.03**	**-0.04**	**-0.003**
	(0.95)	(0.36)	(0.67)	(0.23)	(0.08)	(0.03)	(0.003)
B. Distributed lag ($P_{t+2}, \ldots, P_{t-3}$)							
Year-month FE: coefficient on P_t	**-1.43***	**-0.93****	**-0.50**	**-0.11**	**0.08**	**-0.07***	**-0.006***
	(0.78)	(0.42)	(0.59)	(0.20)	(0.08)	(0.04)	(0.004)
Leads P_{t+1}, P_{t+2}: joint p	0.72	0.22	0.56	0.39	0.42	0.00	0.00
Sum of $P_{t+2}, \ldots, P_{t-3}$	**-2.26**	**-1.07**	**-1.18**	**-0.32**	**0.09**	**-0.12*****	**-0.011*****
	(1.40)	(0.75)	(0.97)	(0.33)	(0.13)	(0.03)	(0.003)
Year-week FE, weather $t \pm 2$: coefficient on P_t	**-0.39**	**-0.48**	**0.09**	**0.06**	**-0.01**	**-0.06***	**-0.006***
	(0.92)	(0.44)	(0.67)	(0.23)	(0.09)	(0.03)	(0.003)
Leads P_{t+1}, P_{t+2}: joint p	0.36	0.48	0.35	0.27	0.04	0.10	0.11
Sum of $P_{t+2}, \ldots, P_{t-3}$	**0.10**	**-1.02**	**1.12**	**0.48**	**-0.15**	**-0.14***	**-0.012**
	(1.75)	(0.66)	(1.34)	(0.44)	(0.12)	(0.08)	(0.008)
C. Two-stage least squares and LIML							
2SLS, wind direction (8 sectors)	**6.52**	**4.94***	**1.58**	**0.17**	**0.07**	**-0.26**	**-0.027**
	(5.32)	(2.56)	(3.46)	(1.13)	(0.47)	(0.22)	(0.022)
First-stage F (Kleibergen–Paap)	3.2	3.2	3.2	3.2	3.2	1.3	1.3
Effective F (Montiel Olea–Pflueger)	2.3	2.3	2.3	2.3	2.3	1.7	1.7
LIML, wind direction	**12.53**	**7.37***	**9.56**	**4.50**	**0.06**	**-0.97**	**-0.129**
	(10.18)	(4.14)	(17.33)	(17.89)	(1.09)	(2.19)	(0.356)
2SLS, thermal inversions	**-8.23****	**-0.86**	**-7.37****	**-2.33****	**0.94****	**0.00**	**-0.001**
	(3.88)	(1.79)	(2.96)	(0.96)	(0.38)	(0.18)	(0.019)
First-stage F (Kleibergen–Paap)	7.8	7.8	7.8	7.8	7.8	3.4	3.4
Effective F (Montiel Olea–Pflueger)	9.5	9.5	9.5	9.5	9.5	4.2	4.2
LIML, inversions	**-8.50****	**-0.88**	**-7.39****	**-2.34****	**0.94****	**0.01**	**0.001**
	(4.07)	(2.01)	(2.97)	(0.97)	(0.39)	(0.21)	(0.023)
2SLS, inversions, year-week FE, weather $t \pm 2$	**-4.43**	**-0.73**	**-3.70**	**-0.97**	**0.62**	**-0.10**	**-0.007**
	(4.06)	(1.57)	(3.10)	(0.98)	(0.41)	(0.24)	(0.025)
First-stage F (Kleibergen–Paap)	5.4	5.4	5.4	5.4	5.4	2.5	2.5
Effective F (Montiel Olea–Pflueger)	8.8	8.8	8.8	8.8	8.8	3.3	3.3
Mean of outcome	206.1	175.2	30.9	14.2	43.5	18.6	1.61

Notes: OLS, distributed-lag OLS, 2SLS and LIML in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: the main OLS estimate in the first column, −1.64, is −0.8% of its mean (206.07); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- With year-month fixed effects, all and offline spending are lower on more polluted days; online spending, the online share and the share of spending outside the home district do not move significantly. Adding weather at $t \pm 2$ changes little; with year-week fixed effects the spending coefficients are close to zero. The estimates are close to the neighborhood draft's, slightly smaller for spending.
- In the distributed lag with year-month fixed effects, the spending leads are jointly insignificant and the same-day coefficients on all and offline spending remain negative; the sum over the window is negative but insignificant.
- Residents spend a smaller share of their time (24 hours) outside the home district and stay closer to home on more polluted days; the mobility leads are jointly significant and the sum over the window is negative and significant; with year-week fixed effects and weather leads, the same-day coefficients remain negative (10% level).
- The wind instrument is weak (effective F 2.3) and the wind 2SLS and LIML estimates are imprecise. The inversion instrument is stronger (effective F 9.5 for spending, 4.2 for mobility) but below the 23.1 critical value; with it, all and online spending and the online share are lower and the share of spending outside the home district higher; LIML gives the same; with year-week fixed effects and weather leads the estimates are smaller and insignificant.

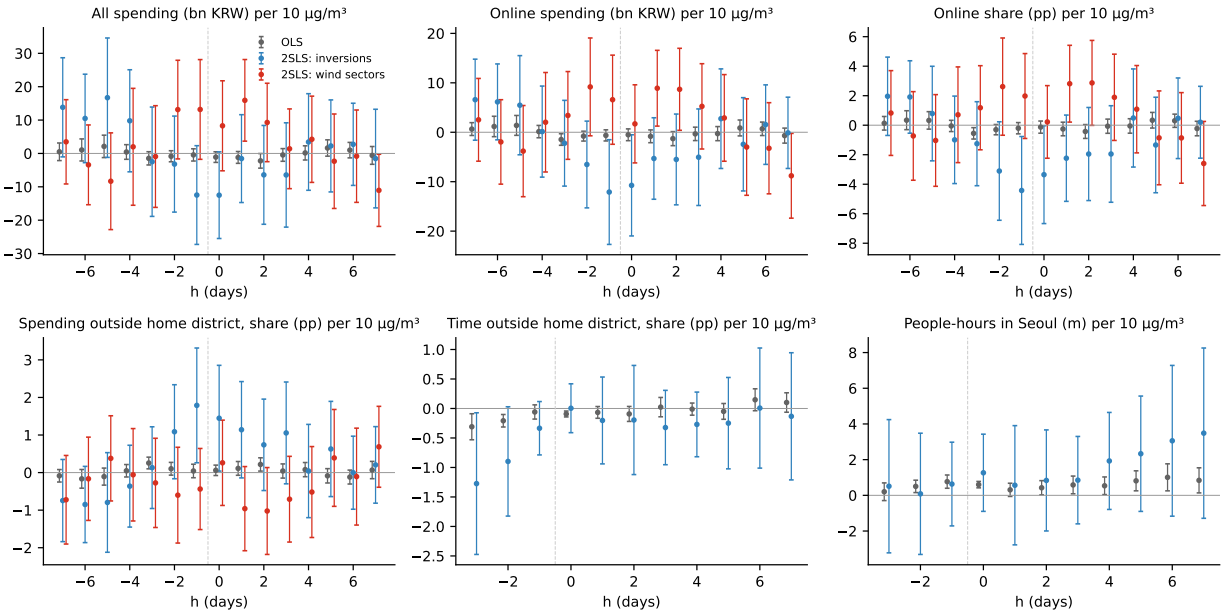


Figure 3: Local projections, city series: β_h from $Y_{t+h} = \beta_h P_t + \gamma_{1h} P_{t-1} + \gamma_{2h} P_{t-2} + \alpha_{\mathrm{dow}(t)} + \alpha_{\mathrm{ym}(t)} + \alpha_{\mathrm{hol}(t)} + \gamma_1 B_t + \gamma_2 S_t + W_t' \delta_h + \varepsilon_{t+h}$, per 10 μg/m^3; OLS and 2SLS (inversions, wind sectors); 95% intervals, year-week clusters.

What the figure shows.

- OLS: all spending is lower two days after the shock; online spending and the online share are lower three days before it; the share of spending outside the home district is higher three

days before and two days after.

- Inversions: online spending and the online share are lower, and the share outside the home district higher, on the day before and the day of the shock. Wind-instrumented paths are imprecise, with spending higher one or two days after the shock.
- Mobility (OLS): the share of time outside the home district is lower two and three days before and on the polluted day; people-hours in Seoul are higher around the shock and in the following week. With year-week fixed effects and weather leads, spending horizons are insignificant and time outside the home district remains lower on the polluted day.

3.2 Weekends and holidays

Specification.

- City (Table 6): $Y_t = \beta P_t + \delta P_t D_t + \alpha_{\text{dow}(t)} + \alpha_{\text{ym}(t)} + \alpha_{\text{hol}(t)} + \gamma_1 B_t + \gamma_2 S_t + W_t'\eta + \varepsilon_t$, $D_t =$ weekend or public holiday; rows β, $\beta + \delta$, δ; second block year-week fixed effects and weather $t \pm 2$.
- City with 2SLS (Table 7): the same equation, P_t and $P_t D_t$ instrumented by Z_t and $Z_t D_t$ (inversions; wind sectors); year-month fixed effects.
- District panel (Table 8): the share regressions of equation (2) and the linear outcomes with $\delta P_{rt} D_t$ added (OLS), and the levels (PPML, with $\log P_{rt}$ and $\log P_{rt} D_t$); Table 9: the share and distance regressions by 2SLS, P_{rt} and $P_{rt} D_t$ instrumented by Z and $Z D_t$. The stacked version is in Appendix Table C.3.

Table 6: Weekends and holidays, city series

	Card spending, 2018–19					Mobility, 2019, 24 hours			
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home	People-hours in Seoul (m)	Time, 60+ (pp)
Year-month FE									
Weekdays	**-1.86****	**-0.87****	**-1.00**	**-0.28**	**0.09**	**-0.05*****	**-0.005*****	**0.370*****	**-0.05*****
	(0.95)	(0.41)	(0.77)	(0.28)	(0.09)	(0.02)	(0.002)	(0.086)	(0.02)
Weekends and holidays	**-1.39**	**-0.47**	**-0.92**	**-0.29**	**0.11**	**-0.15*****	**-0.014*****	**0.688*****	**-0.11*****
	(0.97)	(0.49)	(0.64)	(0.20)	(0.08)	(0.05)	(0.005)	(0.157)	(0.03)
Difference	**0.48**	**0.40**	**0.08**	**-0.00**	**0.02**	**-0.09****	**-0.009****	**0.318****	**-0.06***
	(1.08)	(0.53)	(0.77)	(0.26)	(0.10)	(0.04)	(0.004)	(0.157)	(0.03)
Year-week FE, weather $t \pm 2$									
Weekdays	**-0.26**	**-0.41**	**0.15**	**0.06**	**-0.02**	**-0.03**	**-0.002**	**-0.331*****	**-0.04****
	(0.96)	(0.32)	(0.71)	(0.25)	(0.09)	(0.02)	(0.002)	(0.127)	(0.02)
Weekends and holidays	**0.05**	**0.01**	**0.04**	**0.04**	**-0.02**	**-0.15****	**-0.014***	**-0.326**	**-0.13****
	(1.64)	(0.70)	(1.07)	(0.32)	(0.10)	(0.07)	(0.007)	(0.272)	(0.06)
Difference	**0.31**	**0.42**	**-0.11**	**-0.02**	**0.00**	**-0.12***	**-0.012***	**0.005**	**-0.09***
	(1.60)	(0.69)	(1.05)	(0.32)	(0.11)	(0.07)	(0.007)	(0.245)	(0.05)

Notes: OLS only (no instrument; Table 7 gives 2SLS): $Y_t = \beta P_t + \delta P_t D_t +$ controls, $D_t =$ weekend or public holiday. OLS in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending on weekdays, -1.86, is -0.9% of its mean (206.07); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Spending responds no differently on weekends and holidays than on weekdays.

- Time outside the home district and distance from home fall more on weekends and holidays, also for the 60+ group, and people-hours in Seoul rise more; the differences are significant with year-month fixed effects and, for time outside the home district and distance, at the 10% level with year-week fixed effects and weather leads.

Table 7: Weekends and holidays, city series, 2SLS

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
2SLS, thermal inversions							
Weekdays	**-10.35****	**-2.06**	**-8.29****	**-2.61****	**0.96****	**-0.20**	**-0.019**
	(4.57)	(2.06)	(3.44)	(1.12)	(0.44)	(0.12)	(0.012)
Weekends and holidays	**-7.44****	**-0.21**	**-7.23****	**-2.30****	**0.91****	**-0.28**	**-0.028**
	(3.61)	(1.76)	(2.82)	(0.92)	(0.37)	(0.17)	(0.019)
Difference	**2.91**	**1.85**	**1.06**	**0.32**	**-0.05**	**-0.08**	**-0.010**
	(2.65)	(1.25)	(1.91)	(0.62)	(0.26)	(0.11)	(0.012)
First-stage F (Kleibergen–Paap, both endogenous)	4.1	4.1	4.1	4.1	4.1	1.8	1.8
2SLS, wind direction (8 sectors)							
Weekdays	**8.11***	**4.85****	**3.26**	**0.69**	**-0.31**	**-0.04**	**-0.005**
	(4.64)	(2.33)	(3.00)	(0.96)	(0.40)	(0.09)	(0.009)
Weekends and holidays	**7.37***	**4.66***	**2.71**	**0.74**	**-0.33**	**-0.15**	**-0.019**
	(4.40)	(2.75)	(2.32)	(0.74)	(0.33)	(0.16)	(0.017)
Difference	**-0.74**	**-0.19**	**-0.55**	**0.05**	**-0.03**	**-0.11**	**-0.014**
	(3.66)	(2.23)	(2.10)	(0.67)	(0.29)	(0.12)	(0.012)
First-stage F (Kleibergen–Paap, both endogenous)	3.8	3.8	3.8	3.8	3.8	1.7	1.7

Notes: 2SLS: P_t and P_tD_t instrumented by Z_t and Z_tD_t (D_t = weekend or public holiday), year-month fixed effects and the main weather terms. 2SLS in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending on weekdays, inversions, -10.35, is -5.0% of its mean (205.92); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- With inversions, all and online spending and the online share are lower, and the share of spending outside the home district higher, on more polluted weekdays and on more polluted weekends and holidays alike; the weekend–weekday differences are not significant for any outcome.
- With wind sectors, all and offline spending are higher on more polluted weekdays and weekends (at the 10% level or better), and again the weekend–weekday differences are not significant. Both first stages are weak once the weekend interactions are instrumented (Kleibergen–Paap F about 4 for spending, below 2 for mobility).

Table 8: Weekends and holidays, district panel

Outcome	PM2.5 (weekdays)	PM2.5 × weekend or holiday
Share of spending, home district (pp)	**0.156** (0.119)	**0.159** (0.134)
Share of spending, adjacent districts (pp)	**0.028** (0.038)	**0.082**** (0.037)
Share of spending, other districts (pp)	**0.020** (0.029)	**0.033** (0.038)
Online share (pp)	**-0.204** (0.175)	**-0.274** (0.182)
Share of time, home district (pp)	**0.032*** (0.018)	**0.106***** (0.034)
Share of time, adjacent districts (pp)	**-0.019**** (0.008)	**-0.064***** (0.018)
Share of time, other districts (pp)	**-0.013** (0.011)	**-0.042**** (0.020)
Spending outside home district, share (pp)	**0.048** (0.064)	**0.115*** (0.070)
Time outside home district, share (pp)	**-0.032*** (0.018)	**-0.106***** (0.034)
Km from home	**-0.003** (0.002)	**-0.009**** (0.003)
Time outside home district, 60+ (pp)	**-0.044***** (0.013)	**-0.055**** (0.024)
All spending (PPML)	**-0.010** (0.008)	**-0.006** (0.008)
Offline spending (PPML)	**-0.007** (0.005)	**0.008** (0.007)
Online spending (PPML)	**-0.062*** (0.034)	**-0.052** (0.044)
People-hours in Seoul (PPML)	**0.005***** (0.002)	**0.007**** (0.003)

Notes: Share and distance rows: OLS on PM2.5 in levels (per 10 μg/m^3), weighted by mean spending or people-hours; district-days with no spending (or no people-hours) drop because the share is undefined (observations used 5,861–11,934). Level rows: PPML on log PM2.5 (elasticities), zeros kept, districts always zero dropped as separated (observations used 5,861–11,934). District and date clusters. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: spending outside home district on weekdays, 0.048: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by 0.065, against a mean of 45.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- On weekdays, pollution does not change any band's share of spending or the distance from home; the share of time in the home district is higher and in adjacent districts lower (10% and 5% level), time outside the home district lower (10% level; 1% for the 60+ group); online spending is lower (PPML, 10% level) and people-hours in Seoul are higher on more polluted days.
- On weekends and holidays, the shift of time toward the home district is larger: the weekend–weekday differences in the shares of time in each band, in time outside the home district and in the distance from home are significant. For spending, only the adjacent-district share (5% level) and the share outside the home district (10% level) rise more on weekends; the home-district and online shares do not differ. Spending levels do not differ between weekends and weekdays.

Table 9: Weekends and holidays, district panel, 2SLS

	2SLS, wind		2SLS, inversions		KP F	
Outcome	weekdays	weekend − weekday	weekdays	weekend − weekday	wind	inv.
Spending outside home district, share (pp)	**-0.942**** (0.400)	**0.727**** (0.348)	**0.480*** (0.284)	**0.177** (0.164)	3.6	10.4
Online share (pp)	**3.308***** (0.900)	**-1.856***** (0.621)	**-1.118*** (0.675)	**-0.433** (0.396)	3.6	10.4
Time outside home district, share (pp)	**0.979*** (0.560)	**-0.806** (0.503)	**0.095** (0.152)	**0.064** (0.110)	4.3	3.8
Km from home	**0.023** (0.074)	**-0.008** (0.074)	**0.010** (0.015)	**0.005** (0.011)	4.3	3.8

Notes: 2SLS: P and $P \times D_t$ (PM2.5 in levels, per 10 μg/m^3) instrumented by the instruments and their products with D_t; weighted by mean spending or people-hours; district-days without spending (people-hours) dropped (observations used 5,861–11,934). District and date clusters. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: spending outside home district on weekdays, inversions, 0.480: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by 0.651, against a mean of 45.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- With wind, on weekdays more polluted days have a lower share of spending outside the home district (5% level), a higher online share (1% level) and more time outside the home district (10% level); the weekend–weekday differences in the two spending shares are significant and of the opposite sign. With inversions, the share of spending outside the home district is higher and the online share lower on weekdays (10% level), with no significant weekend difference; mobility does not move.
- The first stages are weak (Kleibergen–Paap F 3.6–10.4), so these estimates are imprecise and sensitive to the instrument.

3.3 Geography within a district-day

Specification.

- Panel A, equation (2): one regression per share, for the shares of all spending in the home district, adjacent districts, other districts and online, and the shares of people-hours (24 hours) in the three bands. OLS; 2SLS and LIML with first stage $P_{rt} = Z'\pi + \alpha_r + \lambda_{\text{ym}} + \lambda_{\text{dow}} + \lambda_{\text{hol}} + \lambda_B + \lambda_S + \lambda_{\text{wc}} + \lambda_{\text{hd}} + v_{rt}$, Z = wind sectors at the district's station or the inversion count and strength.
- Panel B: the same equation with a district-day outcome: the number of transactions, the amount per transaction, the transaction-weighted and spending-weighted distance of offline spending, people-hours in Seoul, the share of time outside the home district and the distance from home; OLS and 2SLS.
- Unit: district × day; spending 25 × 486, mobility 25 × 243. Shares weighted by mean spending (people-hours), counts and people-hours unweighted. District and date clusters.
- The stacked Poisson model of the earlier drafts, its conversion to percentage points and its variants are in Appendix C.

Table 10: Geography within district-day: one regression per band share, and the transaction and mobility outcomes

Outcome	Mean	OLS	2SLS wind	2SLS inversions	LIML wind	KP F wind	KP F inv.
A. Share of each band (pp), one regression per share							
Spending, home district	41.2	**0.201****	**-0.644**	**0.927****	**-1.126**	3.0	17.1
		(0.095)	(0.621)	(0.415)	(1.025)		
Spending, adjacent districts	21.1	**0.051**	**-0.414**	**0.400****	**-0.470**	3.0	17.1
		(0.032)	(0.267)	(0.174)	(0.303)		
Spending, other districts	24.2	**0.029**	**-0.067**	**0.286****	**-0.074**	3.0	17.1
		(0.026)	(0.213)	(0.133)	(0.229)		
Spending, online	13.5	**-0.281****	**1.124**	**-1.613****	**1.642**	3.0	17.1
		(0.142)	(0.975)	(0.666)	(1.373)		
Time present, home district	80.9	**0.056*****	**0.263**	**-0.046**	**0.367**	1.2	5.6
		(0.017)	(0.267)	(0.148)	(0.425)		
Time present, adjacent districts	10.2	**-0.034*****	**0.024**	**0.020**	**0.042**	1.2	5.6
		(0.006)	(0.105)	(0.055)	(0.140)		
Time present, other districts	9.0	**-0.023****	**-0.288**	**0.027**	**-0.402**	1.2	5.6
		(0.011)	(0.229)	(0.093)	(0.368)		
B. Transactions and mobility							
Number of transactions (per district-day)	314,893	**-653.8****	**2316.2**	**549.2**		2.3	16.9
		(264.5)	(2196.6)	(1316.2)			
Amount per transaction (KRW)	25,888	**-197.3*****	**1353.8****	**-862.3****		3.0	17.1
		(71.6)	(590.7)	(378.8)			
Km, offline, transaction-weighted	4.62	**0.002**	**-0.007**	**0.006**		3.0	17.1
		(0.003)	(0.020)	(0.012)			
Km, offline, spending-weighted	4.78	**-0.009***	**0.045**	**-0.036**		3.0	17.1
		(0.005)	(0.035)	(0.023)			
People-hours in Seoul (thousand)	6,818	**14.83*****	**60.78**	**31.13**		1.2	5.5
		(3.81)	(54.07)	(32.46)			
Time outside home district (pp)	19.13	**-0.056*****	**-0.263**	**0.046**		1.2	5.6
		(0.017)	(0.267)	(0.148)			
Km from home	1.62	**-0.005*****	**-0.038**	**0.004**		1.2	5.6
		(0.002)	(0.033)	(0.015)			

Notes: District × day; OLS, 2SLS and LIML on PM2.5 in levels, P_{rt} (per 10 μg/m^3), with district, year-month, day-of-week and holiday fixed effects, weather cells and humidity deciles; shares weighted by mean spending (people-hours), counts and people-hours unweighted. Panel A: each band's share of all spending (online included) or of people-hours, one regression per share, so the spending shares' coefficients sum to zero across the four bands for OLS. Shares and distances are undefined on district-days without spending (people-hours) and dropped; counts and people-hours keep zeros (observations used 5,861–11,934 of 12,150 for spending, 6,075 for mobility). Transactions are counted as reported (Shinhan's count times the month's factor of about 5, absorbed by the year-month fixed effects). District and date clusters. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: the share of spending in other districts (OLS), 0.029: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by 0.040, against a mean of 24.2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Without an instrument, more polluted days have a higher share of spending in the home district and a lower online share (5% level); the adjacent and other-district shares do not change. The share of time in the home district is higher and in adjacent and other districts lower (1% and 5% level).
- With inversions (Kleibergen–Paap F 17.1 for spending, below the critical value), the online share is lower and the shares of all three physical bands higher on more polluted days (5% level); with wind (first-stage F 3.0), no share moves. Instrumented mobility shares are imprecise (wind F 1.2, inversions 5.6).
- By OLS, the number of transactions is lower (5% level), the amount per transaction lower

(1% level), the spending-weighted distance of offline spending slightly shorter (10% level; the transaction-weighted one does not change), people-hours in Seoul higher, and time outside the home district and the distance from home lower (1% level). The amount per transaction is higher with wind and lower with inversions (5% level); the other instrumented estimates are imprecise.

Do we want the instrument here? For the geography and share outcomes the instruments help less than in the city series. Wind direction at the district's station is weak as eight sectors (first-stage F 2.3) and as the modal direction (3.9); the westerly indicator is stronger (9.4 with the held-out arc, 12.3 a priori) but gives imprecise share estimates (Appendix Table B.5). Every first stage is weaker with PM2.5 in levels than it was with log PM2.5 (5.6, 7.7 and 32.2 in the log version), consistent with the instruments moving PM2.5 roughly in proportion to its level; and the city reduced forms show wind moving spending directly (Appendix Table B.4), which casts doubt on the exclusion restriction of every wind instrument. The inversion instrument is city-wide, below the critical value for spending (F 16.9) and weak for mobility (5.5). OLS instead relies on the controls to deal with what moves both pollution and where residents spend; the share regressions have no district $\times$ day fixed effects, but the weather cells, humidity deciles and calendar fixed effects are measured at the district-day. OLS with the full controls is therefore a defensible main specification for the shares, with the instrumented estimates as robustness. This draft keeps the current layout, with OLS and the instrumented estimates side by side, pending that decision.

3.4 The size of purchases

Specification.

- Equation (1) with Y_t = total spending / total transactions (KRW) for all spending, offline, online, without single-transaction cells, by band, by time slot and by category; the district version is in Appendix Table D.4.

Table 11: Average amount per transaction, city series (KRW per 10 μg/m^3)

Amount per transaction	Mean (KRW)	OLS (KRW)	OLS (% of mean)	OLS, year-week, weather $t \pm 2$	DLM, P_t	2SLS inversions	DLM leads p
All	26,094	**-122.2**	**-0.47**	**48.4**	**-57.4**	**-839.2****	0.82
		(94.7)		(111.0)	(83.4)	(426.8)	
Offline	22,737	**-35.0**	**-0.15**	**9.0**	**-25.7**	**44.7**	0.83
		(35.9)		(32.7)	(34.3)	(158.4)	
Online	166,988	**-2902.8**	**-1.74**	**1870.1**	**454.9**	**-41743.9*****	0.22
		(3256.6)		(3945.6)	(3142.8)	(16063.4)	
All, without single-transaction cells	25,523	**-125.2**	**-0.49**	**45.5**	**-55.4**	**-852.3****	0.82
		(93.4)		(109.7)	(82.2)	(424.8)	
Offline, home district	21,875	**-4.8**	**-0.02**	**8.7**	**-14.7**	**230.2**	0.61
		(42.8)		(28.2)	(41.1)	(152.9)	
Offline, adjacent districts	26,908	**-51.3**	**-0.19**	**5.9**	**-26.1**	**-14.1**	0.80
		(40.3)		(44.7)	(43.1)	(183.7)	
Offline, other districts	21,431	**-78.4**	**-0.37**	**5.8**	**-44.1**	**-203.5**	0.96
		(52.7)		(68.8)	(44.7)	(241.4)	
06–10h	19,661	**-126.7**	**-0.64**	**188.4**	**35.1**	**-1844.8***	0.39
		(235.3)		(273.2)	(236.5)	(985.1)	
10–18h	29,625	**-171.1**	**-0.58**	**72.7**	**-88.6**	**-1237.1****	0.88
		(137.4)		(157.2)	(114.0)	(623.4)	
18–22h	24,587	**-64.8**	**-0.26**	**-12.3**	**-45.2**	**-54.4**	1.00
		(41.6)		(48.9)	(42.3)	(231.8)	
22–06h	19,047	**-3.1**	**-0.02**	**-8.9**	**-9.5**	**-112.9**	0.44
		(18.0)		(21.1)	(24.6)	(107.5)	
Health care	46,696	**-216.0**	**-0.46**	**108.1**	**-95.6**	**-2264.8****	0.83
		(180.3)		(234.0)	(170.3)	(946.8)	
Pharmacies (masks)	18,355	**-167.0**	**-0.91**	**410.9**	**156.1**	**-1919.2***	0.93
		(282.4)		(319.4)	(220.6)	(1147.3)	
Air purifiers	204,242	**-4063.5**	**-1.99**	**-2172.2**	**-1475.0**	**-16569.9**	0.75
		(4098.5)		(4487.7)	(4498.8)	(18826.3)	
Outdoor recreation	61,830	**-5.7**	**-0.01**	**252.8**	**-81.2**	**-33.2**	0.72
		(252.9)		(244.3)	(236.1)	(950.4)	
Indoor entertainment	7,420	**-28.9***	**-0.39***	**-15.8**	**-13.7**	**-172.4**	0.51
		(17.4)		(18.5)	(18.5)	(118.7)	
Delivery food	13,860	**4.4**	**0.03**	**2.0**	**-5.3**	**42.5**	0.30
		(10.3)		(13.6)	(15.3)	(53.7)	
Taxi	7,817	**-6.5**	**-0.08**	**-3.2**	**-3.2**	**-43.4**	0.22
		(4.7)		(4.4)	(7.0)	(33.1)	
Bus and subway	8,266	**-10.3****	**-0.12****	**-5.9**	**-11.6***	**-63.2***	0.23
		(4.8)		(5.1)	(6.5)	(35.5)	

Notes: City × day; OLS, distributed-lag OLS and 2SLS (inversions) in levels; the average is total amount / total transactions over the city's cells (a ratio of sums). Zeros: none at the city level. Year-week clusters. Reading: KRW per transaction per 10 μg/m^3 more PM2.5; the second column gives the OLS estimate as percent of the mean. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- The average amount per transaction does not change significantly with pollution in OLS, for all, offline or online purchases, in any band or time slot; among categories, bus and subway transactions are slightly smaller (5% level) and indoor-entertainment transactions slightly smaller (10% level).
- With inversions, the average purchase is smaller, for all purchases and online purchases, in the daytime slots, and in health care and pharmacies.

3.5 Categories that pollution should move

Specification.

- Pre-specified categories and predicted signs: health care up (Barwick et al., 2018; Deryugina et al., 2019); pharmacies up (Zhang and Mu, 2018); air purifiers up (Ito and Zhang, 2020); outdoor recreation down (Neidell, 2009); indoor entertainment ambiguous; delivery food up; taxi up; bus and subway down.
- City columns: OLS (% of mean per 10 μg/m^3), year-week version, PPML, distributed lag.

District columns: $E[Y_{rt}] = \exp(\beta \log P_{rt} + \alpha_r + \lambda_{\text{ym}} + \lambda_{\text{dow}} + \lambda_{\text{hol}} + \lambda_B + \lambda_S + \lambda_{\text{wc}} + \lambda_{\text{hd}})$, PPML and PPML-IV (wind).

Table 12: Pre-specified categories

	Predicted	City × day				District × day		DLM leads
Category	sign	OLS (% of mean)	OLS, year-week (%)	PPML	DLM, P_t (%)	PPML	PPML-IV	joint p
Health care	up	**-0.28**	**0.75**	**-0.003**	**-0.28**	**-0.008**	**0.050***	0.69
		(0.47)	(0.53)	(0.015)	(0.42)	(0.008)	(0.027)	
Pharmacies (masks)	up	**-0.62**	**3.13***	**-0.007**	**1.09**	**-0.023**	**0.240****	0.92
		(1.70)	(1.77)	(0.053)	(1.35)	(0.030)	(0.095)	
Air purifiers	up	**-0.26**	**1.07**	**-0.051**	**-0.09**	**-0.010**	**0.218**	0.29
		(1.76)	(1.49)	(0.045)	(1.92)	(0.029)	(0.139)	
Outdoor recreation	down	**-0.36**	**-0.07**	**-0.005**	**-0.66**	**-0.005**	**0.071*****	0.38
		(0.43)	(0.46)	(0.012)	(0.41)	(0.008)	(0.025)	
Indoor entertainment	ambiguous	**-1.02****	**-1.06*****	**-0.034****	**-0.50**	**-0.027*****	**-0.025**	0.87
		(0.43)	(0.40)	(0.017)	(0.44)	(0.009)	(0.025)	
Delivery food	up	**-0.23**	**-0.31**	**-0.006**	**-0.51**	**-0.006**	**0.023**	0.06
		(0.20)	(0.31)	(0.007)	(0.34)	(0.005)	(0.016)	
Taxi	up	**0.00**	**-0.10**	**0.004**	**0.08**	**0.000**	**0.003**	0.17
		(0.13)	(0.14)	(0.003)	(0.16)	(0.002)	(0.009)	
Bus and subway	down	**-0.25**	**-0.06**	**-0.003**	**-0.21**	**-0.004**	**0.004**	0.36
		(0.18)	(0.20)	(0.005)	(0.20)	(0.003)	(0.010)	

Notes: City columns: OLS and distributed-lag OLS in levels (shown as percent of the outcome's mean per 10 μg/m^3) and PPML (elasticity); zeros kept (486 days). District columns: PPML and PPML-IV (wind); zeros kept; districts with no spending in a category on any day, and other separated observations, are dropped: observations used 11,911–11,934 of 12,150 (dropped 216–239). PPML-IV standard errors corrected for the estimated first stage. Reading: OLS columns in percent of the mean per 10 μg/m^3; PPML columns are elasticities (a 1% rise in PM2.5 changes spending by the coefficient in percent). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- No category moves in its predicted direction in the city series with year-month fixed effects; pharmacy spending rises with year-week fixed effects (10% level); the leads are jointly insignificant at the 5% level for every category.
- Indoor entertainment, whose sign is not predicted, falls in the city series and in the district PPML.
- With the wind control function, health care (10% level), pharmacies and outdoor recreation rise; the outdoor-recreation estimate has the opposite of its predicted sign; delivery food, taxi, and bus and subway do not move in any column.

3.6 Air-quality alerts

Specification.

- PM2.5 alerts: $Y_{ts} = \beta A_{ts} + f(P_{ts}) + \alpha_{s,\text{dow}(t)} + \alpha_{s,\text{ym}(t)} + \alpha_{s,\text{hol}(t)} + \alpha_{s,B(t)} + \alpha_{s,S(t)} + W'_{ts}\delta + \varepsilon_{ts}$, city × day × slot; f = flexible slot pollution controls (omitted in Table 14). Ozone alerts: area × day × slot, May–August, date × slot, area × slot, area × day-of-week and area × year-month fixed effects. OLS and PPML; mobility 2019, all four slots. Date clusters.

Table 13: Alerts, given measured pollution

	Card spending, 2018–19				Mobility, 2019, 24 hours	
	All (bn KRW)	All (%)	Online (%)	Online share (pp)	Time outside home district (pp)	Hours outside home district (%)
PM2.5 alert in effect, city slots	**-1.48**	**-2.4**	**-3.5**	**-1.26**	**-0.16**	**-1.0**
	(1.52)	(2.7)	(10.9)	(1.08)	(0.16)	(0.9)
Ozone alert in effect, area slots (date × slot FE)	**-0.91***	**-3.4**	**-15.9****	**-1.71**	**0.10**	**0.1**
	(0.55)	(2.3)	(7.2)	(1.06)	(0.11)	(0.3)

Notes: OLS and PPML. Slot regressions: zeros kept (slot sums are never zero); city slots 1,944 (486 days × 4), area slots May–August 4,920; mobility 2019 only, all four slots (24 hours). Date clusters. Reading: OLS columns: change in the outcome (billion KRW per slot; shares and time in percentage points) while an alert is in effect, at the same measured pollution; PPML columns: percent change, $100(e^{\beta}-1)$. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

What the table shows.

- Given measured pollution, PM2.5 alerts do not change all spending, online spending, the online share or the share of time outside the home district.
- Ozone alerts lower online spending of the alert area's residents and do not change mobility. The estimate depends on the Downtown area: leaving it out, the online estimate is -4.5% ($p=0.59$), while leaving out any other area it stays between -16% and -32% ($p\leq 0.07$). (Spending outcomes are city or area sums, so these estimates are identical to the neighborhood draft's.)

Table 14: Alerts without pollution controls

	Card spending, 2018–19				Mobility, 2019, 24 hours	
	All (bn KRW)	All (%)	Online (%)	Online share (pp)	Time outside home district (pp)	Hours outside home district (%)
PM2.5 alert, city slots, slot × calendar FE	**-2.71*****	**-5.0*****	**-14.5****	**-1.67****	**-0.32****	**-1.5****
	(1.05)	(1.8)	(6.7)	(0.83)	(0.13)	(0.6)
Ozone alert, area slots, date × slot FE	**-0.76**	**-3.5**	**-16.7****	**-1.81***	**0.16**	**0.2**
	(0.55)	(2.2)	(6.9)	(1.02)	(0.11)	(0.3)
PM2.5 alert day (city × day)	**-10.76*****			**-1.71****	**-0.23**	
	(3.67)			(0.84)	(0.15)	

Notes: OLS and PPML, no pollution controls. Slot regressions: zeros kept (slot sums are never zero); city slots 1,944 (486 days × 4), area slots May–August 4,920; mobility 2019 only, all four slots (24 hours). Date clusters. Alert-day row: OLS, city × day, Newey–West (7 lags). Reading: OLS columns: change in the outcome (billion KRW per slot; shares and time in percentage points) while an alert is in effect, at the same measured pollution; PPML columns: percent change, $100(e^{\beta}-1)$. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

What the table shows.

- Without pollution controls, slots under a PM2.5 alert have lower spending (all and online), a lower online share and fewer people-hours outside the home district; alert days have lower daily spending and online share.
- Ozone alerts lower online spending of the area's residents with or without pollution controls; mobility does not change.

Table 15: Days just above vs. just below the estimated PM2.5 alert cutoff

	Fuzzy RD at the estimated PM2.5 cutoff, bandwidth				2018-07-01 change:
Outcome	MSE-optimal	10 μg/m^3	20	30	80.5–95.5 × after, reduced form
First stage: alert issued that day	**0.63** (0.32)	**0.62** (0.49)	**0.58** (0.29)	**0.56** (0.22)	**0.64***** (0.08)
alert in effect at some time that day	**0.62**** (0.22)	**0.89***** (0.26)	**0.64***** (0.24)	**0.61**** (0.20)	
Days below / above the cutoff	20/18	11/10	35/21	76/26	
All spending (bn KRW)	**-9.737** [0.48]	**2.278** [0.59]	**-6.086** [0.52]	**0.930** [0.54]	**11.882** [0.23]
Online (bn KRW)	**-4.243** [0.64]	**1.856** [0.45]	**-4.098** [0.74]	**2.153** [0.56]	**6.782** [0.33]
Online share	**-0.009** [0.80]	**0.010** [0.39]	**-0.009** [0.89]	**0.005** [0.75]	**0.013** [0.58]
Share outside home district	**0.008** [0.56]	**-0.008** [0.15]	**0.010** [0.70]	**0.005** [0.52]	**0.002** [0.87]
Time outside home district (share)	**0.001** [0.64]	**0.002** [0.52]	**-0.001** [0.10]	**-0.002** [0.88]	**-0.001** [0.67]
People-hours in Seoul (m)	**-0.339** [0.80]	**-4.766***** [0.00]	**-0.264** [0.37]	**0.564** [0.75]	**-0.285** [0.85]
All spending, next day	**5.240** [0.79]	**-24.482** [0.81]	**9.275** [0.21]	**18.619** [0.79]	**28.340***** [0.01]
Time outside home district, next day	**0.010** [0.35]	**-0.000** [0.94]	**0.005** [0.44]	**0.004** [0.34]	**0.002** [0.75]

Notes: City × day, 486 days. Running variable: the day's highest two-hour minimum of area-mean PM2.5 minus the estimated cutoff, the documented one (90 before 2018-07-01, 75 after) plus $\hat{\delta} = 5.5$ μg/m^3 (Appendix Table E.2). Outcomes residualized on day of week, year-month, holiday terms and weather. Fuzzy RD, local linear, triangular kernel (rdrobust); estimate [robust bias-corrected p]. Last column: OLS, days with readings of 80.5–95.5 after vs before the threshold change, relative to days below 80.5 and above 95.5, with separate linear trends in the reading; Newey–West (7); the first row is its first stage (alert issued), the other rows the reduced form (outcome), not divided by the first stage. No zeros (city sums). Reading: the jump in the outcome at the cutoff (outcome units, after the controls are partialled out), for the days that get an alert because they cross it (fuzzy RD: the jump in the outcome divided by the jump in the probability of an alert). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Area-slots just above vs. just below the ozone alert cutoff

Outcome	MSE-optimal	0.01 ppm	0.02 ppm
First stage: ozone alert in effect	**0.76***** (0.14)	**0.84**** (0.19)	**0.76***** (0.13)
All spending (bn KRW)	**-1.4753** [0.42]	**-0.8152** [0.86]	**-1.4213** [0.64]
Online (bn KRW)	**-1.6467** [0.15]	**-1.5324** [0.61]	**-1.5254** [0.43]
Online share	**-0.0394** [0.13]	**-0.0645** [0.41]	**-0.0463** [0.34]
Time outside home district (share)	**0.0032** [0.22]	**0.0015** [0.92]	**0.0031** [0.83]
People-hours in Seoul (m)	**-0.0207** [0.92]	**0.0180** [0.25]	**-0.0169** [0.99]

Notes: Area × day × slot, May–August (4,920). Running variable: the slot's highest hourly ozone at any monitor in the area minus the estimated cutoff (0.12 ppm plus $\hat{\delta}$, -0.001 ppm; Appendix Table E.2). Outcomes residualized on date × slot, area × slot, area × day-of-week and area × year-month fixed effects; fuzzy RD, clustered by date; estimate [robust bias-corrected p]. Reading: the jump in the outcome at the cutoff (outcome units, after the controls are partialled out), for the days that get an alert because they cross it (fuzzy RD: the jump in the outcome divided by the jump in the probability of an alert). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- At the estimated cutoff, the probability that an alert is in effect jumps by about 0.6 in every bandwidth (significant); the jump in an alert issued that day is of the same size but imprecise. Spending, the online share, the share outside the home district and same-day time outside the home district show no jump; people-hours in Seoul are lower with the 10 μg/m^3 bandwidth only. With 10–26 days above the cutoff within the bandwidths, these are at most suggestive.
- The 1 July 2018 threshold change shifted issuance: days with readings between the two estimated cutoffs (80.5–95.5) became more likely to get an alert after the change. On those days, same-day spending, the online share and mobility did not change; next-day spending is higher (1% level).

- Ozone: the probability of an alert jumps by 0.76–0.84 at the estimated cutoff (the documented 0.12 ppm), but spending, the online share, time outside the home district and people-hours in Seoul show no jump.

Pending data

- Disaster text messages by district and the air-quality forecast history; KMA ASOS visibility and yellow-dust records; Seoul's registered population by district, sex and age.

References

Arceo, E., Hanna, R., and Oliva, P. (2016). Does the effect of pollution on infant mortality differ between developing and developed countries? Evidence from Mexico City. *Economic Journal*, 126(591):257–280.

Barwick, P. J., Li, S., Lin, L., and Zou, E. Y. (2024). From fog to smog: The value of pollution information. *American Economic Review*, 114(5):1338–1381.

Barwick, P. J., Li, S., Rao, D., and Zahur, N. B. (2018). The healthcare cost of air pollution: Evidence from the world's largest payment network. Working Paper 24688, National Bureau of Economic Research.

Calonico, S., Cattaneo, M. D., and Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica*, 82(6):2295–2326.

Correia, S., Guimarães, P., and Zylkin, T. (2020). Fast Poisson estimation with high-dimensional fixed effects. *Stata Journal*, 20(1):95–115.

Deryugina, T., Heutel, G., Miller, N. H., Molitor, D., and Reif, J. (2019). The mortality and medical costs of air pollution: Evidence from changes in wind direction. *American Economic Review*, 109(12):4178–4219.

Graff Zivin, J. and Neidell, M. (2009). Days of haze: Environmental information disclosure and intertemporal avoidance behavior. *Journal of Environmental Economics and Management*, 58(2):119–128.

Hansen, B. E. (2000). Sample splitting and threshold estimation. *Econometrica*, 68(3):575–603.

Ito, K. and Zhang, S. (2020). Willingness to pay for clean air: Evidence from air purifier markets in china. *Journal of Political Economy*, 128(5):1627–1672.

Jung, H. (2020). The impact of ambient fine particulate matter on consumer expenditures. *Sustainability*, 12(5):1855.

Kim, S., Koh, K., and Lyou, W. (2023). Spend as you were told: Evidence from labeled COVID-19 stimulus payments in South Korea. *Journal of Public Economics*, 221:104867.

Kim, S. and Trusov, M. (2025). The impact of air pollution on consumer spending. *Journal of Marketing*, 89(3):78–96.

Mink, J. (2025). The healthcare costs of air pollution in france. *Journal of Environmental Economics and Management*, 134:103242.

Murphy, K. M. and Topel, R. H. (1985). Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 3(4):370–379.

Neidell, M. (2009). Information, avoidance behavior, and health: The effect of ozone on asthma hospitalizations. *Journal of Human Resources*, 44(2):450–478.

Santos Silva, J. M. C. and Tenreyro, S. (2006). The log of gravity. *Review of Economics and Statistics*, 88(4):641–658.

Schlenker, W. and Walker, W. R. (2016). Airports, air pollution, and contemporaneous health. *Review of Economic Studies*, 83(2):768–809.

Wooldridge, J. M. (2015). Control function methods in applied econometrics. *Journal of Human Resources*, 50(2):420–445.

Zhang, J. and Mu, Q. (2018). Air pollution and defensive expenditures: Evidence from particulate-filtering facemasks. *Journal of Environmental Economics and Management*, 92:517–536.

Zhang, N., Ren, R., Zhang, Q., and Zhang, T. (2020). Air pollution and tourism development: An interplay. *Annals of Tourism Research*, 85:103032.

A Data and measurement

A.1 Districts, their monitors and stations

Table A.1: Seoul's districts: size, neighbors, monitor and weather stations

District	Code	Area	Legal nbhds.	km^2	Adjacent	Urban monitor	Roadside monitor	KMA station(s)
Jongno	11110	Downtown	87	24.0	5	111123	111125	422
Jung	11140	Downtown	74	10.0	6	111121	111124	419
Yongsan	11170	Downtown	36	21.9	7	111131	111122	415
Seongdong	11200	Northeast	17	16.8	5	111142	111143	421
Gwangjin	11215	Northeast	7	17.0	6	111141		413
Dongdaemun	11230	Northeast	10	14.2	6	111152	111154	408
Jungnang	11260	Northeast	6	18.5	4	111151		409
Seongbuk	11290	Northeast	39	24.6	5	111161	111162	414
Gangbuk	11305	Northeast	4	23.6	3	111291		424
Dobong	11320	Northeast	4	20.7	2	111171		406
Nowon	11350	Northeast	5	35.6	4	111311	111312	407
Eunpyeong	11380	Northwest	11	29.8	3	111181		416
Seodaemun	11410	Northwest	20	17.7	4	111191		412
Mapo	11440	Northwest	26	23.8	6	111201	111202	411
Yangcheon	11470	Southwest	3	17.4	3	111301		405
Gangseo	11500	Southwest	13	41.4	3	111212	111213	404
Guro	11530	Southwest	10	20.1	5	111221		423
Geumcheon	11545	Southwest	3	13.0	2	111281	111282	417
Yeongdeungpo	11560	Southwest	34	24.5	7	111231	111232	418 510
Dongjak	11590	Southwest	9	16.4	5	111241	111242	410 889
Gwanak	11620	Southwest	3	29.6	5	111251		425 509
Seocho	11650	Southeast	10	46.9	4	111262	111264	401
Gangnam	11680	Southeast	14	39.5	5	111261	111263	400
Songpa	11710	Southeast	13	33.8	3	111273		403
Gangdong	11740	Southeast	9	24.5	2	111274	111275	402

Legal nbhds.: legal neighborhoods in the district; Adjacent: districts sharing a boundary; Urban monitor: AirKorea urban-network monitor (used); Roadside monitor: AirKorea roadside monitor (used only in the roadside-inclusive exposure variant); KMA station(s): automatic weather stations located in the district (Jongno's station 422 reports on fewer than 80% of days, so Jongno takes the nearest station). `covariates/13_district_covariates.py`; `output/district/covariates/district_sites.csv`.

Why districts. Legal and administrative neighborhoods split and merge each other, so combining card and phone data at the neighborhood level requires splitting flows by land-area shares; merging every overlap instead gives 176 units of very unequal size (the neighborhood draft, Appendix A.1). Both kinds of neighborhood nest in the 25 districts, which are therefore the smallest units on which both data sets agree without splitting, and Seoul's urban monitoring network has one monitor per district.

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A.2 The card data: contents, scaling, and what they cannot show

Table A.2: What the raw card files contain

Column	Content	Values
TA_YM, APV_TS_D	month and approval date	Jan–Aug 2018–2020 (we use 2018–19)
APV_TS_TM	time slot	06–10, 10–18, 18–22, 22–06h
MCT_BR	merchant category	62 categories, 999 = online
MCT_REG_CD	merchant location	legal neighborhood (Seoul)
SEX_CCD, AGE_CCD	cardholder sex, age group	M/F; 10s, 20s, . . . , 60+
REG_CD	cardholder residence	legal neighborhood (Seoul), district (Gyeonggi), province (elsewhere)
REV_AMT, REV_CNT	amount, number of transactions	Shinhan's own figures × a monthly factor (Table A.3)
`cc_users.txt`: num_ccu	card users, index	residence × sex × age, April and July; first row = 100

The raw data are nine tab-separated files (733 million rows; January–August of 2018–2020), the users' file and the data layout. Each row is a cell of many cardholders; there is no cardholder or card identifier. The layout calls the amounts and counts "corrected values based on Shinhan Card sales". They are Shinhan's own amounts and counts multiplied by one factor per month and rounded: every count up to 60 is a multiple of 5 and the larger counts fit one two-decimal factor per month; with that factor every amount divided by c is a whole number of won (Table A.3). The factor's inverse, 19.7–20.5%, is Shinhan's share of card sales, the scaling described for Shinhan's data products (Kim et al., 2023). Levels are therefore Shinhan's sales scaled to the market; shares, elasticities and amounts per transaction are unaffected; year-month fixed effects absorb the monthly factor; a count of 5 is one Shinhan transaction.

Table A.3: The monthly scaling factor of the card data

Month	Factor c	$1/c$	Counts = round($c\,k$)	Amount/c whole won	c fitted by category
2018-01	4.91	20.4%	100.0%	100.0%	4.908–4.911
2018-02	4.89	20.4%	99.5%	100.0%	4.888–4.890
2018-03	4.92	20.3%	100.0%	100.0%	4.918–4.920
2018-04	4.96	20.2%	100.0%	100.0%	4.948–4.961
2018-05	4.95	20.2%	98.1%	100.0%	4.949–4.952
2018-06	4.94	20.2%	100.0%	100.0%	4.940–4.948
2018-07	4.94	20.2%	100.0%	100.0%	4.939–4.941
2018-08	4.94	20.2%	100.0%	100.0%	4.940–4.948
2019-01	5.04	19.8%	100.0%	100.0%	5.039–5.055
2019-02	5.03	19.9%	100.0%	100.0%	5.030–5.031
2019-03	5.05	19.8%	99.4%	100.0%	5.048–5.052
2019-04	5.07	19.7%	100.0%	100.0%	5.070–5.071
2019-05	5.01	20.0%	99.5%	100.0%	5.000–5.012
2019-06	5.01	20.0%	99.5%	100.0%	5.000–5.011
2019-07	5.06	19.8%	97.8%	100.0%	5.057–5.063
2019-08	5.07	19.7%	100.0%	100.0%	5.064–5.076

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A.3 Who is covered

Shinhan's share of credit-card sales among the specialized card companies was 21.9% in the third quarter of 2019; KT held about 31% of Korean mobile connections. The users' file (`num_ccu`) is an index of card users by residence, sex and age (first row = 100, per `cc_users.xlsx`), so only shares and relative levels are meaningful.

Table A.4: Card users and KT residents by age

	Shinhan card users (num_ccu index)		KT residents present at night
	Share	Number (m)	Share
Ages 10–39	41.2%	2.54	41.6%
Ages 40–59	41.7%	2.12	34.6%
Ages 60+	17.1%	1.46	23.8%
Women	52.6%		
By district: correlation of log card-user index and log KT night population 0.96; relative card coverage (share of card users / share of KT residents): median 0.98 (P10 0.90, P90 1.18)			

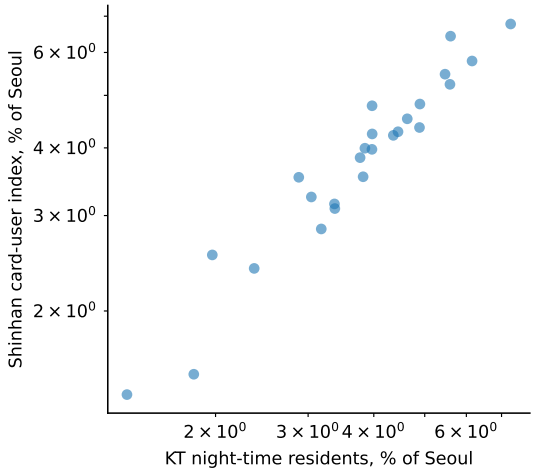


Figure A.1: Each district's share of Seoul's Shinhan card-user index against its share of KT night-time residents (log scales).

What the table shows.

- Card users (index) and KT night-time residents have similar shares aged 10–39; card users are more often aged 40–59 and less often 60+; 53% of card users are women.
- Across districts the two are very highly correlated (0.96 in logs); a district's share of card users relative to its share of KT residents lies between 0.90 and 1.18 for the middle 80% of districts, much closer to one than across neighborhoods.

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A.4 Variation in pollution

Table A.5: Balance of the district pollution panel and variance shares

Check	Value
District-days	12,150 of 12,150 (25 districts × 486 days)
Missing log PM2.5	0
Shares: daily mean, persistent, residual	0.927, 0.013, 0.061 (sum 1.000)

Table A.6: City PM2.5 variation left after fixed effects and weather

Controls	Residual SD of PM2.5 (μg/m^3)	Share of variance left	R^2
2018–19 (486 days)			
Raw	18.0	100%	0.00
Day of week + year-month + holidays	16.1	80%	0.25
+ weather (main controls)	12.6	49%	0.56
Day of week + year-week + holidays	14.6	65%	0.47
+ weather	11.7	42%	0.66
+ weather $t \pm 2$	11.7	42%	0.67
2019 (243 days)			
Raw	19.7	100%	0.00
Day of week + year-month + holidays	17.5	79%	0.29
+ weather (main controls)	14.4	53%	0.54
Day of week + year-week + holidays	15.8	65%	0.49
+ weather	12.3	39%	0.71
+ weather $t \pm 2$	12.3	39%	0.73

What the table shows.

- The city-wide daily mean accounts for 93% of the variance of log PM2.5 across district-days, persistent district differences for 1% and the district-day residual for 6%; identifying variation in the panel with district × day fixed effects is that residual.
- Calendar fixed effects and weather leave about half of the city PM2.5 variance; year-week fixed effects remove more than year-month fixed effects.

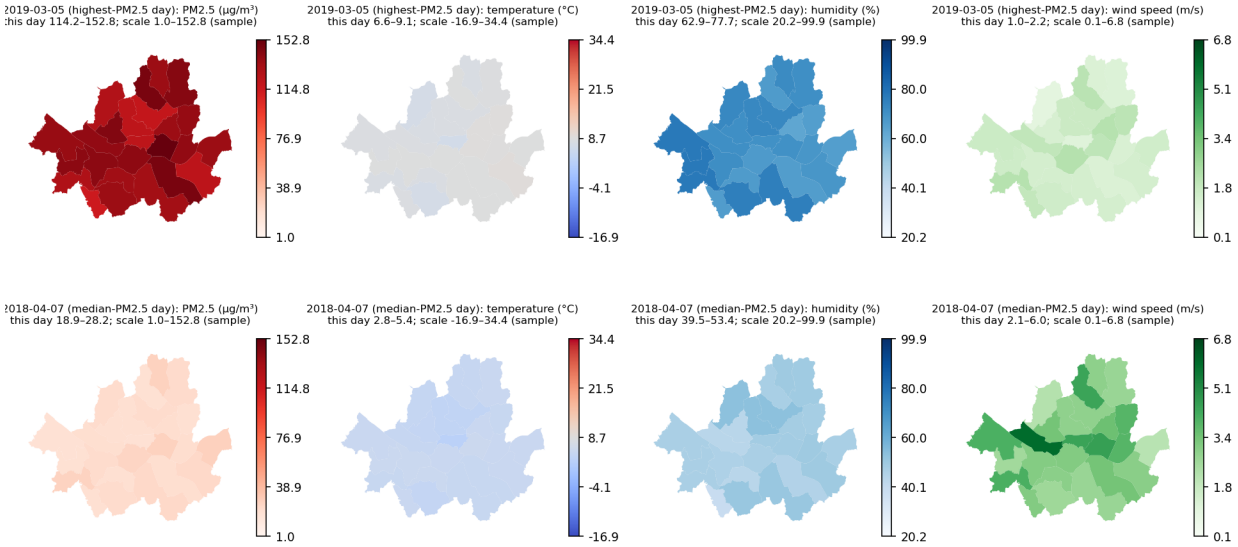


Figure A.2: PM2.5, temperature, humidity and wind speed across districts on the highest-PM2.5 day and on a median day. Each variable has one color scale for both days, from its minimum to its maximum over all districts and all sample days (January–August 2018–19; labeled at both ends), so colors are comparable across the two days; each panel's title gives that day's own range.

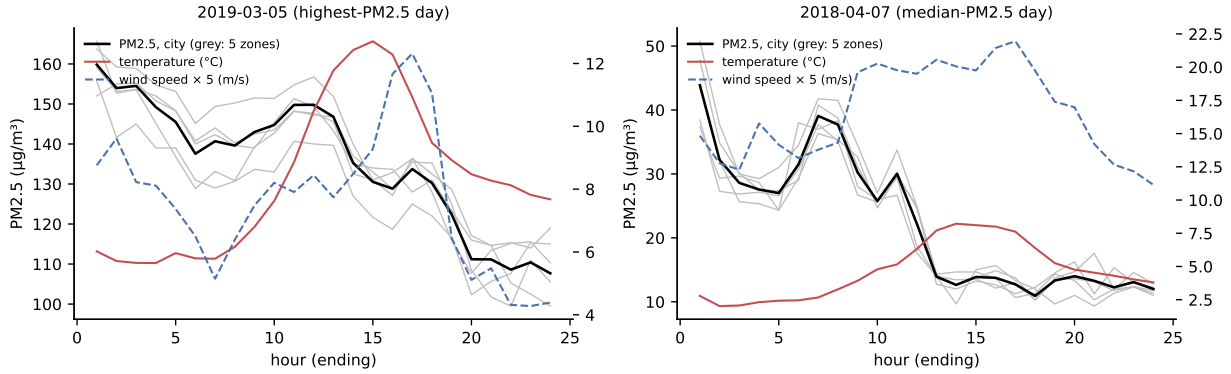


Figure A.3: Hourly PM2.5 (city in black, five areas in grey), temperature and wind speed on the same two days.

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A.5 Destination types

Each merchant district is classified by KT's ratio of daytime to night-time presence of Seoul residents and by the share of its card spending in dining and leisure categories (terciles of the 25 districts).

Table A.7: Destination types

Group (terciles)	Neighborhoods	Median day/night ratio	Median dining-leisure share	Share of card spending
Day/night population ratio: residential	9	0.84	27.1%	24%
Day/night population ratio: mixed	7	0.89	25.0%	25%
Day/night population ratio: commercial	9	1.09	29.5%	51%
Dining and leisure share of card spending: low	9	0.87	21.7%	40%
Dining and leisure share of card spending: middle	7	0.86	27.1%	35%
Dining and leisure share of card spending: high	9	0.97	36.4%	26%

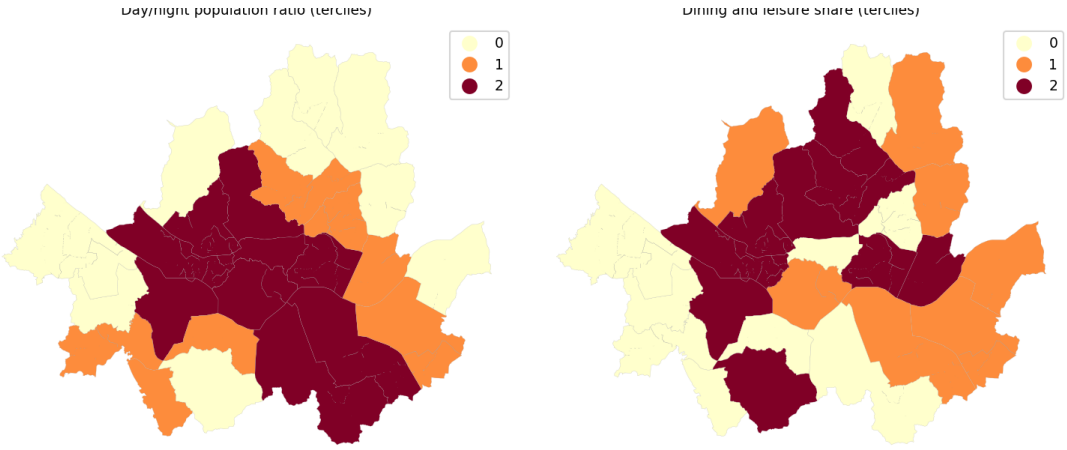


Figure A.4: Destination types by district (terciles).

Table A.8: Spending by destination type: stacked Poisson elasticities relative to the lowest tercile

Destination type (relative to the lowest tercile)	PPML	PPML-IV wind
Day/night ratio: mixed	**0.002** (0.004)	**-0.017** (0.016)
Day/night ratio: commercial	**0.003** (0.005)	**-0.017** (0.016)
Day/night ratio: online	**-0.056*** (0.029)	**0.111** (0.090)
Dining-leisure share: middle	**0.006** (0.004)	**-0.018** (0.014)
Dining-leisure share: high	**-0.003** (0.002)	**-0.020**** (0.009)
Dining-leisure share: online	**-0.057**** (0.029)	**0.115** (0.089)

Notes: Stacked by destination type (terciles of destination districts; online separately). Zeros kept; separated observations dropped (observations used 47,736). PPML-IV SEs corrected. Reading: each coefficient is the change, per log point of PM2.5, in log(spending in the band / spending in the reference band), i.e. roughly the percent change of the band relative to the reference (the lowest tercile). Example: commercial destinations, PPML, 0.003: a typical day-to-day rise in log PM2.5 (0.55 log points) shifts it by about 0.2%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Without an instrument, online spending is lower relative to the lowest tercile; physical destination types do not differ.
- With the wind control function, spending in the districts with the highest dining-and-leisure share is lower relative to the lowest tercile.

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B City time series

B.1 Instruments

Table B.1: Why the wind instrument is weak: first stage under different controls

Controls	8 sectors, HAC F	8 sectors, robust F	Westerly, HAC F	R^2	Days
No controls	5.4	5.5	28.3	0.07	486
Calendar (day of week, year-month, holidays)	2.4	3.7	10.1	0.29	486
+ temperature	2.5	3.2	12.1	0.47	486
+ humidity, precipitation	3.7	2.9	10.7	0.52	486
+ wind speed (= main controls)	3.4	2.8	16.1	0.57	486
Main controls, windier half of days	11.0	9.9	20.3	0.52	243
Main controls, calmer half of days	1.4	0.7	1.4	0.68	243
Main controls, previous day's sectors	3.7			0.57	485

Table B.2: How much of each weather variable wind direction explains

Variable	R^2, calendar	R^2, calendar + sectors	Partial R^2 of sectors
Temperature	0.92	0.93	0.06
Humidity	0.39	0.46	0.11
Precipitation	0.07	0.17	0.11
Wind speed	0.10	0.16	0.06
PM2.5	0.25	0.29	0.05
Share of days by sector (1 = 0–45°, …, 8 = 315–360°; 0 = calm): 1: 5%, 2: 13%, 3: 6%, 4: 1%, 5: 13%, 6: 26%, 7: 29%, 8: 8%			

Table B.3: Instrument strength and 2SLS estimates across instrument sets

Instrument	FS F	KP F	Eff. F	Eff. F, mob.	All	Online	Online sh.	Outside home district sh.	Time outside home district
Wind direction, 8 sectors (daily mean vector)	3.6	3.2	2.3	1.7	**6.52** (5.32)	**1.58** (3.46)	**0.17** (1.13)	**0.07** (0.47)	**-0.26** (0.22)
Modal wind direction, 8 sectors	1.4	1.2	1.2	1.5	**14.79*** (8.78)	**7.02** (5.16)	**2.02** (1.66)	**-0.61** (0.66)	**-0.15** (0.18)
Westerly, 180–360° (a priori)	15.6	13.9	15.6	4.5	**8.50** (6.06)	**1.43** (3.82)	**0.06** (1.27)	**0.21** (0.54)	**-0.07** (0.25)
Westerly, arc chosen on held-out days	11.3	10.1	11.3	2.2	**4.62** (5.81)	**-0.54** (4.10)	**-0.51** (1.40)	**0.24** (0.58)	**-0.28** (0.35)
Speed-weighted sector shares	2.9	2.6	2.5	1.7	**5.99** (4.99)	**1.35** (3.22)	**0.25** (1.04)	**0.06** (0.42)	**-0.02** (0.15)
Sectors × windy day	12.3	10.8	4.4	1.1	**4.72** (3.93)	**1.19** (2.55)	**0.08** (0.85)	**0.03** (0.36)	**0.03** (0.15)
Inversions: count and 03 KST strength	8.8	7.8	9.5	4.2	**-8.23**** (3.88)	**-7.37**** (2.96)	**-2.33**** (0.96)	**0.94**** (0.38)	**0.00** (0.18)
Inversion strength (03 KST)	17.0	15.2	17.0	8.1	**-7.00*** (3.92)	**-6.98**** (3.01)	**-2.23**** (0.99)	**0.93**** (0.40)	**0.02** (0.20)
Surface-based inversion strength	0.4	0.4	0.4	0.3	**-62.96** (92.15)	**-52.09** (77.15)	**-16.31** (24.30)	**6.81** (10.38)	**-0.60** (1.31)
925 minus 1000 hPa temperature	18.3	16.2	18.3	10.3	**-3.32** (2.77)	**-2.95** (1.87)	**-1.08*** (0.62)	**0.33** (0.26)	**0.05** (0.11)
Three inversion strengths	8.3	7.3	9.6	3.7	**-1.71** (1.95)	**-2.38*** (1.38)	**-0.89*** (0.47)	**0.33*** (0.18)	**0.10** (0.11)

Notes: First stage (OLS of PM2.5 on the instruments; HAC Wald F) and 2SLS of the listed outcome. Modal direction: the most frequent hourly sector of the city wind. Westerly (held out): wind from 213.75–326.25° (south-west to north-west), the arc where PM2.5 is significantly higher, chosen on September–December 2018–19 and January–August 2020, days outside the estimation sample (Appendix Figure B.1). 2SLS in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending, inversions, −8.23, is −4.0% of its mean (205.92); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: City series: reduced forms on the instruments

Outcome	Wind sectors: joint F [p]	Inversion count	Inversion strength (°C)
All spending (bn KRW)	1.34 [0.226]	**-1.708** (1.066)	**-1.283** (0.943)
Offline (bn KRW)	2.13 [0.037]	**-0.941**** (0.459)	**0.212** (0.461)
Online (bn KRW)	1.13 [0.338]	**-0.767** (0.836)	**-1.495**** (0.625)
Online share (pp)	1.23 [0.282]	**-0.224** (0.277)	**-0.482**** (0.199)
Outside home district share (pp)	0.80 [0.589]	**0.050** (0.116)	**0.213**** (0.086)
Time outside home district (pp)	1.91 [0.064]	**-0.053** (0.053)	**0.020** (0.060)
Km from home	1.68 [0.110]	**-0.006** (0.005)	**0.002** (0.006)

Notes: Reduced form: OLS of the outcome on the instruments and the controls of equation (1). OLS in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: the change in the outcome (spending in billion KRW; shares in pp) for one more inversion (of three soundings) or one degree of inversion strength. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Specification.

- Equation (1) by 2SLS with the listed instruments, as in the neighborhood draft: wind-sector dummies (direction of the day's mean wind vector); the modal direction (the day's most frequent hourly sector); a westerly indicator chosen a priori (180–360°) and the one chosen on held-out days (213.75–326.25°, Figure B.1); wind-run shares; sectors × windy day; inversion definitions. Year-week clusters.

What the table shows.

- Among the wind instruments, the westerly indicators are the strongest: the a priori one (180–360°; effective F 15.6) and the one chosen on held-out days (11.3), both below the 23.1 critical value; the eight sectors of the mean direction (2.3) and of the modal direction (1.2) are weak. No wind instrument gives a significant 2SLS estimate at the 5% level.
- Among the inversion definitions, the 03 KST strength alone and the 925–1000 hPa difference have the strongest first stages (effective F 17–18); the surface-based strength is weak. With the inversion instruments, all and online spending and the online share are lower and the share outside the home district higher.
- Among the inversion definitions, the 03 KST strength alone and the 925–1000 hPa difference have the strongest first stages (effective F 17–18); the surface-based strength is weak. With the inversion instruments, all and online spending and the online share are lower and the share outside the home district higher.
- Wind direction moves offline spending and, at the 10% level, time outside the home district in the reduced form although its first stage is weak; inversion strength lowers online spending and the online share and raises the share outside the home district.

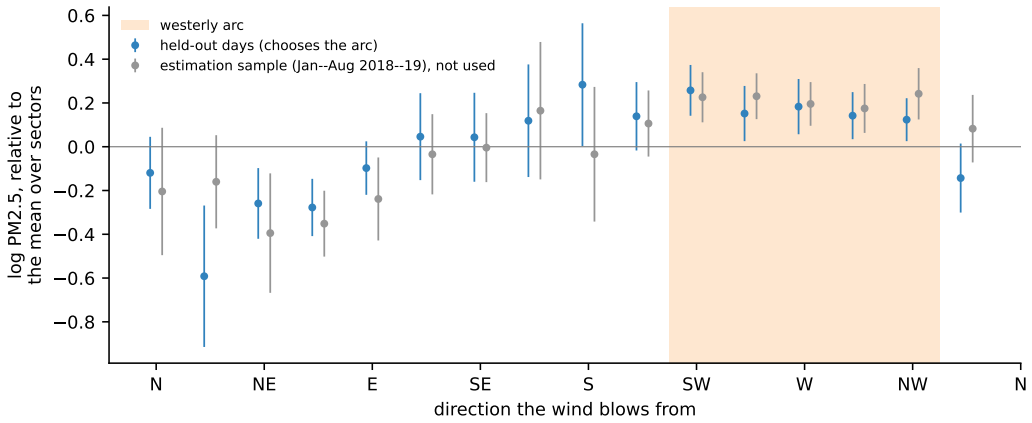


Figure B.1: Which wind directions bring PM2.5: the effect of the day's wind direction (16 sectors of 22.5°) on log city PM2.5, relative to the mean over the 16 sectors, with the weather terms, day-of-week and year-month fixed effects; 95% intervals, Newey–West (7). Blue: held-out days (September–December 2018–19 and January–August 2020, 487 days), which choose the westerly arc (shaded): the longest run of adjacent sectors with a positive effect significant at 5%. Grey: the estimation sample (January–August 2018–19), not used to choose. The profile uses city means of the 25 urban monitors and of the KMA stations, so it is the same in both drafts.

What the figure shows.

- On the held-out days, PM2.5 is significantly higher with wind from the five adjacent sectors between south-west and north-west (213.75–326.25°) and lower with wind from the north-north-east to east-north-east; the estimation sample gives the same pattern, with the same five sectors significantly higher. Wind from due south is also higher on the held-out days (5% level), but the sector between it and the arc is not (10% level), so the rule leaves it out.
- The arc covers 48% of the held-out days and 65% of the estimation days. The a priori indicator (180–360°) also includes the sector just west of north (326.25–348.75°), where PM2.5 is lower on the held-out days.

Table B.5: Instruments in the district panel: first stage and 2SLS of the headline shares

	First-stage F		2SLS (pp per 10 μg/m^3)			KP F		
Instrument	spending	mobility	Outside home district	Online	Time outside home district	Outside home district	Online	Time
Wind direction, 8 sectors (main)	2.3	1.2	**-0.481** (0.426)	**1.124** (0.975)	**-0.263** (0.267)	3.0	3.0	1.2
Modal wind direction, 8 sectors	3.9	1.7	**-0.709*** (0.377)	**0.914** (0.798)	**-0.077** (0.176)	4.0	4.0	2.4
Westerly, 180–360° (a priori)	12.3	1.9	**-0.236** (0.481)	**0.410** (1.111)	**-0.089** (0.311)	13.0	13.0	2.0
Westerly, arc chosen on held-out days	9.4	1.9	**-0.287** (0.468)	**0.153** (1.047)	**0.291** (0.355)	11.8	11.8	2.6
Inversions: count and 03 KST strength	16.9	5.5	**0.686**** (0.279)	**-1.613**** (0.666)	**0.046** (0.148)	17.1	17.1	5.6

Notes: District × day. First stage: OLS of PM2.5 in levels, P_{rt}, on the instruments with the panel's fixed effects and weather terms, cluster-robust Wald F (district and date clusters), on the spending sample (2018–19) and the mobility sample (2019). 2SLS of the share outcomes, weighted by mean spending or people-hours; district-days without spending dropped. Wind instruments at the district's own direction; inversions city-wide. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: spending outside home district, wind, −0.481: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by −0.652, against a mean of 45.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Specification.

- First stage: $P_{rt} = Z'_{rt}\pi + \alpha_r + \lambda_{\text{ym}} + \lambda_{\text{dow}} + \lambda_{\text{hol}} + \lambda_B + \lambda_S + \lambda_{\text{wc}} + \lambda_{\text{hd}} + v_{rt}$, with Z_{rt} at the district's own station (the inversions city-wide); cluster-robust Wald F, district and date clusters. 2SLS of the share outside the home district, the online share and the share of time outside the home district (equation (2)).

What the table shows.

- With PM2.5 in levels, the westerly indicators are the strongest wind instruments (first-stage F 9.4 for the held-out arc, 12.3 a priori, on the spending sample; 1.9 on the 2019 mobility sample), the modal direction (3.9) is somewhat stronger than the eight sectors of the mean direction (2.3), and the inversions are at 16.9 (5.5 for mobility). All are weaker than with log PM2.5 (32.2, 38.7, 7.7, 5.6 and 22.7).
- With the westerly instruments, the share of spending outside the home district, the online share and the share of time outside the home district do not change significantly. With the modal direction, the share outside the home district is lower (10% level); with inversions, the share outside the home district is higher and the online share lower (5% level).

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B.2 Inference, functional form, controls and exposure

Table B.6: City series: standard errors under alternative assumptions

Outcome, estimator	Estimate	Year-week clusters (main)	Newey–West (7 lags)
All spending, OLS	**-1.641**	0.811 [0.04]	0.745 [0.03]
All spending, 2SLS inv.	**-8.227**	3.877 [0.03]	4.205 [0.05]
Online, OLS	**-0.866**	0.641 [0.18]	0.617 [0.16]
Online, 2SLS inv.	**-7.367**	2.960 [0.01]	3.272 [0.02]
Online share (pp), OLS	**-0.238**	0.228 [0.30]	0.222 [0.28]
Online share (pp), 2SLS inv.	**-2.334**	0.965 [0.02]	1.055 [0.03]
Outside home district share (pp), OLS	**0.087**	0.077 [0.26]	0.068 [0.20]
Outside home district share (pp), 2SLS inv.	**0.944**	0.385 [0.01]	0.419 [0.02]
Time outside home district (pp), OLS	**-0.060**	0.022 [0.01]	0.026 [0.02]
Time outside home district (pp), 2SLS inv.	**0.003**	0.183 [0.99]	0.228 [0.99]
Km from home, OLS	**-0.005**	0.002 [0.01]	0.003 [0.03]
Km from home, 2SLS inv.	**-0.001**	0.019 [0.98]	0.024 [0.98]

Notes: Standard error [p-value]. OLS and 2SLS (inversions) in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending, OLS, -1.64, is -0.8% of its mean (206.07); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.7: City series: elasticities, log-log OLS and PPML

Outcome (elasticity)	Log-log OLS	PPML
All spending	**-0.005** (0.012)	**-0.009** (0.012)
Offline	**-0.001** (0.006)	**-0.004** (0.006)
Online	**-0.018** (0.044)	**-0.051** (0.055)
Home district	**0.000** (0.007)	**-0.002** (0.006)
Adjacent districts	**-0.001** (0.006)	**-0.004** (0.006)
Other districts	**-0.005** (0.009)	**-0.007** (0.009)
Transactions	**-0.003** (0.003)	**-0.004** (0.003)
People-hours in Seoul (24 h)	**0.010***** (0.003)	**0.010***** (0.003)
People-hours, home district	**0.012***** (0.004)	**0.012***** (0.003)
People-hours, adjacent districts	**0.002** (0.004)	**0.003** (0.004)
People-hours, other districts	**-0.001** (0.006)	**0.000** (0.005)

Notes: City × day; log-log OLS and PPML (Poisson pseudo-maximum likelihood); no city-day is zero, so both use the same days (486; 243 for people-hours). Year-week clusters. Reading: elasticities: a 1% rise in city PM2.5 changes the outcome by the coefficient in percent. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.8: City series: alternative controls and weather measures

	All (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district (pp)	Time outside home district (pp)
Main controls, OLS	**-1.64****	**-0.87**	**-0.24**	**0.09**	**-0.06*****
	(0.81)	(0.64)	(0.23)	(0.08)	(0.02)
Main controls, 2SLS inversions	**-8.23****	**-7.37****	**-2.33****	**0.94****	**0.00**
	(3.88)	(2.96)	(0.96)	(0.38)	(0.18)
No weather controls, OLS	**0.61**	**0.30**	**0.10**	**-0.03**	**0.01**
	(0.72)	(0.51)	(0.18)	(0.06)	(0.03)
No weather controls, 2SLS inversions	**1.27**	**-0.52**	**-0.21**	**0.14**	**0.15***
	(1.63)	(1.13)	(0.38)	(0.16)	(0.08)
Temperature bins, OLS	**-1.89****	**-1.05**	**-0.30**	**0.11**	**-0.06*****
	(0.87)	(0.66)	(0.23)	(0.08)	(0.02)
Temperature bins, 2SLS inversions	**-6.94**	**-7.04****	**-2.22****	**0.94****	**0.04**
	(4.22)	(3.23)	(1.04)	(0.43)	(0.18)
Lagged PM2.5, OLS	**-1.10**	**-0.51**	**-0.13**	**0.06**	**-0.09*****
	(0.81)	(0.61)	(0.21)	(0.07)	(0.03)
Lagged PM2.5, 2SLS inversions	**-12.51***	**-10.74****	**-3.35****	**1.45****	**0.00**
	(6.62)	(5.22)	(1.69)	(0.72)	(0.21)
Plus the station nearest City Hall, OLS	**-1.79****	**-0.93**	**-0.25**	**0.09**	**-0.07*****
	(0.83)	(0.66)	(0.23)	(0.08)	(0.02)
Plus the station nearest City Hall, 2SLS inversions	**-6.88****	**-6.62*****	**-2.16*****	**0.85*****	**-0.04**
	(2.80)	(2.20)	(0.72)	(0.30)	(0.12)
That station only, OLS	**-1.97****	**-1.01**	**-0.27**	**0.09**	**-0.07*****
	(0.79)	(0.63)	(0.22)	(0.08)	(0.02)
That station only, 2SLS inversions	**-8.67****	**-7.26*****	**-2.23*****	**0.87*****	**-0.02**
	(3.50)	(2.64)	(0.85)	(0.33)	(0.15)
Dropping the 5% coldest and warmest days, OLS	**-1.69****	**-0.91**	**-0.26**	**0.09**	**-0.06*****
	(0.82)	(0.65)	(0.23)	(0.08)	(0.02)
Dropping the 5% coldest and warmest days, 2SLS inversions	**-7.56***	**-7.51****	**-2.40****	**0.94****	**0.03**
	(4.13)	(3.22)	(1.06)	(0.42)	(0.20)

Notes: OLS and 2SLS (inversions) in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending, main controls, OLS, -1.64, is -0.8% of its mean (206.07); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.9: City series: exposure measures

	Card spending, 2018–19					Mobility, 2019, 24 hours	
	All (bn KRW)	Offline (bn KRW)	Online (bn KRW)	Online share (pp)	Outside home district share (pp)	Time outside home district (pp)	Km from home
District's urban monitor (main): OLS	**-1.64****	**-0.78****	**-0.87**	**-0.24**	**0.09**	**-0.06*****	**-0.005****
	(0.81)	(0.36)	(0.64)	(0.23)	(0.08)	(0.02)	(0.002)
District's urban monitor (main): Distributed lag, P_t	**-1.43***	**-0.93****	**-0.50**	**-0.11**	**0.08**	**-0.07***	**-0.006***
	(0.78)	(0.42)	(0.59)	(0.20)	(0.08)	(0.04)	(0.004)
District's urban monitor (main): 2SLS inversions	**-8.23****	**-0.86**	**-7.37****	**-2.33****	**0.94****	**0.00**	**-0.001**
	(3.88)	(1.79)	(2.96)	(0.96)	(0.38)	(0.18)	(0.019)
Neighborhood inverse-distance average: OLS	**-1.64****	**-0.78****	**-0.87**	**-0.24**	**0.09**	**-0.06*****	**-0.005****
	(0.81)	(0.36)	(0.64)	(0.23)	(0.08)	(0.02)	(0.002)
Neighborhood inverse-distance average: Distributed lag, P_t	**-1.39***	**-0.88****	**-0.51**	**-0.12**	**0.07**	**-0.07***	**-0.006**
	(0.78)	(0.42)	(0.59)	(0.20)	(0.08)	(0.04)	(0.004)
Neighborhood inverse-distance average: 2SLS inversions	**-8.19****	**-0.85**	**-7.33****	**-2.32****	**0.94****	**0.00**	**-0.001**
	(3.84)	(1.78)	(2.94)	(0.96)	(0.38)	(0.18)	(0.019)
Including roadside monitors: OLS	**-1.61****	**-0.77****	**-0.84**	**-0.23**	**0.08**	**-0.06*****	**-0.005****
	(0.80)	(0.35)	(0.63)	(0.23)	(0.08)	(0.02)	(0.002)
Including roadside monitors: Distributed lag, P_t	**-1.38***	**-0.88****	**-0.50**	**-0.12**	**0.07**	**-0.07***	**-0.006***
	(0.77)	(0.41)	(0.58)	(0.20)	(0.08)	(0.04)	(0.004)
Including roadside monitors: 2SLS inversions	**-8.17****	**-0.88**	**-7.30****	**-2.31****	**0.93****	**0.00**	**-0.001**
	(3.85)	(1.77)	(2.94)	(0.96)	(0.38)	(0.18)	(0.019)

Notes: OLS, distributed-lag OLS and 2SLS (inversions) in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: all spending, main exposure, OLS, -1.64, is -0.8% of its mean (206.07); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Specification.

- Exposure variants: the mean over districts of the neighborhood inverse-distance PM2.5 (the neighborhood draft's exposure), or of each district's urban and roadside monitors.

What the table shows.

- Newey–West standard errors are close to the year-week clustered ones. Log-log OLS and PPML elasticities are close to zero for spending; people-hours in Seoul and in the home district rise slightly with pollution.
- Without weather controls the OLS spending estimates are positive and insignificant and the inversion estimates disappear; the other control sets change little; the fall in time outside the home district holds in every OLS row.
- The exposure measure (district monitors, the neighborhood interpolation averaged over districts, roadside monitors included) makes no difference.

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B.3 Additional outcomes

Table B.10: City series: additional outcomes

	Offline outside home district (pp)	Other districts share (pp)	Own distr. + online share (pp)	Km, offline (spending-w.)	Km, offline (trans.-w.)	Trans-actions (m)	People-hours in Seoul (m)	Time outside home district, 60+ (pp)	Time outside own distr. (pp)	Time outside home district, 06–22h (pp)
A. Least squares										
Year-month FE (main)	**-0.05**	**0.05**	**-0.09**	**-0.004**	**0.003**	**-0.021*****	**0.433*****	**-0.06*****	**-0.06*****	**-0.05***
	(0.07)	(0.04)	(0.08)	(0.008)	(0.003)	(0.006)	(0.085)	(0.02)	(0.02)	(0.03)
Year-week FE, weather $t \pm 2$	**0.03**	**0.01**	**0.03**	**0.007**	**0.005**	**-0.015****	**-0.337****	**-0.05****	**-0.04**	**-0.03**
	(0.08)	(0.04)	(0.08)	(0.010)	(0.003)	(0.006)	(0.135)	(0.03)	(0.03)	(0.03)
B. Distributed lag ($P_{t+2}, \ldots, P_{t-3}$)										
Year-month FE: coefficient on P_t	**0.02**	**0.05**	**-0.08**	**0.004**	**0.007****	**-0.036*****	**0.452*****	**-0.06****	**-0.07***	**-0.05**
	(0.06)	(0.04)	(0.08)	(0.007)	(0.003)	(0.013)	(0.152)	(0.03)	(0.04)	(0.05)
Leads P_{t+1}, P_{t+2}: joint p	0.75	0.43	0.42	0.92	0.62	0.03	0.34	0.00	0.00	0.00
Sum of $P_{t+2}, \ldots, P_{t-3}$	**-0.09**	**0.05**	**-0.09**	**-0.007**	**-0.000**	**-0.027*****	**0.534*****	**-0.09*****	**-0.12*****	**-0.14*****
	(0.09)	(0.06)	(0.13)	(0.012)	(0.006)	(0.009)	(0.152)	(0.02)	(0.03)	(0.04)
C. Two-stage least squares and LIML										
2SLS, thermal inversions	**-0.34**	**0.45****	**-0.94****	**-0.041**	**0.012**	**-0.039**	**1.145**	**-0.08**	**0.00**	**0.09**
	(0.22)	(0.18)	(0.38)	(0.030)	(0.015)	(0.036)	(1.005)	(0.11)	(0.18)	(0.26)
First-stage F (Kleibergen–Paap)	7.8	7.8	7.8	7.8	7.8	7.8	3.4	3.4	3.4	3.4
Effective F (Montiel Olea–Pflueger)	9.5	9.5	9.5	9.5	9.5	9.5	4.2	4.2	4.2	4.2
Mean of outcome	50.8	23.1	56.5	4.72	4.68	7.87	170.4	14.6	18.6	24.9

Notes: OLS, distributed-lag OLS and 2SLS (inversions) in levels: zeros kept (city sums are never zero). Days used: 486 (spending), 243 (mobility); distributed lags and $t \pm 2$ weather lose the first and last days of each year's window. Year-week clusters. Reading: each coefficient is the change in the outcome per 10 μg/m^3 more PM2.5 that day (spending in billion KRW per day, shares in percentage points, distance in km). Example: the main OLS estimate in the first column, -0.05, is -0.1% of its mean (50.76); a typical day-to-day change in city PM2.5 is 13 μg/m^3 (standard deviation of the daily change), 1.3 times the 10 μg/m^3 the coefficient refers to. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

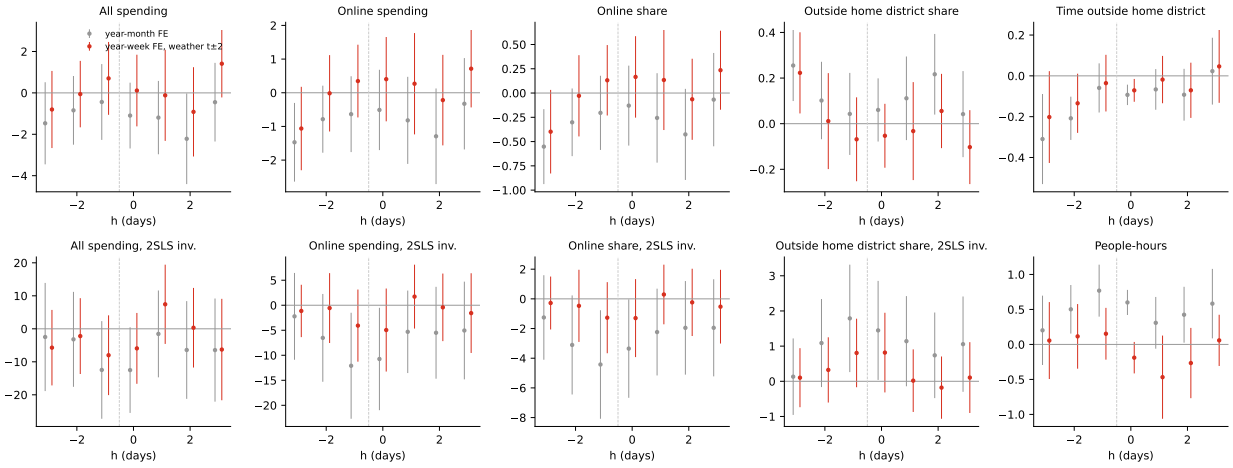


Figure B.2: Local projections, $h = -3..3$, with year-month fixed effects (grey) and with year-week fixed effects and weather at $t \pm 2$ (red); per 10 μg/m^3; 95% intervals.

What the table shows.

- People-hours in Seoul rise with pollution under year-month fixed effects and fall under year-week fixed effects; the 60+ group's time outside the home district is lower, with significant leads and a negative window sum.
- With inversions, the share of spending in other (non-adjacent) districts is higher and the share in the home district plus online lower.
- The distance of offline spending does not change with pollution, whether weighted by transactions or by spending, except a slightly longer transaction-weighted distance on the day in the distributed lag (5% level). The number of transactions is lower on more polluted days,

also with year-week fixed effects and weather leads and summed over the window (its leads are jointly significant at the 5% level); with inversions it is not significant.

Back to Section 3.1.

C The stacked band model

The main geography table (Table 10) regresses each share on pollution directly. The stacked Poisson model of equation (3), the main specification of the earlier drafts, is kept here as robustness: it estimates all bands jointly as a spending-weighted fractional multinomial logit of the shares (Section 2).

Specification.

- Equation (3), reference band: the home district. PPML-IV adds $\sum_{j\neq 0}\rho_j \mathbf{1}[k=j]\hat{v}_{rt}$, $\hat{v}_{rt}$ from $\log P_{rt} = Z'\pi + \alpha_r + \lambda_{\text{ym}} + \lambda_{\text{dow}} + \lambda_{\text{hol}} + \lambda_B + \lambda_S + \lambda_{\text{wc}} + \lambda_{\text{hd}} + v_{rt}$; Z = wind sectors at the district's station (columns 2, 5) or the inversion count and strength (columns 3, 6).
- Unit: district × day × band; spending 25 × 486 × 4, mobility 25 × 243 × 3 (24 hours). District and date clusters; PPML-IV standard errors corrected for the first stage.

Table C.1: Geography within district-day: stacked Poisson elasticities relative to the home district

	Card spending (KRW)			Time present, 24 hours (people-hours)		
	PPML	PPML-IV	PPML-IV	PPML	PPML-IV	PPML-IV
Adjacent districts	**-0.006****	**-0.006**	**-0.010**	**-0.010*****	**0.014**	**0.006**
	(0.003)	(0.010)	(0.010)	(0.004)	(0.019)	(0.020)
Other districts	**-0.011****	**0.021**	**-0.036**	**-0.008**	**0.005**	**0.006**
	(0.005)	(0.016)	(0.024)	(0.006)	(0.030)	(0.033)
Online	**-0.063****	**0.132**	**-0.516*****			
	(0.031)	(0.095)	(0.182)			
Instrument	–	wind	inversions	–	wind	inversions
First-stage F (cluster-robust Wald)		5.6	22.7		1.2	10.4
Effective F (Montiel Olea–Pflueger)		16.8	20.4		2.9	9.5
Observations		46,936–47,736			17,583	

Notes: Residence × day × band; 25 districts × 486 days × 4 bands (spending: 48,600 rows) and × 243 days × 3 bands (mobility: 18,225). PPML keeps zeros; ppmlhdfe drops observations separated by the fixed effects (bands with no spending for a residence in the whole sample, singletons). PPML-IV standard errors corrected for the estimated first stage (two-step; Appendix Table D.8). Observations dropped as separated: 864–1,664 (spending), 642 (mobility). District and date clusters. Reading: each coefficient is the change, per log point of PM2.5, in log(spending in the band / spending in the reference band), i.e. roughly the percent change of the band relative to the reference (home district). Example: other districts, PPML, −0.011: a typical day-to-day rise in log PM2.5 (0.55 log points) shifts it by about −0.6%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Without an instrument (PPML), spending in adjacent districts, in other districts and online is lower relative to spending in the home district on more polluted days, and so is time present in adjacent districts.
- With the wind control function, no band moves significantly relative to the home district. With inversions, whose first stage is close to the critical value for spending (effective F 20.4), online spending falls relative to the home district and the physical bands do not move. The wind first stage is weak for mobility (effective F 2.9) and the inversion first stage below the critical value; instrumented mobility estimates are imprecise.

Table C.2: The stacked estimates as changes in shares, next to the linear share regressions

Change in the share, pp per 10 μg/m^3 of PM2.5	No instrument	Wind	Inversions
Spending: share outside home district, from the stacked model	**0.075**	**-0.245**	**1.052**
the same share, OLS / 2SLS share regression	**0.080** (0.055)	**-0.481** (0.426)	**0.686**** (0.279)
Spending: online share, from the stacked model	**-0.287**	**0.621**	**-2.454**
the same share, OLS / 2SLS share regression	**-0.281**** (0.142)	**1.124** (0.975)	**-1.613**** (0.666)
Time present (24 hours): share outside home district, from the stacked model	**-0.054**	**0.056**	**0.032**
the same share, OLS / 2SLS share regression	**-0.056***** (0.017)	**-0.263** (0.267)	**0.046** (0.148)

Notes: Stacked rows: $\Delta s_j = s_j(\beta_j - \sum_l s_l \beta_l)$ with $\beta_0 = 0$ per log point, times $10/\bar{P}$ to express it per 10 μg/m^3 at the mean PM2.5 ($\bar{P} = 26.2$), evaluated at the pooled shares (spending, home district, adjacent districts, other districts, online: 41.8, 20.1, 23.0, 15.0%); point estimates only. Share rows: the linear share regressions of Appendix Table D.2 (OLS next to the stacked PPML, 2SLS next to the stacked PPML-IV with the same instrument). The two coincide when the share regression's weights (mean spending) and linearity do not matter. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Converted to percentage points per 10 μg/m^3 at the mean PM2.5, the stacked estimates have the same signs as the linear share regressions: without an instrument they nearly coincide (the share of spending outside the home district near zero, the online share and the share of time outside the home district slightly lower); with inversions, both show a higher share of spending outside the home district and a lower online share, the stacked conversion about half as large again; with wind, both are insignificant.

Table C.3: Weekends and holidays, stacked model

Band	$\log P$ (weekdays)	$\log P\times$ weekend or holiday
Spending (reference: home district): Adjacent districts	**-0.002** (0.004)	**-0.012***** (0.005)
Spending (reference: home district): Other districts	**-0.004** (0.007)	**-0.021**** (0.009)
Spending (reference: home district): online	**-0.050** (0.034)	**-0.072** (0.046)
Time present, 24 hours (reference: home district): Adjacent districts	**0.001** (0.005)	**-0.040***** (0.011)
Time present, 24 hours (reference: home district): Other districts	**0.006** (0.008)	**-0.056***** (0.018)

Notes: Stacked PPML (equation (3)) with band-specific $\log P \times D_t$ terms; zeros kept, separated observations dropped (observations used 17,583–47,736 of 48,600 and 18,225). District and date clusters. Reading: each coefficient is the change, per log point of PM2.5, in log(spending in the band / spending in the reference band), i.e. roughly the percent change of the band relative to the reference (home district). Example: other districts on weekdays, -0.004: a typical day-to-day rise in log PM2.5 (0.55 log points) shifts it by about -0.2%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Specification.

- Equation (3) with $\sum_{j \neq 0} \delta_j \mathbf{1}[k = j] \log P_{rt} D_t$ added, D_t = weekend or public holiday (PPML); district and date clusters.

What the table shows.

- On weekends and holidays, more polluted days have less spending and less time present in adjacent and other districts relative to the home district; these weekend–weekday differences are significant.

D District panel

Table D.1: Stacked estimates with district-border rings and with daytime mobility

	Ring 1 (adjacent) / adjacent band	Ring 2 / other districts	Ring 3+	Online
Spending, PPML	**-0.006**** (0.003)	**-0.010**** (0.005)	**-0.011*** (0.006)	**-0.063**** (0.031)
Spending, PPML-IV wind	**-0.006** (0.010)	**0.019** (0.016)	**0.025** (0.019)	**0.132** (0.095)
Spending, PPML-IV inversions	**-0.010** (0.010)	**-0.029** (0.020)	**-0.044** (0.029)	**-0.516***** (0.182)
Spending, home = own + online, PPML	**0.007** (0.008)	**0.003** (0.006)	**0.002** (0.007)	
PPML-IV wind	**-0.048**** (0.023)	**-0.022** (0.021)	**-0.015** (0.019)	
Spending, bands, online pooled with home district, PPML	**0.007** (0.008)	**0.003** (0.006)		
PPML-IV wind	**-0.048**** (0.023)	**-0.019** (0.019)		
Time present (24 h), PPML	**-0.010***** (0.004)	**-0.009** (0.006)	**-0.007** (0.006)	
PPML-IV wind	**0.014** (0.019)	**0.026** (0.026)	**-0.021** (0.036)	
PPML-IV inversions	**0.006** (0.020)	**0.008** (0.030)	**0.003** (0.035)	
Time present 06–22h, bands, PPML	**-0.007*** (0.004)	**-0.004** (0.006)		
PPML-IV wind	**0.016** (0.018)	**0.004** (0.029)		

Notes: Stacked PPML and PPML-IV as Table C.1; rings count district borders (reference: own district). Zeros kept; separated observations dropped (observations used 17,583–59,670). PPML-IV SEs corrected. Reading: each coefficient is the change, per log point of PM2.5, in log(spending in the band / spending in the reference band), i.e. roughly the percent change of the band relative to the reference (home district). Example: ring 3+, PPML, −0.011: a typical day-to-day rise in log PM2.5 (0.55 log points) shifts it by about −0.6%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.2: Panel: shares and distance

Outcome	OLS	2SLS wind	2SLS inversions	KP F wind	KP F inv.
Spending outside home district (pp)	**0.080** (0.055)	**-0.481** (0.426)	**0.686**** (0.279)	3.0	17.1
Offline spending outside home district (pp)	**-0.079**** (0.039)	**0.107** (0.261)	**-0.221** (0.144)	3.0	17.1
Online share (pp)	**-0.281**** (0.142)	**1.124** (0.975)	**-1.613**** (0.666)	3.0	17.1
Own district + online share (pp)	**-0.080** (0.055)	**0.481** (0.426)	**-0.686**** (0.279)	3.0	17.1
Km, offline (spending-weighted)	**-0.009*** (0.005)	**0.045** (0.035)	**-0.036** (0.023)	3.0	17.1
Km, offline (transaction-weighted)	**0.002** (0.003)	**-0.007** (0.020)	**0.006** (0.012)	3.0	17.1
Time outside home district, 24 h (pp)	**-0.056***** (0.017)	**-0.263** (0.267)	**0.046** (0.148)	1.2	5.6
Time outside home district, 06–22h (pp)	**-0.053**** (0.023)	**-0.480** (0.369)	**0.151** (0.220)	1.2	5.6
Km from home	**-0.005***** (0.002)	**-0.038** (0.033)	**0.004** (0.015)	1.2	5.6
Time outside home district, 60+ (pp)	**-0.057***** (0.012)	**-0.014** (0.175)	**0.010** (0.087)	1.2	5.6
Time outside own district (pp)	**-0.056***** (0.017)	**-0.263** (0.267)	**0.046** (0.148)	1.2	5.6
LIML, wind: spending outside home district (pp)		**-0.527** (0.462)			
LIML, wind: time outside home district (pp)		**-0.367** (0.425)			

Notes: District × day; OLS, 2SLS and LIML on PM2.5 in levels, P_{rt} (per 10 μg/m^3), weighted by mean spending or people-hours. Shares and distances are undefined on district-days with no spending (offline spending; people-hours), which are dropped: observations used 5,861–11,934 of 12,150 (spending) or 6,075 (mobility). District and date clusters. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: spending outside home district (OLS), 0.080: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by 0.109, against a mean of 45.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.3: Panel: levels, Poisson and log elasticities

	Poisson (PPML)			Log outcome, least squares		
Outcome (elasticity)	PPML	PPML-IV wind	PPML-IV inversions	Log OLS	Log 2SLS wind	Log 2SLS inversions
All spending	**-0.012***	**0.070*****	**-0.094****	**-0.010**	**0.064*****	**-0.064**
	(0.007)	(0.024)	(0.042)	(0.007)	(0.022)	(0.039)
Offline	**-0.005**	**0.045*****	**-0.006**	**-0.004**	**0.045*****	**-0.000**
	(0.004)	(0.014)	(0.023)	(0.005)	(0.014)	(0.029)
Online	**-0.070****	**0.176***	**-0.578*****	**-0.025**	**0.173****	**-0.278****
	(0.031)	(0.101)	(0.204)	(0.025)	(0.077)	(0.132)
Home district	**-0.001**	**0.040*****	**0.010**	**-0.000**	**0.040*****	**0.013**
	(0.004)	(0.014)	(0.024)	(0.005)	(0.015)	(0.030)
Adjacent districts	**-0.005**	**0.041*****	**-0.008**	**-0.006**	**0.039*****	**-0.002**
	(0.004)	(0.014)	(0.023)	(0.005)	(0.014)	(0.030)
Other districts	**-0.010***	**0.059*****	**-0.032**	**-0.010***	**0.060*****	**-0.024**
	(0.005)	(0.019)	(0.028)	(0.006)	(0.018)	(0.031)
People-hours in Seoul (24 h)	**0.007*****	**0.003**	**0.011**	**0.007*****	**0.003**	**0.012**
	(0.002)	(0.011)	(0.012)	(0.002)	(0.011)	(0.012)
People-hours, home district	**0.009*****	**0.003**	**0.010**	**0.009*****	**0.002**	**0.010**
	(0.002)	(0.014)	(0.015)	(0.002)	(0.015)	(0.015)
People-hours, adjacent districts	**-0.001**	**0.014**	**0.016**	**-0.002**	**0.020**	**0.020**
	(0.003)	(0.014)	(0.015)	(0.003)	(0.015)	(0.017)
People-hours, other districts	**0.001**	**0.006**	**0.016**	**0.001**	**0.007**	**0.021**
	(0.004)	(0.020)	(0.025)	(0.005)	(0.027)	(0.032)

Notes: District × day, one regression per outcome; same fixed effects and weather terms in every column. PPML and PPML-IV (control function): zeros kept, separated observations dropped (observations used 5,861–11,934; dropped 216 for spending, 214 for people-hours). Log OLS and log 2SLS: zeros dropped because log 0 is undefined (observations used 243–11,934 of 12,150 for spending, 6,075 for people-hours); weighted by mean spending or people-hours. PPML-IV SEs corrected for the first stage. District and date clusters. Reading: elasticities: a 1% rise in PM2.5 changes the outcome by the coefficient in percent. Example: all spending, PPML, −0.012: a typical day-to-day change in a district's log PM2.5 (0.55 log points, one standard deviation) changes it by about −0.7%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- Do we still need PPML-IV (Section 2)? With wind, PPML-IV and log 2SLS give nearly the same elasticities for every spending and people-hour level. With inversions they agree for offline spending, the three bands and people-hours, but PPML-IV gives a larger fall in online spending (and so in all spending) than log 2SLS; without an instrument, PPML and log OLS likewise agree except for online spending. Where they agree, log 2SLS would do; the gap for online spending is the case where heteroskedasticity (or the weighting by level) matters.

Table D.4: Panel: average amount per transaction

Amount per transaction (KRW per 10 $\mu g/m^3$)	Mean	OLS	2SLS wind	2SLS inversions
All	25,888	**-197.3***** (71.6)	**1353.8**** (590.7)	**-862.3**** (378.8)
Offline	22,589	**-57.2**** (28.1)	**760.8***** (280.6)	**-23.5** (135.2)
Online	167,123	**-4838.2**** (2154.4)	**22636.7** (15369.8)	**-36781.3***** (12382.5)
All, without single-transaction cells	25,272	**-200.9***** (71.5)	**1335.7**** (584.9)	**-888.3**** (381.9)

Notes: District × day; OLS and 2SLS on PM2.5 in levels, weighted by mean spending. The average is undefined on district-days without transactions (or without online transactions), which are dropped: observations used 11,734–11,934 of 12,150. District and date clusters. Reading: KRW per transaction per 10 $\mu g/m^3$ more PM2.5; a typical day-to-day change in a district's PM2.5 (14 $\mu g/m^3$, one standard deviation) moves the average by 1.4 times the coefficient. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.5: Panel: robustness of the headline estimates

	Stacked PPML-IV wind			Outside home district
	Adjacent districts	Other districts	Online	2SLS (pp)
Main specification	**-0.006** (0.010)	**0.021** (0.016)	**0.132** (0.095)	**-0.481** (0.426)
Exposure: AQI (log in the stacked columns; per 10 points in the share column)	**-0.009** (0.018)	**0.041** (0.029)	**0.279** (0.171)	**-0.210** (0.137)
Controls: no weather controls	**0.005** (0.004)	**0.026**** (0.012)	**0.146*** (0.083)	**-0.482**** (0.232)
Controls: max × min temperature cells	**-0.003** (0.008)	**0.019** (0.016)	**0.038** (0.103)	**-0.151** (0.440)
Controls: weather at $t \pm 1, t \pm 2$	**-0.008** (0.009)	**0.018** (0.015)	**0.104** (0.097)	**-0.452** (0.465)
Controls: wind direction at $t \pm 1, t \pm 2$	**-0.016** (0.011)	**0.010** (0.021)	**0.055** (0.119)	**-0.286** (0.570)
Controls: PM2.5 at $t-1, t-2$	**-0.002** (0.013)	**0.040*** (0.021)	**0.241*** (0.131)	**-0.871** (0.534)
Sample: 2018	**-0.010** (0.007)	**0.006** (0.015)	**0.045** (0.075)	**-0.240** (0.286)
Sample: 2019	**-0.007** (0.028)	**0.069** (0.045)	**0.404** (0.264)	**-0.542** (0.966)
Sample: excluding holidays	**-0.004** (0.009)	**0.021** (0.016)	**0.122** (0.095)	**-0.306** (0.413)
Sample: weekdays	**-0.009** (0.012)	**0.027** (0.021)	**0.116** (0.111)	**-0.457** (0.635)
Sample: weekends	**0.016** (0.012)	**0.036*** (0.021)	**0.098** (0.107)	**0.281** (0.227)
Sample: smaller districts	**-0.009** (0.013)	**0.019** (0.017)	**0.119** (0.101)	**-0.426** (0.504)
Sample: larger districts	**-0.016** (0.015)	**0.022** (0.019)	**0.149** (0.105)	**-0.500** (0.372)
Unweighted (shares)				**-0.495** (0.450)

Notes: Stacked PPML-IV (wind) on log PM2.5: zeros kept, separated observations dropped; share column 2SLS (wind) on PM2.5 in levels (pp per 10 μg/m^3): district-days without spending dropped. PPML-IV SEs corrected. Reading: each coefficient is the change, per log point of PM2.5, in log(spending in the band / spending in the reference band), i.e. roughly the percent change of the band relative to the reference (home district). Example: other districts, main specification, 0.021: a typical day-to-day rise in log PM2.5 (0.55 log points) shifts it by about 1.2%. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.6: Panel: standard errors under alternative clustering

Clustering	Stacked Adjacent districts	Stacked other districts	Outside home district (2SLS, pp)	Time outside home district (2SLS, pp)
District and date (main)	0.0097 [0.54]	0.0160 [0.18]	0.4258 [0.26]	0.2667 [0.32]
Date	0.0096 [0.54]	0.0154 [0.17]	0.4303 [0.26]	0.2636 [0.32]

Notes: Standard error [p-value]: stacked PPML-IV (wind; SEs corrected for the first stage) and 2SLS (wind) share regressions under the listed clustering. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7: Panel: exposure measures

	Stacked PPML			Stacked PPML-IV wind			Outside home district (pp)		Time
	Adjacent districts	Other districts	online	Adjacent districts	Other districts	online	OLS	2SLS wind	OLS (pp)
District's urban monitor (main)	**-0.006**** (0.003)	**-0.011**** (0.005)	**-0.063**** (0.031)	**-0.006** (0.010)	**0.021** (0.016)	**0.132** (0.095)	**0.080** (0.055)	**-0.481** (0.426)	**-0.056***** (0.017)
Neighborhood inverse-distance average	**-0.004** (0.003)	**-0.011*** (0.006)	**-0.065*** (0.034)	**-0.006** (0.010)	**0.022** (0.016)	**0.138** (0.098)	**0.090** (0.056)	**-0.458** (0.421)	**-0.060***** (0.017)
Including roadside	**-0.006**** (0.003)	**-0.012**** (0.005)	**-0.062*** (0.032)	**-0.007** (0.010)	**0.021** (0.016)	**0.135** (0.096)	**0.077** (0.054)	**-0.487** (0.423)	**-0.059***** (0.016)

Notes: Stacked PPML and PPML-IV (wind) as Appendix Table C.1, on log PM2.5; share columns OLS and 2SLS (wind) as Appendix Table D.2, on PM2.5 in levels (per 10 μg/m^3); with the listed exposure measure. Zeros and observations as in those tables. Reading: each coefficient is the change in the share, in percentage points, per 10 μg/m^3 more PM2.5 in the district that day; distances in km, counts and amounts in the outcome's units, per 10 μg/m^3. Example: spending outside home district (OLS, main exposure), 0.080: a typical day-to-day change in a district's PM2.5 (14 μg/m^3, one standard deviation of the daily change) moves it by 0.109, against a mean of 45.3. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8: PPML-IV standard errors: uncorrected, two-step corrected, and bootstrap

Stacked PPML-IV	Coefficient	SE, uncorrected	SE, two-step (reported)	SE, bootstrap
Spending, wind: Adjacent districts	**-0.0059**	0.0097	0.0097	0.0112
Spending, wind: Other districts	**0.0213**	0.0163	0.0160	0.0177
Spending, wind: online	**0.1320**	0.0952	0.0947	0.1073
Spending, inversions: Adjacent districts	**-0.0102**	0.0105	0.0105	0.0131
Spending, inversions: Other districts	**-0.0362**	0.0236	0.0243	0.0267
Spending, inversions: online	**-0.5164**	0.1651	0.1818	0.1841
Time present, wind: Adjacent districts	**0.0135**	0.0178	0.0195	0.0298
Time present, wind: Other districts	**0.0053**	0.0293	0.0297	0.0409
Time present, inversions: Adjacent districts	**0.0055**	0.0199	0.0198	0.0267
Time present, inversions: Other districts	**0.0057**	0.0334	0.0331	0.0468

Notes: Uncorrected: ppmlhdfe's two-way clustered SE, treating the control function as data. Two-step: the same, with the influence of the first-stage estimates added (Murphy–Topel). Bootstrap: dates resampled with replacement, both stages re-estimated (199 draws); clusters on dates only. Coefficients as in Appendix Table C.1.

Specification.

- As the corresponding tables of the neighborhood draft, with districts: shares and distance by OLS, 2SLS and LIML on PM2.5 in levels (per 10 μg/m^3), weighted by mean spending or people-hours; levels by PPML, PPML-IV and log-log OLS on log PM2.5; robustness: exposure (AQI; log AQI in the stacked model), controls, samples ("smaller/larger" = districts below/above the median area), weights; inference: district and date clusters (main) vs date only.

What the table shows.

- Rings by district borders give the pattern of the bands: without an instrument, spending in every ring and online is lower relative to the home district; with inversions, online spending falls. Pooling online with the home district, the wind control function puts spending in adjacent districts lower relative to home.
- Shares: without an instrument, the share of offline spending outside the home district and the online share are lower (5% level), the share of time outside the home district lower (1% level; also 06–22h and for the 60+ group), the distance from home and the spending-weighted distance of offline spending shorter; with inversions, the share of all spending outside the home district is higher and the online share lower; wind gives insignificant shares, with LIML close to 2SLS.
- Levels: without an instrument, all and online spending and spending in other districts are slightly lower and people-hours in Seoul and in the home district slightly higher on more polluted days, with log-log OLS agreeing for people-hours and for spending in other districts. With wind, spending is higher in every band; with inversions, all and online spending are lower.
- Amount per transaction: lower without an instrument (all, offline and online purchases; 5% level or better); higher with wind (all and offline) and lower with inversions (all and online).
- The stacked wind estimates and the 2SLS (wind) share outside the home district are insignificant in nearly every robustness row (the share is lower, 5% level, without weather controls);

clustering by date alone gives the same standard errors as district and date; the exposure measure (district monitor, neighborhood average, roadside-inclusive) makes no difference.

- The two-step correction changes the PPML-IV standard errors by at most about 10% (online spending, inversions). The date-cluster bootstrap (199 draws) gives standard errors 1–25% above the two-step ones for spending and 35–53% above for time present; the fall in online spending with inversions remains significant at the 1% level, and no time-present estimate is significant with either.

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E Air-quality alerts

E.1 Alerts and our readings

Issuance rules: a PM2.5 advisory when an area's hourly mean reaches 75 μg/m^3 for two hours (90 before 1 July 2018), a warning at 150 (180 before); PM10 at 150 and 300; ozone at 0.12 ppm (hourly). Seoul issued all PM alerts city-wide in 2018–19 and ozone alerts by area. Readings: the mean of each area's urban monitors and, for ozone, the highest monitor.

Table E.1: Alerts issued and the issuance rules applied to our readings

Alert type	Alerts issued (2018–19)	Rule met in our readings before issuance	Days: rule met / alert in effect both	rule only	alert only
PM25 advisory	23	23	42	12	5
PM25 warning	2	2	3	1	1
PM10 advisory	13	13	19	10	1
PM10 warning	1	1	1	1	0
O3 advisory	83	27	12	0	12
O3 advisory, area maximum over monitors	83	83			

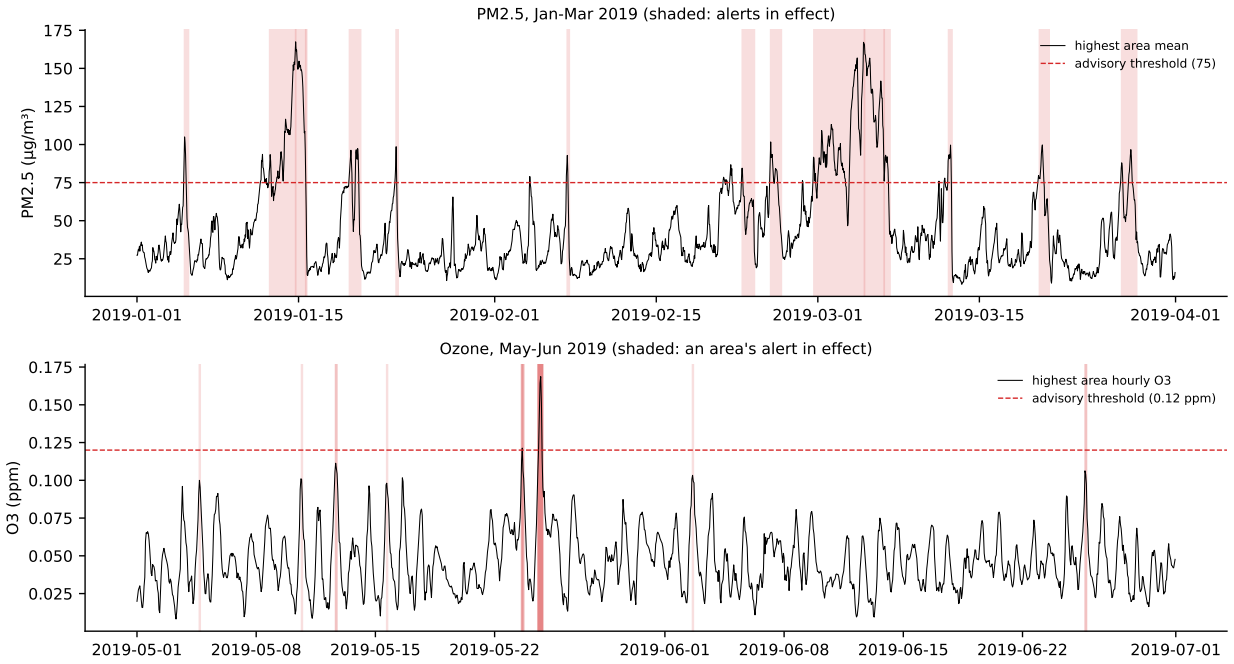


Figure E.1: Hourly readings, thresholds and alerts in effect: PM2.5 (highest area mean), January–March 2019; ozone (highest area), May–June 2019.

Table E.2: Where alert issuance actually jumps: the estimated cutoffs

Sample	Documented cutoff	Shift $\hat{\delta}$	95% set	P(alert), below	P(alert), above	Obs.	Obs. above
PM2.5, all days	90 / 75	**+5.5**	[+5.5, +11.0]	0.000	0.611	486	36
PM2.5, before 2018-07-01	90	**+10.0**	[+10.0, +11.0]	0.000	0.667	181	9
PM2.5, from 2018-07-01	75	**+5.5**	[+5.5, +7.0]	0.000	0.615	305	26
Ozone, area-slots May-Aug	0.12 ppm	**-0.001**	[-0.001, -0.001]	0.003	0.978	4,920	92

Notes: Least-squares threshold: $D = a + b\,\mathbf{1}[\text{reading} - \text{documented cutoff} \geq \delta] + e$, with δ chosen on a grid (0.5 μg/m^3; 0.001 ppm) to minimize the sum of squared residuals (Hansen, 2000); 95% set: the δ whose likelihood-ratio statistic $n(\text{SSR}(\delta) - \text{SSR}_{\min})/\text{SSR}_{\min}$ is at most 7.35. PM2.5: city × day, January–August 2018–19, D = a PM2.5 alert issued that day, reading = the day's highest two-hour minimum of area-mean PM2.5 (μg/m^3). Ozone: area × slot, May–August, D = an ozone alert in effect, reading = the slot's highest hourly ozone at any monitor of the area (ppm). Reading: $\hat{\delta}$ is how far above the documented cutoff, by our readings, issuance jumps; the comparison of Table 15 is centered there.

Specification.

- For each candidate shift δ (grid of 0.5 μg/m^3, or 0.001 ppm for ozone), OLS of the alert indicator on $\mathbf{1}[\text{reading} - \text{documented cutoff} \geq \delta]$; $\hat{\delta}$ minimizes the sum of squared residuals (the lowest such δ when several do); 95% set: the δ whose likelihood-ratio statistic is at most 7.35 (Hansen, 2000). The readings and alerts are city- or area-level, so the table is the same in both drafts.

What the table shows.

- By our readings, PM2.5 issuance jumps 5.5 μg/m^3 above the documented cutoff (95% set 5.5–11): no alert was issued below that, and 61% of the days above it got one. Before 1 July 2018 the jump is at 10 above the 90 cutoff, from then at 5.5 above the 75 cutoff.
- Ozone alerts follow the documented 0.12 ppm at the area's highest monitor ($\hat{\delta} = -0.001$ ppm): 98% of the area-slots above it are under an alert, 0.3% of those below.

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E.2 Alert tables and event studies

Table E.3: Alerts and spending: further specifications

	PM2.5 alert, city slots					Ozone alert, area slots	
	OLS	PPML (%)	OLS, days with PM2.5 $\geq$ 50	OLS, no pollution controls	OLS, PM2.5 or PM10 alert	OLS	PPML (%)
All spending	**-1.480**	**-2.4**	**-2.458**	**-2.715*****	**-2.390**	**-0.909***	**-3.4**
	(1.523)	(2.7)	(1.693)	(1.049)	(1.462)	(0.552)	(2.3)
Offline	**-0.730**	**-1.7**	**-1.054**	**-1.333****	**-1.131**	**0.011**	**-0.4**
	(0.710)	(1.5)	(0.764)	(0.546)	(0.692)	(0.214)	(0.7)
Online	**-0.751**	**-3.5**	**-1.403**	**-1.382****	**-1.259**	**-0.920****	**-15.9****
	(0.993)	(10.9)	(1.129)	(0.691)	(0.949)	(0.438)	(7.2)
Own district + online	**-1.174**	**-3.1**	**-1.923**	**-1.999****	**-1.856***	**-0.775***	**-4.4**
	(1.181)	(3.6)	(1.318)	(0.802)	(1.124)	(0.467)	(3.1)
Own district	**-0.423**	**-2.1**	**-0.519**	**-0.617****	**-0.596***	**0.145**	**0.8**
	(0.348)	(1.5)	(0.385)	(0.290)	(0.340)	(0.108)	(0.7)
Transactions (m)	**-0.033****	**-1.6****	**-0.025***	**-0.038*****	**-0.038*****	**0.003**	**0.0**
	(0.014)	(0.7)	(0.014)	(0.010)	(0.014)	(0.004)	(0.1)
Online share	**-0.013**		**-0.018**	**-0.017****	**-0.015**	**-0.017**	
	(0.011)		(0.014)	(0.008)	(0.010)	(0.011)	
Home share	**-0.009**		**-0.010**	**-0.010****	**-0.010***	**-0.005**	
	(0.006)		(0.008)	(0.004)	(0.005)	(0.007)	

Notes: OLS and PPML. Slot regressions: zeros kept (slot sums are never zero); city slots 1,944 (486 days $\times$ 4), area slots May–August 4,920; mobility 2019 only, all four slots (24 hours). Date clusters. Reading: OLS columns: change in the outcome (billion KRW per slot; shares and time in percentage points) while an alert is in effect, at the same measured pollution; PPML columns: percent change, $100(e^{\beta}-1)$. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

Table E.4: Alerts and mobility (2019)

	PM2.5 alert, city slots				Ozone, area slots	PM2.5 alert day
	OLS, 24 h	OLS, 06–22h	OLS, no pollution controls	PPML (%)	OLS	OLS (city $\times$ day)
People-hours in Seoul (m)	**-0.1623**	**0.0696**	**0.0566**	**-0.2**	**-0.0270**	**-0.3087**
	(0.4057)	(0.3572)	(0.2469)	(0.9)	(0.0355)	(1.3352)
People-hours outside home district (m)	**-0.1297***	**-0.2206****	**-0.1560*****	**-1.0**	**-0.0759**	**-0.3270**
	(0.0711)	(0.0995)	(0.0504)	(0.9)	(0.0883)	(0.2651)
Share outside home district	**-0.0016**	**-0.0031**	**-0.0032****		**0.0010**	**-0.0002**
	(0.0016)	(0.0021)	(0.0013)		(0.0011)	(0.0013)
Share outside own district	**-0.0016**	**-0.0031**	**-0.0032****		**0.0010**	**-0.0002**
	(0.0016)	(0.0021)	(0.0013)		(0.0011)	(0.0013)
Share outside own district, 60+	**-0.0015**	**-0.0023**	**-0.0031*****		**0.0002**	**-0.0000**
	(0.0012)	(0.0014)	(0.0010)		(0.0008)	(0.0010)

Notes: OLS and PPML. Slot regressions: zeros kept (slot sums are never zero); city slots 1,944 (486 days $\times$ 4), area slots May–August 4,920; mobility 2019 only, all four slots (24 hours). Date clusters. Reading: OLS columns: change in the outcome (billion KRW per slot; shares and time in percentage points) while an alert is in effect, at the same measured pollution; PPML columns: percent change, $100(e^{\beta}-1)$. * $p<0.10$, ** $p<0.05$, *** $p<0.01$.

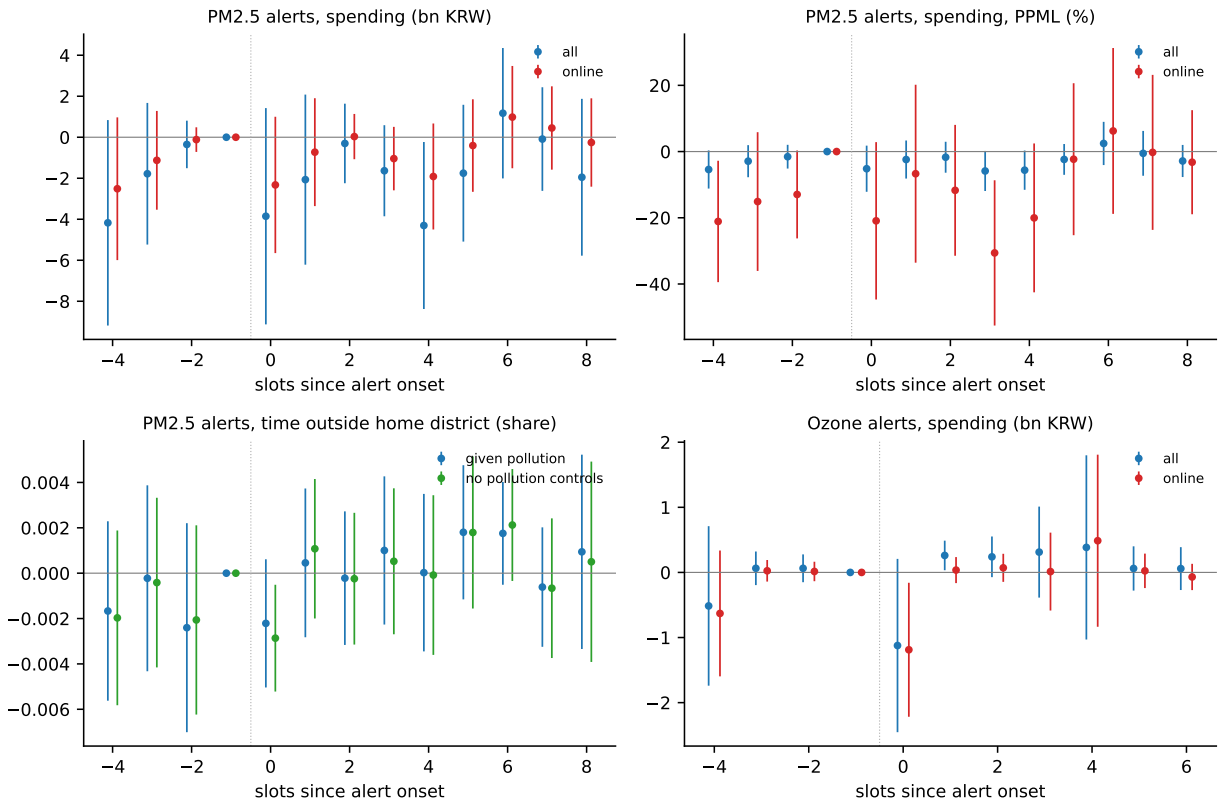


Figure E.2: Event studies around alert onsets (slots $k = -4..8$, $k = -1$ omitted; 95% intervals).

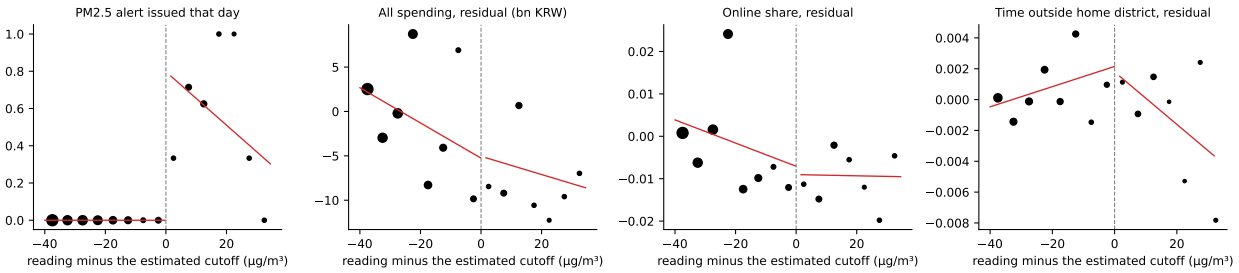


Figure E.3: Alert issuance and outcomes (net of calendar and weather) against the day's PM2.5 reading minus the estimated cutoff (the cutoff in force plus 5.5 μg/m^3); 5 μg/m^3 bins, linear fits on each side.

What the table shows.

- Given measured pollution, PM2.5 alert slots have fewer people-hours outside the home district (10% level over 24 hours, 5% in daytime slots); the share outside the home district does not move given pollution but is lower without pollution controls, also for the 60+ group. Ozone alerts and PM2.5 alert days do not change mobility.
- Event studies (spending outcomes are those of the neighborhood draft): PM2.5 spending coefficients are mostly negative and imprecise before and after onset; ozone alerts lower online

spending in the onset slot only.

- Figure E.3: around the estimated cutoff, the share of days with an alert issued rises from zero below it to about 60% above; the outcomes show no visible jump.

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F Replicating Kim and Trusov (2025)

Kim and Trusov (2025) relate the daily card spending of 3,199 Seoul cardholders aged 20–30 (85% men; July 2017–June 2019) to the Ministry of Environment's AQI in the city (district) where each purchase is made, by individual, ten category groups and day: $\ln(\text{Spending}_{iczt}) = \beta\,\text{AQI}_{zt} + \gamma' X_{z't} + \alpha_i + \mu_z + \tau_t + \varepsilon_{iczt}$, with province weather X (temperature, humidity, precipitation; Seoul is one province), individual, city, year-month, day-of-month and weekend fixed effects, and a control function whose first stage uses an indicator that the day's most frequent wind direction is 270–360° ("northwester") *and the day's maximum wind speed* as excluded instruments. In their first stage (Web Appendix Table B1; city-days, robust standard errors), wind speed carries the instrument: −7.61 (0.107) against 2.47 (0.336) for the northwester indicator. They find $\beta = 0.0242$ (0.0081) per 100 AQI points with the control function (bootstrapped standard errors) and 0.0045 (0.0015) by OLS (standard errors clustered by consumer), larger in hedonic categories (AQI × HEDUT 0.0091); no standard error is clustered by date, the level at which pollution and wind vary. Our data are cells, not consumers, so the closest analog uses residence district × sex × age cells in place of consumers, their ten category groups (our mapping of the 62 merchant categories, following the subcategory examples of their Web Appendix Table A1; their Table 4 hedonic scores) and the merchant district's own monitor; the period is January–August 2018–19, and online purchases are excluded, as in their main analysis. Table F.1 starts from their specification and changes one element at a time toward ours.

Table F.1: From Kim and Trusov's specification to ours, one change at a time

Step	OLS, robust	OLS, date clusters	CF, robust	CF, date clusters	First-stage F (robust / date)	Obs.
R0: Their specification: AQI/100; wind speed an instrument; province weather; year-month, day-of-month, weekend FE	**0.0181***** (0.0030)	**0.0181*** (0.0110)	**0.0043** (0.0108)	**0.0043** (0.0387)	523.6 / 23.9	4,261,158
R1: Wind speed moved from the instruments to the controls	**0.0182***** (0.0032)	**0.0182** (0.0115)	**-0.1035***** (0.0326)	**-0.1035** (0.1285)	109.9 / 5.3	4,261,158
R2: Our calendar (day of week, holidays) and weather terms	**0.0019** (0.0034)	**0.0019** (0.0057)	**-0.0505** (0.0486)	**-0.0505** (0.0886)	55.0 / 2.7	4,261,158
R3: Log PM2.5 instead of AQI	**0.0027** (0.0020)	**0.0027** (0.0036)	**-0.0194** (0.0187)	**-0.0194** (0.0340)	147.8 / 6.8	4,261,158
R4: All ages instead of 20–29	**0.0025**** (0.0012)	**0.0025** (0.0034)	**0.0210*** (0.0109)	**0.0210** (0.0346)	147.8 / 6.8	13,255,828
R5: Our instrument: eight wind sectors at the merchant district	**0.0025**** (0.0012)	**0.0025** (0.0034)	**0.0403***** (0.0041)	**0.0403***** (0.0097)	145.5 / 11.5	13,255,828
Their equation (3), AQI × HEDUT (r0 terms)	**0.0028**** (0.0013)	**0.0028** (0.0056)	**0.0028**** (0.0013)	**0.0028** (0.0056)		
AQI × HEDUT with our controls (r2 terms)	**0.0028**** (0.0013)	**0.0028** (0.0058)	**0.0028**** (0.0013)	**0.0028** (0.0058)		

Notes: Outcome: log in-store spending of residents aged 20–29 (all ages from R4) by residence district × sex × age group × merchant district × category group (their ten, our mapping) × day, Jan–Aug 2018–19, positive spending only (zeros drop because of the log); coefficient on AQI/100 (R0–R2) or log PM2.5 (R3–R5) at the merchant district. Fixed effects: cell, merchant district and the listed time terms. CF: control function, first stage at merchant district × day; second-stage SEs not corrected for the first stage. "Robust": heteroskedasticity-robust, close to their unclustered bootstrap; "date clusters": clustered by day, the level at which pollution and wind vary. Kim and Trusov report 0.0242 (0.0081) per 100 AQI points with the control function and 0.0045 (0.0015) by OLS (clustered by consumer), July 2017–June 2019; their first-stage coefficient on wind speed is −7.61 (0.107). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

What the table shows.

- In our data, their specification (R0) gives a positive OLS estimate, 0.018 per 100 AQI points (their OLS 0.0045, their control function 0.024). It is highly significant with heteroskedasticity-robust standard errors, close to their unclustered inference, but significant only at the 10% level when the standard errors are clustered by date, the level at which pollution varies. Their control-function estimate is small and insignificant in our data.
- Their strong first stage comes from using the day's maximum wind speed as an excluded instrument, with unclustered inference: the first-stage F is 524 with robust standard errors and 24 clustered by date. Moving wind speed into the controls (R1) leaves the northwester indicator alone, with a date-clustered first-stage F of 5, and the control-function estimate turns negative (significant only with robust standard errors).
- With our calendar and weather terms (R2), with PM2.5 (R3) and with all ages (R4), the OLS and control-function estimates are close to zero and insignificant with date-clustered standard errors. With our instrument, wind sectors at the merchant district (R5), the control-function estimate is positive and significant, as with the wind instrument in Table C.1; the reduced forms (Appendix Table B.4) suggest that wind direction moves spending other than through pollution.
- The hedonic interaction is positive (0.003 per 100 AQI points per unit of hedonic score; theirs 0.009), significant with robust standard errors and not with date clusters.
- In sum, the positive and significant estimate requires unclustered inference, and the strong first stage requires wind speed as an excluded instrument; with date clusters, wind speed as a control and fuller weather and calendar controls, our data show no effect of AQI on young cardholders' in-store spending. Their period (July 2017–June 2019) includes autumn and winter months that ours does not, and they observe individuals rather than cells.

G Fixed effects in related studies

Table G.1: Time and location fixed effects in related studies

Study	Unit	Fixed effects and controls	Inference
Deryugina et al. (2019), wind, mortality	county × day	county, state × month, month × year; flexible weather	county clusters; population weights
Schlenker and Walker (2016), airports, admissions	zip code × day	zip code, year, month, weekday, holidays with surrounding days; weather quadratics	clusters by zip code and day
Arceo et al. (2016), inversions, infant mortality	municipality × week	municipality and week (OLS); municipality, year, municipality trends (IV)	week clusters
Barwick et al. (2018), card spending on health	city × day	city, week of sample, day of week, holidays and working weekends, city trends and seasonality; weather	city clusters; KP F
Kim and Trusov (2025), card spending	consumer × category × day; spending located in the 25 districts	to be checked against the paper	to be checked
This draft, city series	city × day	day of week, year-month (main) or year-week, one FE per holiday, bridge days, days around Seollal; flexible weather	year-week clusters
This draft, panel	district × day	district, year-month, day of week, the same holiday terms, weather cells, humidity deciles	district and date clusters

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H Estimators, zeros and observations

Table H.1: Estimator, zero handling and observations, by table

Table	Estimator	Zeros	Observations used	Full sample
Table 5; App. city tables	OLS, distributed-lag OLS, 2SLS, LIML (levels, shares)	zeros kept; no city-day is zero	243–486	486 / 243
Table 6	OLS	zeros kept	243–486	486 / 243
Table 7	2SLS	zeros kept	243–478	486 / 243
App. Table B.7	log-log OLS; PPML	log needs > 0: no city-day is zero	243–486	486 / 243
Table 10	OLS, 2SLS, LIML (band shares, counts, distances)	shares undefined without spending or people-hours: dropped; counts keep zeros	5,861–11,934	12,150 / 6,075
Tables 8, 9	OLS, 2SLS (shares); PPML (levels)	shares: dropped without spending; PPML: separated dropped	11,734–11,934	12,150 / 6,075
App. Tables C.1, D.1	PPML, PPML-IV (stacked bands)	zeros kept; separated observations dropped	17,583–47,736	48,600 / 18,225
App. Table D.2	OLS, 2SLS, LIML (shares, distance)	undefined without spending or people-hours: dropped	5,861–11,934	12,150 / 6,075
App. Table D.3	PPML, PPML-IV (levels)	zeros kept; separated dropped	5,861–11,934	12,150 / 6,075
App. Table D.3	log OLS, log 2SLS (levels)	zeros dropped (log 0)	486–11,734	12,150
Table 12 (district columns)	PPML, PPML-IV	zeros kept; separated dropped	11,911–11,934	12,150
App. Table D.4	OLS, 2SLS (amount per transaction)	undefined without transactions: dropped	11,934	12,150
Table 13, App. alert tables	OLS, PPML (slots)	zeros kept (slot sums are not zero)	1,944 / 4,920	slots
Table 15	fuzzy RD, local linear (rdrobust)	no zeros; days within the bandwidth	see table	486 days

Notes: What a zero is: in the card data, a cell with no Shinhan transaction (cells with one transaction are reported, as count 5 after scaling; Appendix A.2); in the KT data, a cell below about 2.9 estimated people is suppressed and read as zero. Separated observations: ppmlhdfe drops observations whose fixed-effect groups have only zero outcomes (e.g., a residence with no online spending in the sample) and singletons; they carry no information on the slope.

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